Journal of Cleaner Production 525 (2025) 146523

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Review

Cleaner processing techniques for rubidium extraction and recovery: A critical review

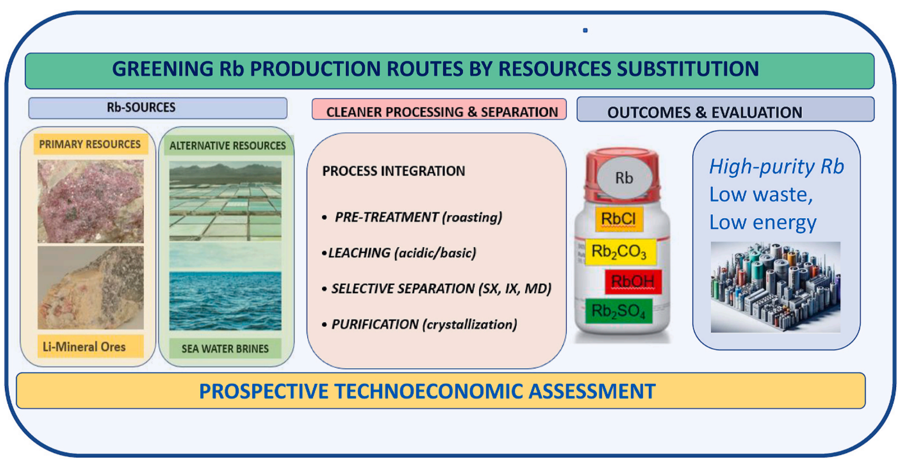
Anil K. Pabby [a,*], Ana María Sastre [b,c], José Luis Cortina [b,c,d,**]

[a] Ex-Nuclear Recycle Board, BARC, Tarapur, Distt. Palghar, 401502, Maharashtra, India
[b] Chemical Engineering Department, Universitat Politècnica de Catalunya (UPC)-BarcelonaTECH, Barcelona, Spain
[c] Barcelona Research Center for Multiscale Science and Engineering, Barcelona, Spain
[d] Water Technology Center (CETaqua), Cornellà de Llobregat, Spain

HIGHLIGHTS

- **Recovery of rubidium from different mineral resources is critically assessed.**
- **Salt lake and seawater brine are sustainable alternative sources for Rb recovery.**
- Integration of sorption and membrane-based process are assessed.
- Analysis of alternative processing routes for Rb from alternative sources.
- **Prospective techno economic analysis of alternative resources for rubidium recovery.**

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Rubidium
Salt lake brine
Extraction
Seawater brine
Membrane absorption
Separation
Prospective
Technoeconomic assessment

ABSTRACT

Rubidium (Rb), an important strategic rare alkali metal, has received significant attention in the last few years due to its applications in biomedical research, solar cells, atomic clocks, electronics, aerospace, quantum computing, laser materials and chemical industry. However, the extraction and processing of this metal are complex, since mineral resources containing rubidium are very scarce and are characterised by low rubidium content. For this reason, several technologies have recently been put forward for cleaner recovery processes of rubidium from various resources. Rb compounds are typically co-produced from silicate minerals, such as pollucite or lepidolite, when they are processed for the production of Li and Cs by-products. Seawater and salt lakes have been identified as promising and more sustainable alternative sources This approach could avoid the highly chemical- and energy-intensive routes associated with the processing of minerals containing Rb. As can be observed from the published literature, researchers have employed various techniques such as solvent extraction,

* Corresponding author.
** Corresponding author. Chemical Engineering Department, Universitat Politècnica de Catalunya (UPC)-BarcelonaTECH, Barcelona, Spain.
E-mail addresses: dranilpabby@gmail.com (A.K. Pabby), jose.luis.cortina@upc.edu (J.L. Cortina).

https://doi.org/10.1016/j.jclepro.2025.146523
Received 26 March 2025; Received in revised form 22 August 2025; Accepted 30 August 2025
Available online 12 September 2025

membrane sorption processes (hybrid mode), precipitation and adsorption to recover Rb. Very few reviews of Rb extraction have been published in peer-reviewed journals, meaning that Rb has received less attention than other alkali metals. In the last few years, hybrid processes involving membrane sorption have attracted attention, but this aspect has not been covered in previously published reviews. A more in-depth examination of membrane separation technology and other separation/recovery techniques is needed for selective Rb recovery from different resources, with a particular focus on the potential for recovery of Rb via membrane sorption, ion exchange and adsorption processes. Hence, this state-of-the-art review covering the period 2010–2024 (up to December 2024) highlights the progress of Rb recovery technologies, addresses the challenges involved, and suggests future avenues for the extraction of rubidium. We also discuss the fact that these cleaner routes, based on the valorisation of waste streams (e.g. the mineral processing stages of lithium production from lake and seawater brines), face thermodynamic barriers due to low Rb concentrations, thus requiring the development of advanced materials (e.g. with high selectivity separation factors over other dominant monovalent ions such as Na or K). Such thermodynamic barriers currently limit both the technical and economical feasibility of these new, cleaner alternatives for Rb recovery.

GLOSSARY.

AA: Acrylic acid.	NF: nanofiltration
AIBN: 2,20-azobisisobutyronitrile.	NIP: non-imprinted membranes
Aliquat 336: Tricaprylmethylammonium chloride	NIPS: nonsolvent induced phase separation.
AMP: Ammonium molybdophosphate.	NF: nanofiltration
AMP@MOF-76(Sm)): grafted ammonium molybdophosphate (AMP) onto MOF-76 (Sm).	NIP: non-imprinted membranes
AWP: Ammonium phosphowolframate	NIPS: nonsolvent induced phase separation.
BPC6: Bis(2-propyloxy)calix [4]crown-6 ether	PES: polyethersulfone.
18C6: 18-crown-6 ether	PDMS, polydimethylsiloxane
CML: Critical Minerals List	PIM: Polymer inclusion membranes
CMP: caesium molybdophosphate.	PMA: phosphomolybdic acid.
CoHCF: Cobalt ferricyanide	PMMA: poly methyl methacrylate
CRM: Critical Raw Material	PP: popropylene (PP)
CTAB: cetyltrimethylammonium bromide	Psf: Polysulfone
CuHCF: Copper ferricyanide	PVDF: poly-(vinylidene fluoride).
DDA: dodecylamine.	P-Yeast-PAA: porous polymer adsorbents.
DMAc: N,N-dimethylacetamide	Rb-ISMs: Rb ion separation membranes
EDTA: ethylendiaminetriacetic acid.	R&D: research and development.
EGDMA:ethylene glycol dimethacrylate	REE: Rare earth elements.
GWe: electric giga wats	RMP: rubidium molybdophosphate
HIPE: Pickering-High internal phase emulsions.	RO: Reverse Osmosis
HKUST: 1,3,5-benzenetricarboxylate struts	SRM: Strategic Raw Material
HMIM: methylimidazole.	SAFHP: sulfonic acid-functionalized hypercrosslinked polymer
HPC: hydroxypropyl cellulose.	SMD: submerged membrane distillation.
IL: ionic liquid.	SMSR: submerged membrane sorption reactor
IIP: imprinted membranes	SX: Solvent Extraction.
IX: Ion Exchange	SWRO: seawater reverse osmosis.
KCuFC-PAN: encapsulated KCuFC KFeHCF in PAN.	t-BAMBP: 4-tert-butyl-2-(a-methylbenzyl)-phenol
MAA: methacrylic acid.	TOPO: trioctylphosphine oxide.
Mag-KCoFC: magnetic potassium cobalt hexacyanoferrate composite	UF: ultrafiltration.
MAHS: Membrane adsorption hybrid system.	UiO-66@iPCC5: hybridization of 25,27-bis(iso-propoxyl)-calix [4] arene-26,28-crown-5 (iPCC5) into the pores of UiO-66 Zr-MOFs.
MD: Membrane distillation.	USGS: United States Geological Survey
MF: microfiltration.	XRD: X-Ray Diffraction.
MIIP: Magnetic Ion-Imprinted Polymers	ZIF: zeolitic imidazole frameworks ZIF.
MOF- Metal Organic Framework.	ZnHCF: Zinc ferricyanide
NaOL: sodium oleate.	ZP: zirconium phosphate.

1. Introduction

Rubidium (Rb), an important rare alkali metal, was discovered in 1861 by German chemists R.W. Bunsen and G. R. Kirchhoff using flame spectroscopy. It is a silvery-white element with unusual physical properties including softness, malleability, and a low melting point (39 °C) (De, 2003). The first industrial applications were centred around photoelectric cells, electronics and chemistry (Thompson, 2011)**.** Due to its chemical and physical properties, which are closely related to those of caesium, Rb has often been use as a component in caesium-based compounds. Over the last decade, there has been an increasing growth in the industrial usage of Rb (Shi et al., 2017)**,** with biomedical applications being the primary drivers of demand for Rb(I) (Awwad et al., 2021). In this context, the annual consumption of this high-priced and commercially scarce metal has exceeded about one metric tonne in the United States, although its current consumption may now be as high as two metric tonnes (Butterman and Reese, 2003; USSG, 2023; USSG, 2020). Table 1 summarises the various applications of Rb and its compounds in different industries.

In 2023, the United States Geological Survey (USGS) published the USGS Critical Minerals List (CML) to enable users to understand and anticipate potential supply chain disruptions by defining and quantitatively evaluating the criticality of each mineral. A total of 54 mineral commodities were quantitatively analysed; although rubidium was not assessed, due to a lack of sufficient data, it was not removed from the CML based on a qualitative evaluation of its supply and demand (Fortier et al., 2021). In the EU, a further list of 34 critical minerals (critical raw materials, CRMs) from the 70 raw materials evaluated was published in 2023. However, unlike the USGS list, rubidium was considered neither a CRM nor a strategic mineral (strategic raw material, SRM), although the expected growth in demand over the next decades was considered

Table 1
Applications of rubidium and its compounds.

Type of application	Reference(s)
Optoelectronics	Huang et al. (2018a); Zhang et al., 2016; Harikesh et al., 2016; Saliba et al., 2016
Catalysis	Wright et al. (1994); Zhong et al., 2007
Quantum computing	Hosseini et al. (2011)
Lasers, energy conversion, electrical energy generation	Hosoe et al. (1988); Naidu et al. (2018); Qian et al. (2022); Momiera et al. (2023)
Glasses for electrical conductivity, telecommunication systems and night-vision equipment	Jandova et al. (2012); Buttermann et al., 2003
Raw materials for the preparation of atomic clocks for precise time measurement in aerospace	Bauch, 2003; Cernigliaro et al., 2013
Biomedical applications: soporifics, sedatives and treatments for epilepsy	Wagner (2011); Hosseini et al., 2011

(Grohol and Veeh, 2023).

Before delving into the details of this review, it is important to clarify the definitions of secondary and alternative resources, as these terms are used frequently throughout the article. Although they are closely related within the context of mining and mineral recovery, they are not synonymous. Both play key roles in modern strategies aimed at achieving a sustainable supply of raw materials, particularly in support of the circular economy and CRM security, but they differ in terms of their origin, definition, and application. Secondary resources are materials that: (i) are not directly extracted from primary natural ore bodies; (ii) are typically recovered through recycling, reprocessing, or revalorisation; (iii) play a crucial role in resource recovery and waste management; and (iv) contribute to reducing environmental impacts by minimising the need for new mining operations. Examples include mine tailings, metallurgical slags and dross, spent catalysts and batteries, e-waste (end-of-life electronic devices), industrial residues, and various end-of-life products such as vehicles and appliances.

In contrast, alternative resources are non-traditional or unconventional sources that can supplement or replace conventional ore bodies as sources of metals and minerals. This category includes both previously unused natural deposits and processed waste streams. Examples include geothermal and oilfield brines, seawater, deep-sea polymetallic nodules, industrial byproducts, and agricultural or fertiliser-related waste streams. Their key features are as follows: (i) they offer alternative extraction pathways when conventional ores are economically or environmentally less viable; and (ii) they support diversification of supply, particularly for CRMs and SRMs.

In two of the latest published reviews on rubidium, they are associated with ores such as lepidolite, pollucite, carnallite, potash feldspar and brines, which are typically mined to produce lithium and caesium compounds (Sharma et al., 2023; Xing et al., 2021). Rubidium generally coexists with other metals such as lithium, caesium, beryllium, tantalum and niobium in granite pegmatites, including lepidolite, zinnwaldite, pollucite biotite, and feldspar. The main industrial producers are located in Germany, Japan, Russia, Denmark and the United Kingdom, and these process rubidium-bearing ores for the production of high-purity rubidium. In addition, manufacturers in different chemical industries produce various grades of derivative rubidium compounds. However, the industry and market associated with rubidium are extremely small compared with those for other metals (Butterman and Reese, 2003; USSG, 2023; USSG, 2020).

In this context, the recovery of rubidium from alternative resources is important in order to improve the environmental and economic performance of rubidium production. Seawater offers potential for recovery of some minerals, since it covers approximately 70 % of the earth's surface and receives substantial amounts of important elements due to weathering processes. It is well known that valuable elements are present in seawater, although in most cases (as for Rb) in very low concentrations (e.g. on the order of mg/L) (Sharma et al., 2023; Xing et al., 2021; Sharkh et al., 2022). As can be observed from the literature, approximately 84 % of the total number of plants operating are based on reverse osmosis (RO) desalination (Bello et al., 2021). A water recovery ratio of about 45–50 % is maintained in all desalination processes, which in turn results in large quantities (124–130 Hm^3/d at a global scale) of concentrated brine, which is currently disposed of (Sharma et al., 2023; Vu et al., 2013; Yan et al., 2012a; Vieceli et al., 2018). Enormous amounts of total dissolved solids and various important elements, salts, and ions are present in this brine, which could be successfully recovered if a valorisation approach were to be applied. Salt lake brines offer another potential source of Rb (Xie et al., 2024). The rubidium content of salt lake brine has been reported to be in the range of 7–9 mg/L, whereas the content in seawater brines is on the order of 0.1 mg/L (Loganathan et al., 2017; Vicari et al., 2022). Other alternative resources for Rb include oil-field brines (Sharma et al., 2023), geothermal brines, saltwork bitterns (Xing et al., 2021), and gold mine waste (Xie et al., 2024).

The global market for rubidium is expected to rise by 4 % between 2022 and 2027, from its present estimated value of 2.57 kt (Ogunbiyi et el.,2021; Mordor intelligence, 2022). Currently, the market price of Rb is approximately 14720 USD/kg. Since the recovery of Rb from rock ores is so difficult, it is anticipated that the price of Rb will rise in the future. Reports in the literature have extensively documented the gradual increase in the price of Rb in the United States (Butterman and Reese, 2003; USSG, 2023; USSG, 2020; Sharma et al., 2023; Mordor intelligence, rubidium market, 2021). Fig. 1 shows the ratios between the economic importance/vulnerability (in terms of the consequences of disruption to supply) and the price for elements (USD/kg) obtained from global mining production.

As can be seen from Fig. 1, Rb has the highest price, approaching 100,000 USD/kg, although its economic importance/vulnerability ratio is lower than for Sr, Pb and U.

In response to market challenges and the need for access to new mineral resources, process intensification is of great importance in the field of mining engineering for the extraction and recovery of metals from different resources. In the frameworks proposed by Stankiewicz and Moulijn (Stankiewicz and Moulijn, 2000) **and** Pabby et al. (2020), process intensification is defined as any development in mineral process technologies that contributes to significant improvements in process efficiency and durability, and which is ultimately expected to result in a more sustainable technology. In this context, emerging techniques for separation and concentration processes (e.g. adsorption, ion exchange and solvent extraction, and membrane techniques, among others) can provide an integrated solution to recover rare elements such as Rb from secondary sources (Davoodbeygi et al., 2023). Hence, this review places special emphasis on the reported progress in these techniques for recovering/extracting Rb from different resources. The objective of this review is to cover the entire value chain, from the resources themselves (e.g. minerals or brines), to leaching (when necessary), the separation and concentration stages, and finally the production of the most commercially relevant rubidium compounds.

Some of the challenges associated with the separation of alkali metals (Cs^+, Rb^+, K^+ and Na^+) arise from their physicochemical properties, which limit the basic mechanisms of separation. In practice, the values for the crystal ionic radii and ionic radii for Cs^+, Rb^+, K^+ and Na^+ in the completely hydrated state are extremely close (see Fig. 2).

It is therefore necessary to adopt a strategy/process to achieve the required separation factors from aqueous solutions. This is particularly the case when dealing with secondary resources such as salt lake brine, seawater or seawater brine, where the presence of competing ions in the largest contents is the main problem that needs to be overcome (Li et al., 2023a). Few most relevant examples have been reported related to the accumulation of these elements in soils and plants (Kot, 2018).

1.1. Secondary and alternative resources in mining

To demonstrate the increasing research interest in this area, a bibliometric analysis was carried out using the Scopus and Web of Science databases, using the following search terms: rubidium (Rb) adsorption; rubidium (Rb) sorption; rubidium (Rb) extraction; rubidium (Rb) recovery; rubidium (Rb) removal; rubidium (Rb) separation; rubidium (Rb) membranes; (Rb) minerals; rubidium (Rb) concentration; rubidium (Rb) purification. The publications found from this search were mainly associated with peer-reviewed journals, book chapters and books. As depicted in Fig. 3, the increasing numbers of publications on Rb between 2000 and 2023 indicates growing research interest in the various aspects of Rb recovery. However, there is still a lack of comprehensive discussion and a systematic approach to understanding the mechanism of Rb recovery from different resources. Notably, previously published reviews on rubidium have not covered membrane techniques, despite the significant number of papers published in this field.

The conventional method of extraction of rubidium from land-based ore minerals (primarily lithium-rich silicates such as lepidolite,

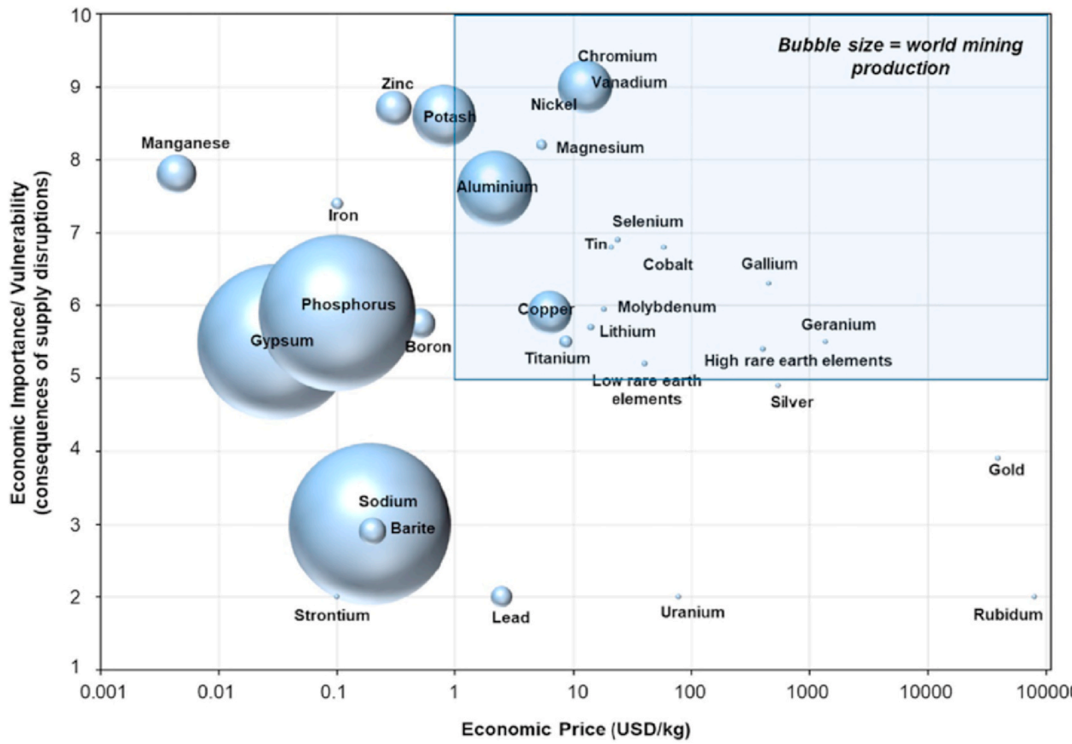


Fig. 1. Comparison of the economic importance/vulnerability (in terms of the consquences of supply disruption) as a function of the economic price (USD/kg) (with production quantity represented as the sizes of the circles) (Ogunbiyi et el., 2021).

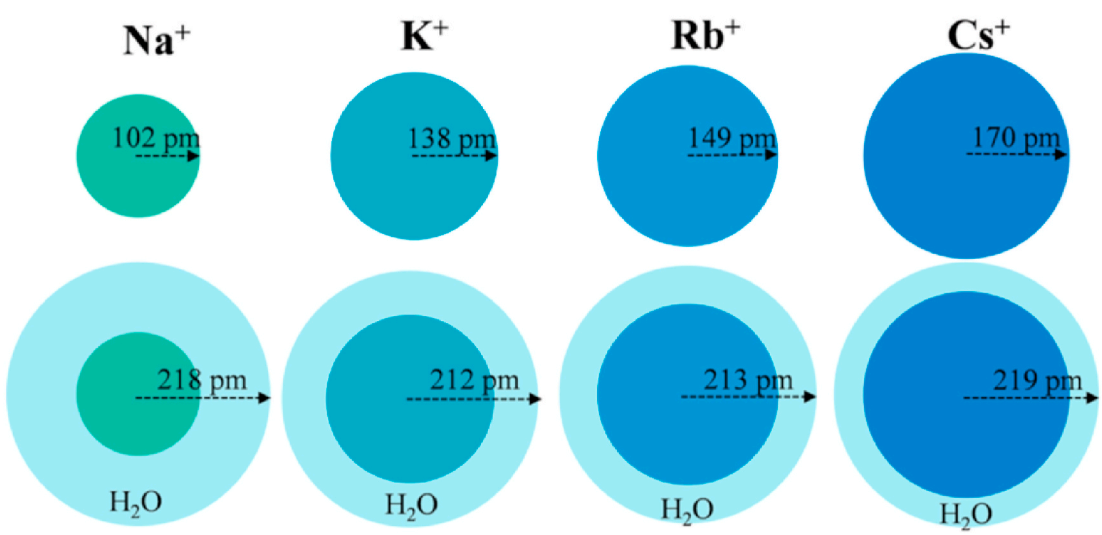


Fig. 2. Comparison of the ionic radii in the crystalline and hydrated states (light blue) of alkaline elements (Li et al., 2023a). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

zinnwaldite, and pollucite) is subject to several technical and economic challenges that limit the widespread recovery of this strategic element. These minerals typically contain rubidium as a minor constituent (0.5–2 %), and it is rarely the primary target in mining operations; instead, it is usually present as a byproduct of lithium or caesium extraction processes, and current industrial flowsheets are not optimised for rubidium recovery. One of the key technical limitations lies in the complex mineralogy and similar chemical behaviour of rubidium relative to other alkali metals, such as potassium and caesium. This similarity complicates the separation and purification processes, often requiring multiple extraction and crystallisation steps, which increase both the energy consumption and processing costs. Moreover, commonly used methods such as roasting and acid digestion often lead to the co-leaching of unwanted elements (e.g. aluminium), thus further complicating downstream purification and reducing the efficiency of the process. From an economic standpoint, the relatively low market demand for rubidium limits the incentive to develop a dedicated recovery infrastructure. Industrial operations tend to prioritise higher-value or higher-volume elements such as lithium, which leaves rubidium underutilised or discarded in residues. In addition, environmental concerns related to the generation of solid and liquid waste from mineral processing also pose regulatory and sustainability challenges. Given these constraints, attention is increasingly turning toward alternative and more sustainable sources of rubidium, and particularly seawater brines and

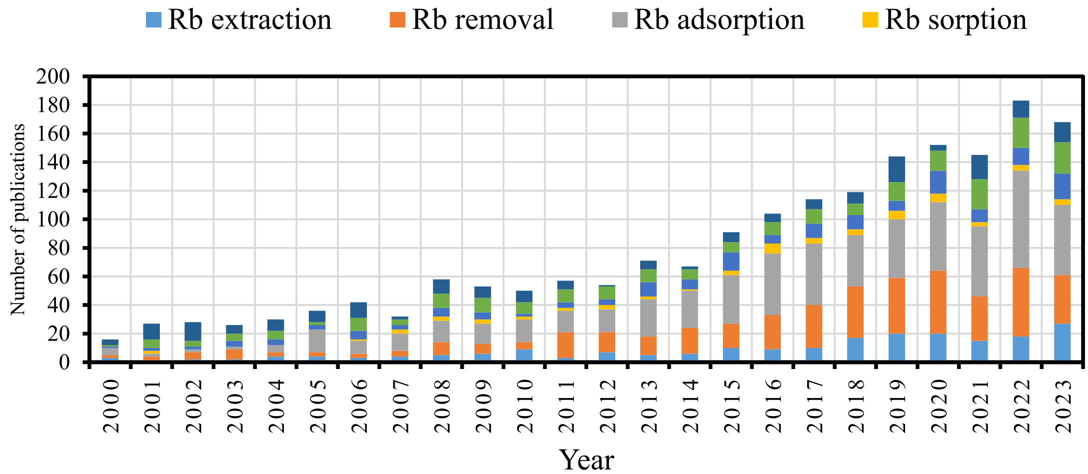


Fig. 3. Number of publications on Rb between the years 2000 and 2023 in the Scopus and Web of Science databases. The eligibility criterion for the language was publication in English. References were divided based on keywords associated with Rb processing technologies.

desalination concentrates. Although rubidium is present in seawater at low concentrations (~120 μg/L), its global abundance and the massive volumes processed in seawater desalination plants and saltworks offer promise for large-scale recovery. These brines often contain Rb(I) alongside other alkali and alkaline-earth metals, and are typically regarded as waste streams, making them an underexploited and low-cost resource. Furthermore, brines offer advantages such as: (i) continuous availability from the existing desalination and salt production infrastructure; (ii) lower generation of solid residues; (iii) the potential for integration with circular economy and zero-liquid discharge (ZLD) approaches; and (iv) a reduced environmental footprint compared to conventional mining. By addressing the limitations of conventional ore-based Rb recovery processes and leveraging the untapped potential of seawater brines, future research and technological development can promote more sustainable and economically viable pathways for the supply of rubidium.

The present review aims to provide state-of-the-art information on the recovery of Rb from secondary resources. Specifically, this review focuses on novel R&D technological solutions for the extraction/recovery of Rb from minerals, brine resources and alternative secondary sources. Recent technological developments involving membrane technologies and adsorbents are also considered. Finally, a review of techno-economic feasibility analyses conducted on the most promising Rb resources is presented.

2. Extraction of rubidium from mineral resources

As described previously, Rb generally co-exists with other metals, such as lithium, caesium, beryllium, tantalum and niobium, and occurs in granite pegmatites such as lepidolite (KLi_2 [$(F,OH)_2|Si_4O_{10}$]), zinnwaldite ($KLiFeAl(AlSi_3)O_{10}(F,OH)_2$), pollucite ($(Cs,Na)_2Al_2Si_4O_{12}\cdot H_2O$), spodumene ($LiAl(SiO_3)_2$), biotite (K(Mg, Fe_3^{II} [$(OH)_2$ (Al, Fe^{III})Si_3O_{10}]) and feldspar ((K,Na,Ca,Ba,NH_4) $(Si,Al)_4O_8$).

One of the most important mica minerals containing lithium is lepidolite, which has a monoclinic structure and a chemical composition that shifts from polylithionite ($KLi_2Al(Si_4O_{10})$ $(F,OH)_2$) to trilithionite (K ($Li_{1.5}Al_{1.5}$) ($AlSi_3O_{10}$) $(F,OH)_2$) in a solid solution series. It contains relatively high amounts of lithium (3.0–7.7 % (w/w)) and other important elements such as Rb and Cs (0.3–5 % (w/w) Rb as Rb_2O and 0.3–2 % (w/w) Cs as Cs_2O) (Zeng et al., 2021; Lepidolite,). The crystal structure of lepidolite ore is shown in Fig. 4(a).

One type of lepidolite with a high iron content, known as zinnwaldite [$KLiFeAl(AlSi_3)O_{10}(F,OH)_2$], has also been investigated as a ource of lithium and rubidium (Jandova et al., 2008; Siame and Pascoe, 2011; Jandova et al., 2010). Zinnwaldite is a mineral mica that is rich in Rb,

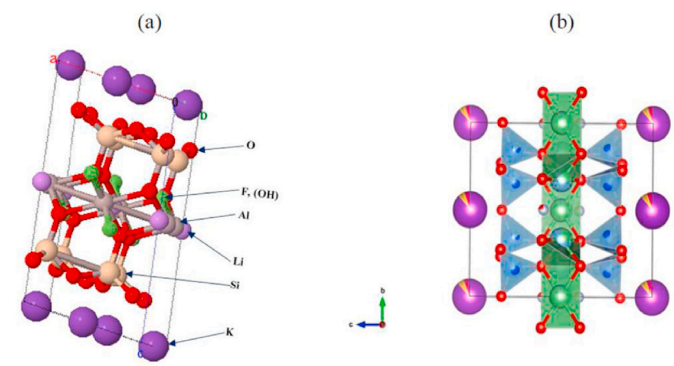


Fig. 4. Representations of the crystal structure of lepidolite: (a) image from (Guggenheim, 1981), (b) image from (Zinnwaldite,).

with values of 0.5–1 % (crystal structure shown in Fig. 4(b)). It is usually present in greisen and pegmatite, and its extraction typically involves roasting followed by leaching (Zhong et al., 2024; Siame and Pascoe, 2011). Pollucite ($(Cs,Na)_2Al_2Si_4O_{12}\cdot H_2O$) is a colourless zeolite mineral with rubidium and potassium as substituting elements (crystal structure shown in Figure (5b)). It is frequently contained in granite pegmatites and coexists with spodumene, albite ($NaAlSi_3O_8$) and lepidolite. The values for the content of Li, Rb and Cs in wt% are around 0.25, 0.27 and 23.6, respectively (Xing et al., 2021). The main producers of pollucite are Zimbabwe and Canada, whereas the main processing countries are Japan, Germany, United Stated and Russia (Butterman and Reese, 2003; USSG, 2023; USSG, 2020).

Among these minerals, lepidolite has the highest rubidium content compared to other sources. As a common practice, rubidium is typically obtained as a byproduct of lithium processing, as in the case of some lepidolite processing facilities. The recovery of rubidium is achieved from process liquors or from the alum generated after the precipitation of lithium carbonate (Lenher et al., 1924; Jiao et al., 2024). The market need to increase the production of lithium carbonate and lithium hydroxide in the short term is generally limiting the deployment of rubidium recovery processing routes.

Mineral leaching is the most critical step in this process, and is commonly achieved using the direct acid leaching method, based on the fact that minerals such as lepidolite and pollucite can be dissolved by strong mineral acids (e.g. sulphuric acid). Lepidolite is used in China and Russia to produce lithium salts and for the recovery of K, Rb, and Cs as byproducts from sulphuric acid leaching solutions (Xing et al., 2021; Macedonio et al., 2023), while pollucite is used in the United States,

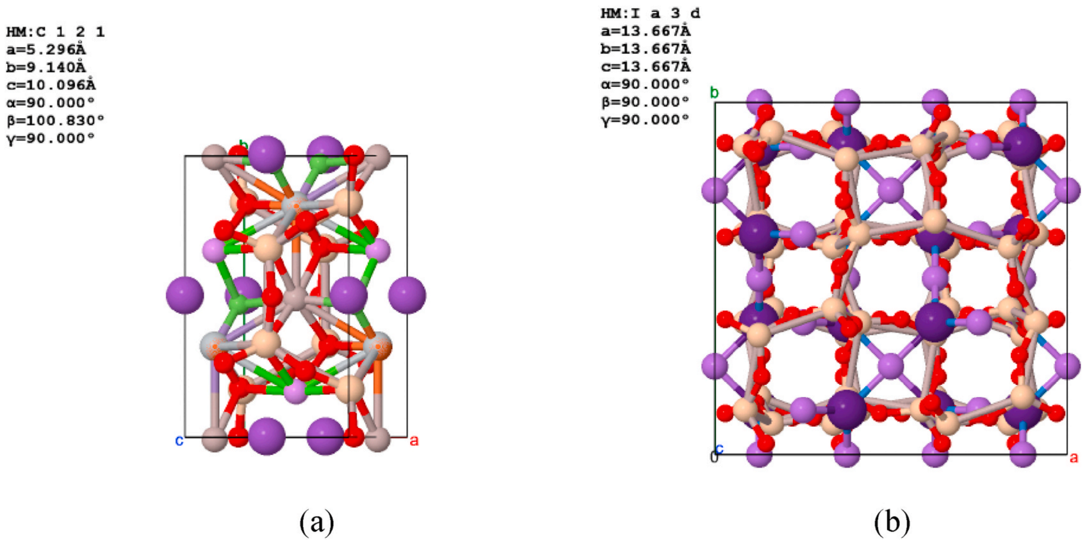


Fig. 5. Crystal structures of (a) zinnwaldite [$KLiFeAl(AlSi_3)O_{10}(F,OH)_2$] and (b) pollucite ($(Cs,Na)_2Al_2Si_4O_{12}\cdot H_2O$) (American Mineralogist).

Canada, and Japan (Butterman and Reese, 2003; USSG, 2023; USSG, 2020). Pre-treatments such as roasting and mechanical activation are also typically required (Wang et al., 2021a; Krys et al., 2013). Once the leaching process is complete, lithium, caesium and rubidium are separated and concentrated by solvent extraction (SX) and finally recovered as marketable products through precipitation and/or crystallisation. Table 2 summarises the results of a literature survey of some of the most relevant publications dealing with Rb extraction from different mineral resources, where the main effort was allocated to evaluating the leaching performance as a function of the mineral properties. The next section deals with mineral processing by the flotation technique for the recovery of rubidium.

2.1. Concentration of rubidium-containing minerals by flotation

Flotation is a widely used mineral processing technique that exploits the differences in the surface properties between minerals. It involves adding reagents to a slurry of crushed ore, causing hydrophobic minerals to rise as a froth that can be skimmed off while the hydrophilic ones remain in the slurry (Wills and Finch, 2015). Bian et al. (2023) achieved a reduction in the operational cost associated with large chlorination furnaces by selective kaolin concentration through the use of a batch flotation process. To accomplish the reverse flotation desilication, sodium oleate (NaOL) and dodecylamine (DDA) were employed as mixed collectors. This increased the lithium and rubidium content while simultaneously lowering the quantity of mineral that needed to be chlorinated. At the flotation stage, a Li_2O grade of 5 % with a 63 % recovery and a Rb_2O grade of 2 % with a 51 % recovery were obtained. Through physical adsorption and charge neutralisation of the mica surface particles, the mixed collector used at the flotation stage affected the hydrophobicity of quartz. The muscovite mica was processed by flotation using DDA, NaOL and DDA–NaOL mixtures (Xu et al., 2013; Jordens et al., 2016; Liu et al., 2019) to compensate for its negative charge over a pH range of 2–12. The recovery rates varied from approximately 80 % (at pH 2) to 50 % (at pH 11) using only DDA, whereas when mixtures of DAA/NaOL were used, the recovery rate increased from 80 % (at pH 2) to 90 % (at pH 11). In these two component mixtures, the sorption of both the cationic and anionic collectors was increased thanks to co-sorption phenomena. In general, the use of flotation for Li-, Rb- and Cs-containing 1 minerals is unusual, even at the research level, as the liberation results for Li, Rb and Cs have been demonstrated to be limited at present. In addition, the pressure to increase lithium (Li^+) production to secure the supply chain for electric vehicle battery manufacturing has led to flotation processes often being overlooked. Instead, direct leaching methods are typically applied after size reduction (and roasting if necessary), regardless of the specific nature of the Li, Rb-, or Cs-bearing ore. Other mineral processing methodologies will be discussed in detail in the next two sections.

2.2. Leaching procedures for primary minerals

2.2.1. Direct acid leaching routes

In this section, we consider only those papers which describe schemes with some feasibility at an industrial scale. From the discussion of the floation technique in Section 2.1, it is clear that other mineral techniques such as direct acid leaching and basic leaching (Section 2.2.2) need to be adopted to achieve promising performance. Acids such as sulphuric and hydrochloric acids are known to dissolve lepidolite effectively, producing a liquid that contains a mixture of lithium, potassium, rubidium, and caesium. The separation and purification of rubidium and caesium in solution is then achieved by repeated fractional crystallisation, due to the low solubility of rubidium and caesium alums at low temperature (Sharma et al., 2023; Xing et al., 2021; Loganathan et al., 2017; Macedonio et al., 2023). The leaching equilibrium and kinetic data for lepidiolite in H_2SO_4 solutions have been thoroughly evaluated (Liu et al., 2019). Under optimal conditions, with a mass ratio of lepidolite (<180 μm)/sulphuric acid (1:2), a leaching temperature of 140 °C, a liquid−solid ratio of 2.5:1, and a leaching time of 10 h, recovery ratios of 94 % (Li), 94 % (K), 92 % (Rb) and 89 % (Cs) could be achieved. Improvements in the leaching efficiency were made using a mechanical activation stage as the initial stage, to change its crystal structure and make it more amorphous and chemically reactive, as proposed by Vieceli et al. (2017) and shown in Fig. 6. This work centred around the recovery of lithium carbonate, although the leaching ratios and process kinetics for rubidium were also improved. The redundancy of the process was not discussed by the authors, although this would have important implications for scaling up the process.

To ensure the high-quality recovery of Li and Rb, after mechanical activation of the lepidolite in a disc mill and the leaching stage with sulphuric acid, Al is removed from the leachate via crystallisation of alums by reducing the temperature to 5 °C. The crystallisation kinetics of aluminium is shown in Fig. 7. After a further 5 h, 25 % of the Al (from 20 to 15 g/L) and more than 99.6 % (from 1.3 to 0.027 g/L) of the Rb are precipitated, while the Li concentration in the solution is kept constant. The recovery of Rb, which is not described in this paper, could be carried out from the precipitated $KAl(SO_4)_{2(s)}$. The leachate is treated with

Table 2
Processes for extracting Rb from mineral resources.

Extraction procedure	Metal extraction efficiencies	Ref.
Lepidolite ($K(Li,Al)_3(Al,Si,Rb)_4O_{10}(F,OH)_2$)		
Direct leaching		
Sulphuric acid (85 %w/w), temperature 200 °C, acid/lepidolite concentration ratio (1.7:1 for 4 h), water leaching (85 °C)	>90 % Li, Rb and Cs	Jiao et al. (2024)
Mechanical activation and leaching with sulphuric acid: acid:ore mass ratio of 0.66 at 130 °C for 15 min	90 % Li	Vieceli et al. (2018)
Lime (CaO) milk (for defluorination) at150 °C, lime: ore mass ratio 1:1, liquid:solid ratio 4:1 for 1 h	73 % Rb, 99 % Li	Yan et al. (2012d)
Roasting/leaching		
Roasting: mass ratio of lepidolite: Na_2SO_4: $CaCl_2$ 1:0.5:0.3, 880 °C, 0.5 h	>90 % Li, Rb and Cs	Yan et al. (2012a)
Roasting: $CaSO_4\cdot 2H_2O$:Na_2SO_4 mass ratio 3:1, 850 °C, 2 h	~80 % Li and Rb	Vieceli et al. (2018)
Roasting: SO_4/Li and Ca/F molar ratios of 3:1 and 1:1, 850 °C, 1.5 h	93.3 % Li, 33.2 % Rb	Luong et al. (2014)
Roasting: mass ratio of lepidolite: NaCl:$CaCl_2$ 1:0.6:0.4, 880 °C	~93 % Li, Rb and Cs	Yan et al. (2012b)
Roasting: mass ratio of lepidolite: Na_2SO_4:K_2SO_4/CaO 1:0.5:0.1:0.1, 850 °C, 0.5 h; water leaching: L/S ratio 2.5:1, 0.5 h	92 % Li, 75 % Rb	Yan et al. (2012c)
Zinnwaldite ($KLiFeAl(AlSi_3)O_{10}(OH,F)_2$)		
Roasting-leaching (high temperature): (i) $CaCl_2$, (ii) chlorination of the alkali metals between 500 and 650 °C, and (iii) chlorination CaCl2(s) and β-spodumene (decomposition product of zinnwaldite)	77 % Li, 90 % K, and 90 % Rb	Kristanova et al.,2023
Sintering zinnwaldite with $CaCO_3$ powders: (i) decomposition of zinnwaldite at 800 °C, (ii) formation of new phases in the range from 750 to 835 °C, and (iii) formation of glass phase at above 835 °C and extraction of Li and Rb	84 % Li and 91 % of Rb	Vu et al. (2013)
Other rubidium-bearing minerals and secondary resources		
Polylithionite **Alkaline pressure leaching** (250 °C) NaOH: 600 g/L, leaching duration: 3 h, liquid:solid ratio 7:1 mL/g.	Leaching efficiencies of lithium and rubidium: >95 %	Lv et al. (2020)
α-spodumene **Roasting with Na_2CO_3** (1100 °C) for 30 min with 45 % Na_2CO_3, water quenching and strengthening leaching	96 % Li, 90 % Rb, 1.5 % Al	Zhou et al. (2023)
Biotite **Chlorination roasting/water leaching**: mass ratio ($CaCl_2$ versus minerals) of 0.4 at 900 °C for 10 min **Direct acid leaching:** sulphuric acid concentration 5.52 mol/L, 80 °C and liquid:solid ratio 3	Leaching efficiency of Rb: 97 % For direct acid leaching method: yield 96 %	Zeng et al. (2019)
Porcelain stone **Alkaline roasting** with Na_2SO_4 and $CaCl_2$ mixtures, mass ratio ore (1)/Na_2SO_4(0.2)/$CaCl_2$(0.2), at 850 °C for 60 min, water leaching conditions liquid:solid ratio of 1:1 at room temperature for 60 min	Leaching efficiencies: 99 % Li, 50 % Na, 38 % K, 97 % Rb and 98 % Cs	Wang et al. (2020)

limestone to neutralise the acidity and to remove remaining metal impurities (Fe(III), Al(III) and Mn(II) prior to the recovery of lithium carbonate (Vieceli et al., 2018). The authors performed all of these steps in a stagewise manner, although the Rb precipitate and its characteristics (precipitate filtration quality, particle size, etc.) were not studied, which would be helpful when designing large-scale production processes.

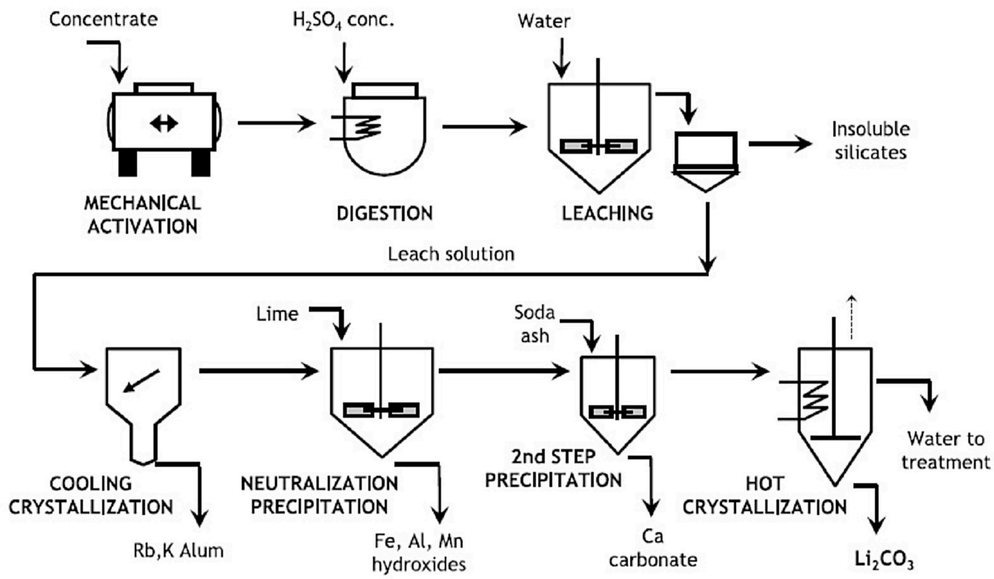
Zhang et al. (2019) evaluated the influence of the sulphuric acid concentration in the leaching stages with lepidolite minerals. Concentrated sulphuric acid solutions of 85 % H_2SO_4 (w/w) were used after the water leaching stage to yield leaching ratios of 92 %, 32 % and 79 % for Li, Rb and Cs, respectively (Fig. 8).

For the pre-treatment stage with H_2SO_4, experiments were carried out at 180 °C, with a mineral:sulphuric acid solution ratio of 1.7:1, a leaching time of 8 h, and lepidolite that was milled with 50 % particles being below 74 μm. For the leaching stage with water at 20 °C, the contact time was 1 h and the solid:liquid ratio was 1:3 **(**Zhang et al., 2019**).** With an increase in the concentration of H_2SO_4 up to 98 % (w/w), the leaching efficiencies were reduced to 10 %, 7 % and 25 % for Li, Rb and Cs, respectively. Although the leaching temperature was higher than 85 °C, these results are different from those of Vieceli et al. (2017), who reported higher leaching efficiencies (more than 90 % for Li, Rb, and Cs). The minerals reported after treatment with hot sulphuric acid (98 % (w/w) were $KAl(SO_4)_2$(s), Li_2SO_4(s) and $Al(SO_4)$ $(OH)_5H_2O$ (s)), while the Rb and caesium were postulated to be in the forms $RbAl(SO_4)_2$(s) and $CsAl(SO_4)_2$(s), respectively. An optimised protocol that included the use of H_2SO_4 (85 % (w/w)) at 200 °C at a solid to liquid ratio of 1:1.7 for 4 h and mineral particles with a particle size distribution below 74 μm, followed by a water leaching stage at 85 °C, was suggested after a study of the temperature dependence of the baking and water leaching stages. Global leaching efficiencies were increased compared to the values reported by Vieceli et al. (2017), with extraction rates for Li, Rb, and Cs of 97 %, 96 %, and 95 %, respectively.

In the case of lepidolite minerals containing fluoride, modifications of the leaching processes need to be introduced. A two-stage process that included acid leaching with HCl and a thermal pyrolytic process was introduced by Liu et al. (2020) to achieve the removal of fluoride ions as HF(g) and partial removal of the excess HCl. The thermally treated mineral was calcined at 363 °C, in order to further remove the residual contents of HCl and HF, and dissolved in water to produce a rich solution of Li, Rb, K and AlF_6^{-3}. As described previously, rubidium and caesium were separated and purified by recrystallisation after 30 stages (Butterman and Reese, 2003). To increase the leaching efficiency of the process, Yan et al. (2012d) proposed a processing route that included a fluoride removal stage followed by a CaO(s)/water slurry leaching stage. The extraction rates were 73 % for Rb and 81 % for Cs under optimised operational conditions with a leaching temperature of 15 °C, a leaching time of 1 h and a solid:liquid ratio of 1:4. However, a cost analysis was not carried out to evaluate the economic aspects of the defluorination process in comparison with traditional processes of direct acid leaching with H_2SO_4.

Using a different approach, Nie et al. (2021) investigated the recovery of Rb from a low-grade aluminosilicate mineral using H_2SO_4-CaF_2(s) mixtures, as only the mica structure is dissolved in the presence of H_2SO_4. The addition of CaF_2(s) together with a rise in temperature and an increase in the H_2SO_4 concentration from 0.5 to 4 mol/L increased the leaching of Rb. Similar approaches have been reported for the processing of pollucite, with different acid leaching reagents. In the case of hot hydrochloric acid digestion of pollucite, rubidium and caesium can be recovered from the resulting liquor by precipitation with iodide or chloride salts (e.g. antimony trichloride, stannic chloride, and iodine chloride) **(**Zhang et al., 2016**) .** We note that the authors of this reference **(**Zhang et al., 2016) selected a low-grade aluminosilicate mineral, and this study lacked a cost analysis, which would determine whether these type of minerals are worth processing.

Finally, the use of H_2SO_4 baking, reductive decomposition, and alkaline leaching to extract Rb from Rb-containing granitic rock



Y

Fig. 6. Processing flow-sheet for the recovery of lithium from lepidolite, including a pre-stage of mechanical activation, followed by sulphuric acid leaching and recovery of lithium carbonate (Vieceli et al., 2018).

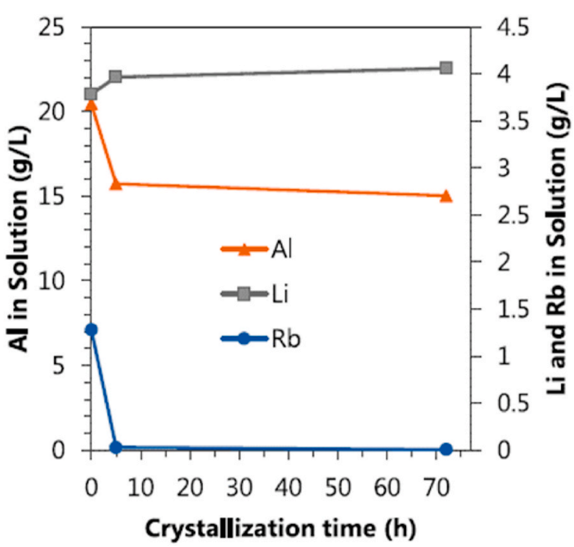


Fig. 7. Variation in the concentrations (g/L) of Al, Li and Rb versus time (h) throughout the removal stage of Al by crystallisation at 5 °C (Vieceli et al., 2018).

revealed a significant alteration in the mineral phase of K-feldspar and micas, with high leaching ratios for Rb and K of 94 % and 92 %, respectively (Xing et al.,2018).

2.2.2. Direct basic leaching routes

Basic leaching methods have been explored to enhance the extraction of lithium and rubidium from minerals such as lepidolite and polylithionite. Mulwanda et al., (2021) demonstrated that adding lime (CaO(s)) to the leaching process (2 h) at high temperatures (250 °C) in alkaline media (350 g/L NaOH) improved the leaching ratios of Li (94 %), K (98 %), Cs (90 %) and Rb (96 %) by precipitating fluorine and silicon as CaF_2(s), $NaCaHSiO_4$(s), and $Ca_3Al_2SiO_4(OH)_8$(s). An alkaline pressure leaching flowsheet for lepidolite in the recovery of rubidium was discussed by the authors of this paper. Lv et al. (2020) described pressure-based basic leaching of lithium and rubidium from polylithionite, along with a recovery stage that included solvent extraction

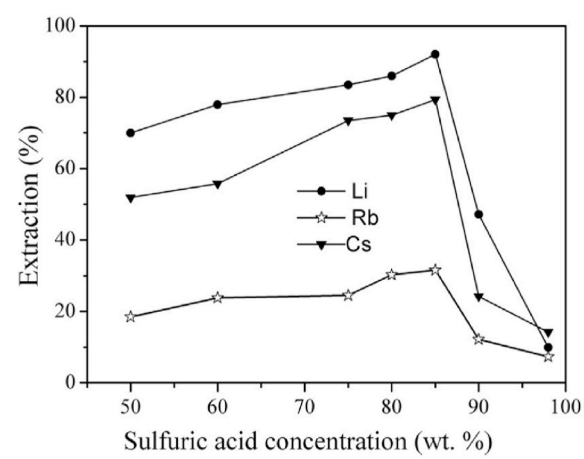


Fig. 8. Results for the extraction efficiency (after a global process of pretreatment with H_2SO_4 and subsequent water leaching) of Li, Rb and Cs from lepidolite mineral as a function of the concentration of the leaching reagent (H_2SO_4) (Zhang et al., 2019).

and final precipitation. When employing a solid:liquid ratio (7 kg/L) at 250 °C, a NaOH concentration of 600 g/L, and leaching durations of 3 h, the leaching ratios obtained for lithium and rubidium were greater than 95 %. Another option that was evaluated using basic leaching routes involved Rb-containing granitic rock containing muscovite, biotite, quartz, and K-feldspar, which is typically not considered as a viable source due to a lack of leaching data (Naidu et al., 2016b; Zhang et al., 2020a). **Xing et al.** (Xing et al., 2018) presented a new method for extracting Rb from granitic ores based on alkaline leaching using CaO (s), desilication, and SX. Since the phase change of the mica mineral involving zeolite and cancrinite structures during the alkaline hydrothermal treatment favours the dissolution of Rb, the leaching efficiency of Rb reached values higher than 95 % at leaching temperatures of 230 °C. The final residue could be used again in geopolymer applications, meaning that this approach lowered the energy consumption and production of waste. A detailed leaching flowsheet for extracting Rb from granitic ores is evaluated in a step-by-step manner by the authors of reference (Xing et al., 2018). All of the studies mentioned above involving the basic leaching route show potential for the efficient recovery of Rb, but comparisons between the acid leaching route and the basic leaching route were not presented by authors in terms of their mineral processing cost.

2.2.3. Roasting-leaching processes

Several authors have highlighted the technique of roasting-leaching and have compared this with the acid leaching and basic leaching routes (described in Sections 2.2.1 and 2.2.2). For this reason, the roasting-leaching method has been widely utilised for lithium extraction, and has also gained attention for rubidium extraction. The roasting process involves the addition of different chemicals (e.g. sodium sulphate, calcium chlorides and limestone, among others) to facilitate the solubilisation of rubidium and lithium. In fact, Rb has been successfully extracted using this method from a variety of mineral sources (Wang et al., 2021a; Krys et al., 2013). As a representative case, a flowsheet for the sulphation roasting-leaching process of lepidolite is discussed in depth by Yan et al. (2012c) using a stagewise approach.

Most of the works considered here focused on the extraction of Rb from lepidolite using sulphate salts as additives. For instance, Yan et al. (2012a) investigated salt roasting with a mixture of Na_2SO_4 (s) and $CaCl_2$(s) for the leaching of lepidolite. Extraction yields of a (s)bove 90 % were achieved for Li, Rb and Cs at mass ratios (value in brackets) of lepidolite (10), Na_2SO_4(s) (5), $CaCl_2$(s) (3) with a roasting time of 0.5 h at 880 °C. To crystallise Na_2SO_4•$10H_2O$(s) (92 %) and NaCl (4 %), the solution was cooled to −5 °C for 2 h at the final stage of the dissolution process in water. Li was later precipitated as Li_2CO_3 (99.5 % purity) using Na_2CO_3, and Rb and Cs were recovered from the filtrate. In the same vein, Vieceli et al. (2017) evaluated the use of Na_2SO_4(s) and $CaSO_4$(s) as additives for lepidolite. The extraction rates for lithium and rubidium improved from 70 % to 85 % as the temperature was increased from 750 °C to 850 °C and the content of additives increased. The thermodynamics associated with the expected reactions of lepidolite decomposition along the roasting stage with Na_2SO_4 were explored by Luong et al. (2013) using HSC Chemistry Software (HSC Chemistry software, 2011). Raising the leaching temperature and subsequently the liquid:solid ratio was expected to favour the formation of soluble mineral phases such as $LiKSO_4$ and LiF. Remarkably, a similar pattern was observed for the leaching of Rb. In a similar study, Yan et al. (2012c) reported a roasting process that could reduce operating costs and increase the lithium leaching ratio by introducing a mixture of sulphates (Na_2SO_4, K_2SO_4) and alkalis (CaO), which gave a positive effect. During the sulphation roasting process, the lepidolite was transformed into $NaSi_3AlO_8$(s), $KAlSi_2O_6$(s) and $CaAl_2Si_2O_8$(s), according to the XRD patterns of the reaction byproducts. Other works have evaluated the extraction of Rb and Cs using $FeSO_4{\cdot}7H_2O$(s) in lepidolite during roasting, as indicated in work by Yan et al. (2012c) . A thermodynamic study indicated that these elements can be extracted at lower temperatures (<600 °C), with extraction efficiencies of 93 %, 87 %, 83 %, and 86 % for Li, Rb, Cs, and K, respectively. The suggested technique, which was carried out at a lower temperature, demonstrated more potential for large-scale industrial applications than industrial recovery processes from lepidolite.

In the same vein, Luong et al. (2014) studied the use of $FeSO_4{\cdot}7H_2O$(s) with and without CaO(s) as an additive mixture for the roasting of lepidolite. HSC Chemistry Software was used to perform a thermodynamic analysis to predict the reactions during the roasting stage, and to assess the effects of temperature dependency and the molar ratios of SO_4/Li and Ca/F on Li recovery. The optimal sulphonation temperature for Li/Rb extraction was 800–900 °C, and the SO_4/(Li/Rb) molar ratio was greater than 2.5:1. Furthermore, a Ca/F molar ratio greater than 2:1 decreased the production of HF, but also promoted lower production of SO_2(g)/SO_3(g), which is necessary for the leaching of Li and Rb. Under optimum conditions, the roasting tests produced calcinates that were used for the subsequent leaching of Li. The formation of Li_2SO_4 and Rb_2SO_4 was higher, favouring their solubility throughout the leaching stage. The XRD patterns for the calcine and residue after leaching (Fig. 9) showed that they mainly contained Fe_2O_3(s) and other major insoluble compounds including $CaSiO_3$(s), $CaSO_4$(s) and CaF_2.(s). Li_2SO_4(s) and Rb_2SO_4(s) were the major soluble Li and Rb species in the calcine, and were completely dissolved during leaching.

Other studies have reported alternative approaches using acid leaching. Liu et al. (2022a) developed a process involving the preparation of calcium sulphate whiskers, followed by grinding and leaching of lepidolite ore with H_2SO_4. The leachate was then cooled with water and thermally activated. This method achieved lithium and rubidium recovery rates of 99.8 % and 99.6 %, respectively. The leach residues were rich in sulphate and were recycled to produce gypsum whiskers. The same authors proposed a two-stage sulphuric acid roasting process characterised by more selective leaching of lithium, rubidium, and caesium from lepidolite minerals, with limited leaching of iron or aluminium (Liu et al., 2022b). Sulphonation of lepidolite ores took place in the first roasting step, and partial breakdown of the sulphates occurred in the second roasting phase. In the subsequent steps, recycling of sulphuric acid and separation of rubidium and caesium were accomplished, thus improving the impact on the consumption of reagents.

For spodumene and lepidolite mixtures, a combination of thermal activation followed by alkali leaching was proposed by Yang et al. (2022) to enhance the extraction of Li and Rb. Through thermal activation, this processing method promoted phase transformation of the primary minerals. Fig. 10 depicts the extraction ratios of Li, Rb, Si, and Al under different operating conditions. High leaching ratios of 93 % and 88 % for Li and Rb, respectively, were achieved under optimal conditions (see Fig. 10). An increase in the leaching temperature to above 100 °C had a significant detrimental effect on the strong leaching efficiency of Li, while the efficiency of Rb leaching was maintained up to 150 °C. The leaching process was accompanied by a low dissolution ratio for Al, with values below 5 %, and moderate dissolution of Si, with values below 40 %.

A similar study was carried out by **Huidong** Zhou et al. (2023), in which alkaline roasting technologies using carbonates (e.g. Na_2CO_3(s)) were evaluated using α-spodumene. More precisely, the authors reported the direct extraction of lithium from α-spodumene using

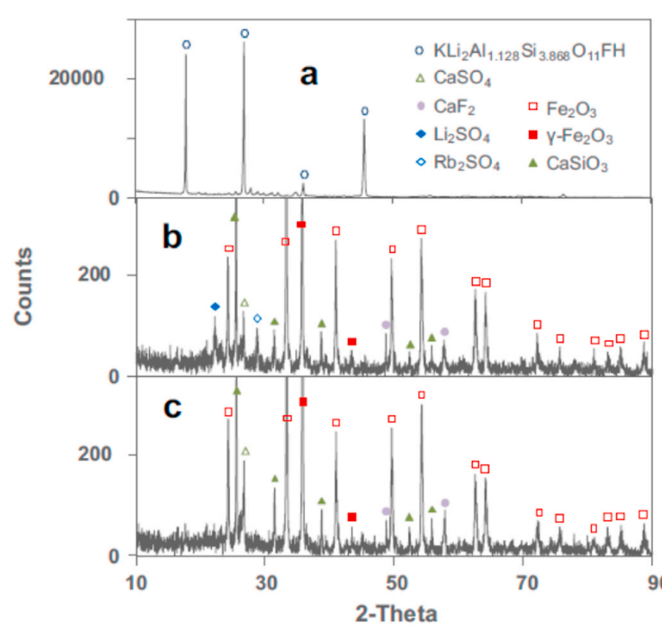


Fig. 9. Identification of the roasting mechanism via analysis of the XRD patterns for (a) lepidolite ore, (b) calcined lepidolite ore and (c) undissolved solids from the calcination stage of the lepidolite ore (Luong et al., 2014; HSC Chemistry software, 2011).

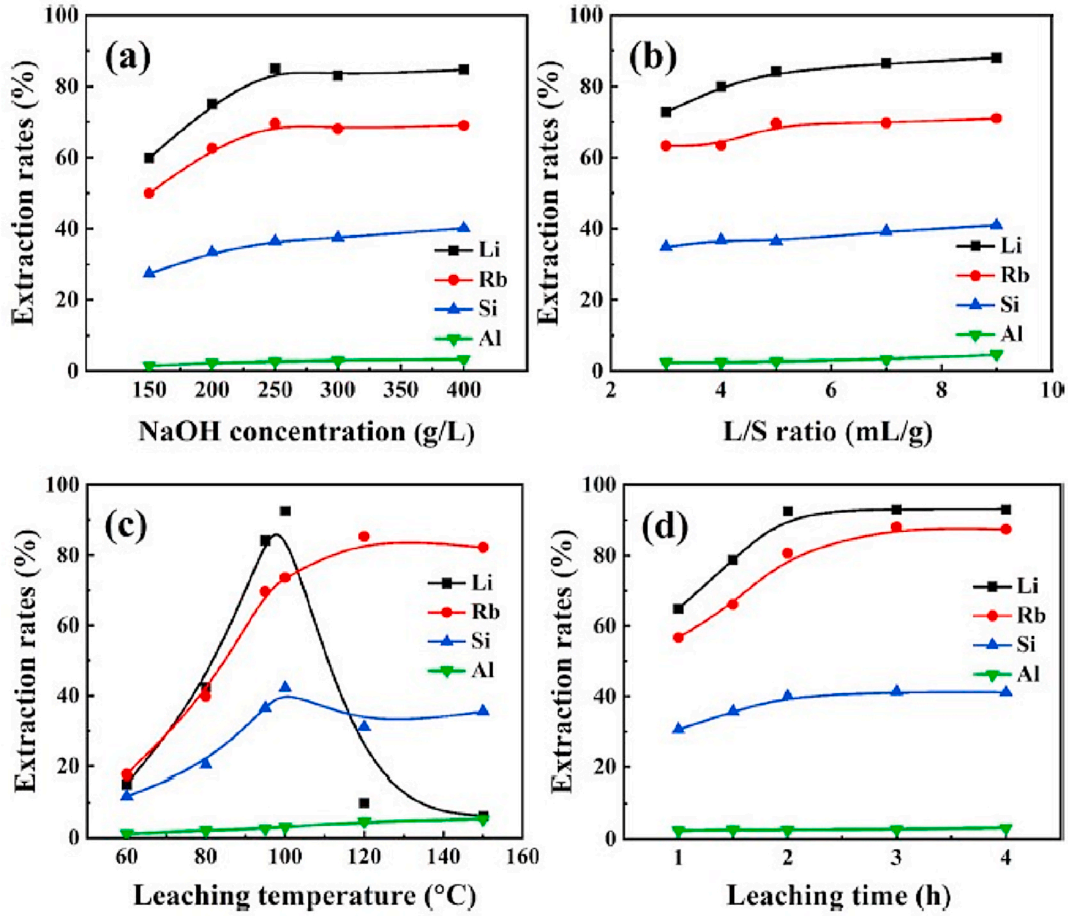


Fig. 10. Experimental evaluation of the extraction ratios of Li, Rb, Si and Al from mineral ores consisting of spodumene and lepidolite as a function of (a) concentration of NaOH, (b) L/S ratio, (c) leaching temperature, and (d) time (Yang et al., 2022).

Na_2CO_3(s) roasting followed by water leaching. The influence of the Na_2CO_3(s)/α-spodumene ratio in the roasting stage is shown in Fig. 11 (a).

The findings showed that at high temperatures, α-spodumene reacts with Na_2CO_3(s) to generate Li_2SiO_3, $NaAlSiO_4$, and Na_2SiO_3. Fig. 11(b) depicts the XRD patterns of the α-spodumene at the leaching residue at various Na_2CO_3(s). The results indicated that α-spodumene reacts with Na_2CO_3(s) at high temperatures to form Li_2SiO_3, $NaAlSiO_4$, and Na_2SiO_3. When working with mixtures of Na_2CO_3(s) (45 %)/spodumene (55 %), extraction efficiencies of 96 % and 90 % were achieved for Li and Rb, respectively, while the extraction rate of Al(III) was only 1.5 % under these conditions. This processing route for α-spodumene is considered promising compared to acidic leaching routes.

Kristianova et al. (2023) examined the extraction efficiency from zinnwaldite involving a roasting stage with $CaCl_2$(s). With a roasting period of 0.25 h and a heating rate of 10 °C/min, these authors achieved

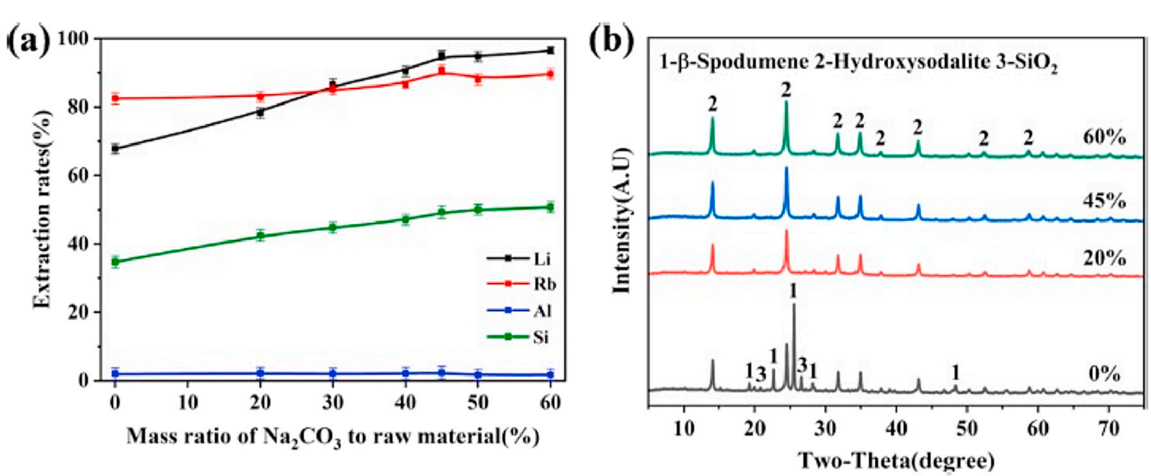


Fig. 11. Results for the calcination stage of α-spodumene with Na_2CO_3(s): (a) as a function of the Na_2CO_3(s) dosage, (b) XRD patterns for the leaching residue at varying proportions of Na_2CO_3(s) for the extraction of lithium (Zhou et al., 2023).

extraction efficiencies of 77 %, 90 %, and 90 % for lithium, potassium, and rubidium, respectively, at 1000 °C. To increase the efficiency of the process, Siame and Pascoe (Siame and Pascoe, 2011) adopted a different strategy with a tailing containing 0.8 % Li_2O, which was concentrated to 2.1 % Li_2O. This concentration stage involved froth flotation, and a wet high-intensity magnetic separator, with dodecylamine as the collector. The use of Na_2SO_4(s) increased the extraction efficiency for lithium to 97 % at 850 °C. Although this paper dealt with lithium, this idea can be exploited for the recovery of rubidium under optimum conditions. Similarly, Vu et al. (2013)(Vu et al., 2013) evaluated the roasting of zinnwaldite using $CaCO_3$(s) followed by water leaching of the obtained calcinates. The calcination process was described as a combination of the following stages: (i) transformation of zinnwaldite at temperatures of up to 800 °C; (ii) formation of new mineral phases from 750 °C to 835 °C; and (iii) formation of amorphous glassy phases above 835 °C. The first two stages favoured an increase in the extraction rates for lithium and rubidium, while the last stage of the formation of a glassy phase probably hindered lithium extraction while not affecting the extraction of rubidium. The results showed that maximum extraction efficiencies of 84 % and 91 % could be achieved for lithium and rubidium, respectively (temperature: 825 °C; leaching temperature: 95 °C). Lower efficiencies for lithium and rubidium extraction, also using $CaCO_3$(s) for the roasting of a zinnwaldite concentrate, have been reported by Jandova et al. (2010) **[50]**. An analysis of the calcinates by XRD identified the formation of a lithium and rubidium mineral called eucryptite, which has low solubility in water.

With a focus on other Rb-bearing kaolin minerals, **Zheng et al. (**Zheng et al., 2016**)** developed a chlorination roasting process with $CaCl_2$ (s) or NaCl(s) as alternative chlorinating reagents for Rb-containing kaolin. The leaching ratio for Rb reached a maximum value of 97 % when the chlorination roasting conditions were a mass ratio of kaolin (2):$CaCl_2$(s) (1) at 800 °C for 0.5 h. Zhou et al. (2015) compared direct H_2SO_4 leaching with chlorination roasting–water leaching of Rb from kaolin clay waste, and obtained similar extraction values. Hydrochloric acid, sulphuric acid, CaF_2(s), $CaCl_2$(s) and NaCl(s) were used as leaching and chlorination agents. For both acids, the Rb leaching efficiency decreased as a function of the chlorination reagents used, as follows: CaF_2(s) > NaCl(s) > $CaCl_2$(s). Rb leaching efficiencies of 95.5 % and 93 % were achieved using sulphuric acid and CaF_2(s) and clay waste/$CaCl_2$(s)/NaCl(s), respectively (temperature: 900 °C; leaching time: 60 min). For muscovite, a 97 % Rb leaching efficiency was accomplished by Tian et al. (2020) via a roasting process at 800 °C, with a muscovite (2):$CaCl_2$(s) (1) mass ratio and leaching in water at 60 °C for 0.5 h. During the chlorination roasting process, the muscovite was converted into $Ca(Al_2Si_2O_8)$(s) and Ca_2Al $[(AlSi)O_78$(s). Another study evaluated the roasting of an Rb-containing muscovite mineral incorporating the same chlorination reagents (Shan et al., 2013): when using a combination of NaCl(s) and $CaCl_2$(s) with a mass ratio of muscovite (1): NaCl(s) (0.25):$CaCl_2$(s) (0.25), and roasting at 850 °C for 30 min, a Rb leaching yield of 90 % was achieved. RbCl was subsequently obtained through SX using t-BAMBP, followed by re-extraction with hydrochloric acid and purification.

For pollucite, a roasting-water leaching method has been evaluated using single reagents (e.g., Na_2CO_3(s), $CaCl_2$(s)) and mixtures of Na_2CO_3(s)-NaCl(s) or $CaCO_3$(s)-$CaCl_2$(s), at temperatures of between 800 °C and 900 °C. In the final stage of leaching with water, similar extraction efficiencies were achieved compared to those reported for lepidolite (Butterman and Reese, 2003; USSG, 2023; USSG, 2020; Burt, 1993). Zeng et al. (2019) used a related methodology to recover Rb from biotite, and assessed both the direct H_2SO_4 leaching and chlorination roasting–water leaching pathways. Rb extraction rates from both leaching methods were comparable, at 97 % and 96 %, respectively. The potassium-to-rubidium ratio was lower in the acid-leaching leachate, which would make Rb purification easier.

Rb extraction from various unconventional sources has also been studied. For example, mixtures of Na_2SO_4(s) and $CaCl_2$(s) were evaluated with lithium-containing porcelain stone (Wang et al., 2020). The extraction efficiencies were measured for Li (99 %), Na (50 %), K (38 %), Rb (97 %), and Cs (98 %) using ore (1):Na_2SO_4(s) (0.2):$CaCl_2$(s) (0.2) mass ratios at 850 °C for 1 h, although a liquid:solid ratio of 1:1 and 60 min mixing time were ideal for water leaching. An XRD analysis of the calcined and leaching wastes identified a relevant cation-exchange extraction mechanism between the alkali ions in the mineral with Na^+ ions from Na_2SO_4(s), and Ca^{2+} ions from $CaCl_2$(s). The measured concentrations in the leachate were 3.4 g/L for Li, 25.5 g/L for Na, 11.0 g/L for K, 1.4 g/L for Rb, and 0.6 g/L for Cs. Extraction of Rb from gold-bearing waste via a three-stage process, involving acid washing, salt roasting, and water leaching, was examined by Mohammadi et al. (2015). In order to eliminate impurities, several strong acids (HCl, H_2SO_4, and HNO_3) were evaluated for the acid washing step, and 5 M HNO_3 at 85 °C for 5 h was found to produce the highest impurity removal ratios. A response surface methodology was used to optimise the roasting stage, achieving 91 % Rb leaching with the gold-containing waste using Na_2SO_4(s)/$CaCl_2$(s) combinations (temperature: 910 °C; leaching duration: 30 min). Scanning Electron Microscopywas used to confirm that an increase in the roasting temperature from 750 °C to 910 °C promoted phase transformation by increasing the leaching reaction rate.

Through the abovementioned studies, researchers have developed the confidence to initiate work on the large-scale processing of minerals to recover high-quality rubidium. The most Rb-enriched lithium minerals include lepidolite, a lithium mica rich in Rb and Cs; zinnwaldite, another lithium mica with significant Rb content; and spodumene and petalite, which are less commonly used for Rb recovery due to their lower Rb concentrations. For all of these groups of minerals, Rb is usually substituted for K^+ in the crystal lattice due to its similar ionic radius. In general, it is difficult to achieve selective leaching and separation of Rb, since Rb, Li, and K are chemically similar alkali metals and the required leaching factors could not be be achieved with the most widely applied leaching processes. Acid or alkaline leaching of roasted lithium mica concentrates (e.g. with H_2SO_4 or HCl) generates leachates containing Rb, and typically also Cs and K. The key challenges faced by these processing routes involve differentiating Rb^+ from K^+ and Cs^+, due to their similar chemical behavior. The studies evaluated here indicate that salt roasting (e.g. with Na_2CO_3(s), $CaCO_3$(s), $CaCl_2$, Na_2SO_4(s), NaCl(s) or CaF_2(s)) followed by water leaching can selectively liberate Rb^+ ions. Alkaline pressure digestion, i.e. the digestion of lepidolite/zinnwaldite with CaO(s), NaOH(s) or Na_2CO_3(s) under pressure, enhances Rb dissolution.

2.3. Concentration and separation of rubidium

It is essential to explore advanced extraction systems for Rb recovery. In this field, the focus of researchers in recent years has been on phenols, substituted phenols, and crown ethers. The solutions generated in the leaching stages of the minerals described above (in Sections 2.2.1, 2.2.2 and 2.2.3) are characterised by aqueous solutions with:

(i) A high acidity/basicity content, depending on whether the leaching stage has been performed with acids or alkali;
(ii) A high content of cationic species such as Na(I), K(I), Ca(II) and Mg(II) ions, which are either originally contained in the mineral or incorporated in the leaching and roasting stages (e.g. Al(III), Fe (III), Si(IV) species);
(iii) A higher content of Li^+ than Rb^+ ions, since the main driver for the processing of these mineral sources is the recovery of lithium.

The most successful recovery technologies for both elements (Li and Rb) involve the production of marketable products, which are Li_2CO_3(s) and LiOH(s) in the case of lithium, and RbCl(s) and Rb_2CO_3(s) for rubidium. All of these salts are characterised by relatively high solubilities (e.g. 910 g/L for RbCl(s) at 20 °C, and 2493 g/L at 25 °C for

Rb_2CO_3(s)), and efforts are therefore focused on concentrating the Li and Rb streams to reach the conditions needed for crystallisation. Concentration processes are typically carried out in two stages: (i) SX or ion exchange is used to selectively extract rubidium and to increase its concentration to the highest feasible level (e.g. greater than 100), and (ii) evaporation is applied to further concentrate the solution.

To date, the lithium industry has not shown much interest in the production of Rb salts, giving rise to limited efforts toward the development of SX reagents and ion exchange resins for Rb recovery. However, new inorganic ion exchangers have been developed to recover lithium for the production of batteries. In the case of Rb, inorganic sorbents industrially developed for caesium extraction from aqueous streams show a high level of selection over other alkaline and alkaline earth ions. Of this, we note the K-hexaferrocyanide sorbent commercialised by Fortum (Finland) to remove caesium during the reprocessing and management of nuclear wastes (Cs-Treat®). This substance has high levels of sorption and selectivity factors, but the non-reversible exchange mechanism for Cs^+ and Rb^+ limits the application of this material to recovery and concentration schemes.

A similar situation is observed for the case of SX reagents, as all the research efforts are focused on the use of 4-tert-butyl-2-(a-methylbenzyl)-phenol (t-BAMBP), whose molecular structure is shown in Fig. 12.

The extraction mechanism of t-BAMBP benefits from a chelating process in which the Rb^+ ions are exchanged with the H^+ of the phenol structure via the general extraction reaction in Eq. (1):

$$Rb^+ + n\ [(ROH)_2]_{org} \leftrightarrow [(RbOR)\cdot(ROH)_{2n-1}]_{org} + H^+ \quad (1)$$

Although no significant efforts have been made to evaluate the SX mechanism for Rb^+ extraction, some studies have described the formation of dimer species of t-BAMBP in the organic solvents used for this application ($(ROH)_{2,org}$), and extraction is postulated with the formation of solvated species with n values of 2 (e.g. $RbOR\bullet 2ROH_{(o)}$) (Lv et al., 2022). The regeneration reaction of t-BAMBP is later achieved by reversing the extraction reaction, using either a strong acidic solution or a strong basic solution, followed by the conversion of the anionic phenolate (RO^-) to the protonated form (ROH). The performance of t-BAMBP for alkali metal extraction follows the order $Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$ (Cheng et al., 2024)·

One of the main sources of interference when recovering Rb^+ and Cs^+ using t-BAMP is the presence of K^+ at much higher concentrations. The limitations on the potential for separation were evaluated by Zhang et al. (2020b) using solutions of t-BAMBP in sulphonated kerosene, as shown in Fig. 13. Separation factors for the couple Rb^+/K^+ ($\beta(Rb^+/K^+)$) remained constant at around 12 in the alkalinity range evaluated. The measured selectivity factors were high enough to enable the integration of a SX stage using t-BAMBP in the recovery of Rb^+ and Cs^+ from rich K^+ solutions.

2.3.1. Recovery and concentration of rubidium using solvent extraction

Originally, the solvent extraction technique was used for the trace level estimation of Rb from aqueous streams. In this vein, an extractive separation method for trace level rubidium was developed by **Mohite et al. (**Mohite and Burungale, 1999**)** using dibenzo-24-crown-8 from a picric acid medium. Rubidium was quantitatively extracted from 0.0006 to 0.05 M picric acid with 0.0006–0.01 M dibenzo-24-crown-8 in nitrobenzene as a diluent. From the organic phase, rubidium was quantitatively stripped with 1–10 M HNO_3, 1–8 M HBr, 0.5–10 M HCl and 10 M $HClO_4$. Rubidium was estimated by flame emission spectroscopy at 780 nm. Separation of rubidium was carried out from a number of binary mixtures. Most of the elements from the s-block, p-block and d-block showed high tolerance limits. It was possible to separate rubidium from the associated elements in multicomponent mixtures. This method is worth noting, as it was extended to the determination of rubidium in various rock samples. After this work, several studies were published on solvent extraction with various extractants, and it proved to be an important predecessor.

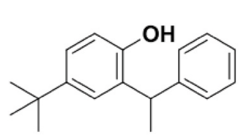


Fig. 12. Chemical structure of 4-tert-butyl-2-(a-methylbenzyl)-phenol (t-BAMBP).

Using another approach, Tang et al. (2018) tested a combined process involving water leaching of calcined samples of sintering dust, followed by solvent extraction using t-BAMBP. Purification with sodium carbonate and SX using t-BAMBP in sulphonated kerosene allowed for the selective extraction of Rb from the K^+-rich solution, with extraction and back-extraction efficiencies of 92 % and 90 %, respectively. The quantified separation factor (Rb/K) was 16. In the final stage, after two continuous counter-current extraction stages, a RbCl product with 99.5 % purity was achieved. By precipitating it as $Cs_3Bi_2I_9$, Cs^+ was extracted from coexisting solutions of Cs^+, Rb^+, and K^+ (Li et al., 2023a). For a Cs^+/Rb^+ molar ratio of 1.3, a maximum recovery rate for Cs^+ of 99 % could be achieved. In the $Cs_3Bi_2I_9$ precipitate, the Rb^+ content was found to be 0.2 %, confirming that the evaluating processing stage was effective for separating rubidium and caesium from each other (e.g. with a Cs/Rb separation factor of 85). $Cs_3Bi_2I_9$(s) was subsequently converted to CsI(s) (purity >99 %) and BiI_3(s) by applying vacuum thermal decomposition.

Lv et al. (2020) used solvent extraction as a processing methodology for lithium and Rb recovery from polylithionite via alkaline pressure leaching (95 % Li and Rb). This approach has also been applied to granitic ores: after alkaline leaching using CaO(s) and the removal of silicon, a three-stage solvent extraction using t-BAMBP in xylene (O/A (3/1) phase ratio provided a recovery yield of 99 % for Rb. Efficient stripping was performed using 1 mol/L HCl in two stages at a phase ratio of 10:1.

In view of the higher selectivity of t-BAMBP for Cs than Rb, Lv et al. (2022) evaluated the separation and purification of caesium from rubidium chloride leaching liquor, and reported an extraction efficiency for caesium of above 99 % after five stages of counter-current extraction (initial alkalinity: 10 g/L NaOH; t-BAMBP concentration: 1 mol/L; O/A ratio: 1:4). Following a four-stage counter-current scrubbing procedure, Rb and K were successfully scrubbed (>99.8 % for Rb^+ and K^+) from the organic phase using 0.1 mol/L HCl, despite the co-extraction of Rb and K as impurities in the organic phase (16 % Rb and 0.5 % K).

One of the most detailed schemes for the recovery of Rb by solvent extraction was reported for lepidolite mineral ores in China. The most critical stages of leaching were widely evaluated, including: (a) acid leaching (Liu et al., 2019; Vieceli et al., 2017), (b) sulphuric acid baking (Vieceli et al., 2018) **[76]**, (c) sulphate roasting (Yan et al., 2012c), (d) chlorination roasting (Zhang et al., 2020a), and (e) alkali leaching (Yan et al., 2012d). Of the abovementioned processing routes, the use of H_2SO_4 baking followed by water leaching was selected as a route for the recovery of lithium, potassium, caesium, and rubidium. The generated leachate was characterised as containing up to 5.1 g/L of Li^+, 2.7 g/L of Rb^+, 0.5 g/L of Cs^+, 17.9 g/L of K^+, and 30.5 g/L Al^{3+}. A recovery process of Rb(I) and Cs(I) based on a selective SX separation stage using t-BAMBP and the selectivity factors for both elements was postulated by Zhang et al., 2019, 2020b. A simplified flow-sheet for the recovery scheme of Li(I) as Li_2CO_3(s), Cs(I) as Cs_2SO_4(s) and Rb(I) as Rb_2SO_4(s) is shown in Fig. 14.

After proper filtration, the leachate undergoes an evaporation and crystallisation stage at 35 °C to give a blend of alum of $RbAl(SO_4)_2.12H_2O$(s), $CsAl(SO_4)_2.12H_2O$, $KAl(SO_4)_2.12H_2O$ to separate Rb (I) and Cs(I) from Li(I). After a new solid/liquid separation stage, the Li (I)-containing streamis neutralised with lime and treated with soda ash to precipitate Li_2CO_3(s). In this process, which takes place at 900 °C, Cs_2SO_4(s), Rb_2SO_4(s), K_2SO_4(s), and Al_2O_3(s) are the products of the

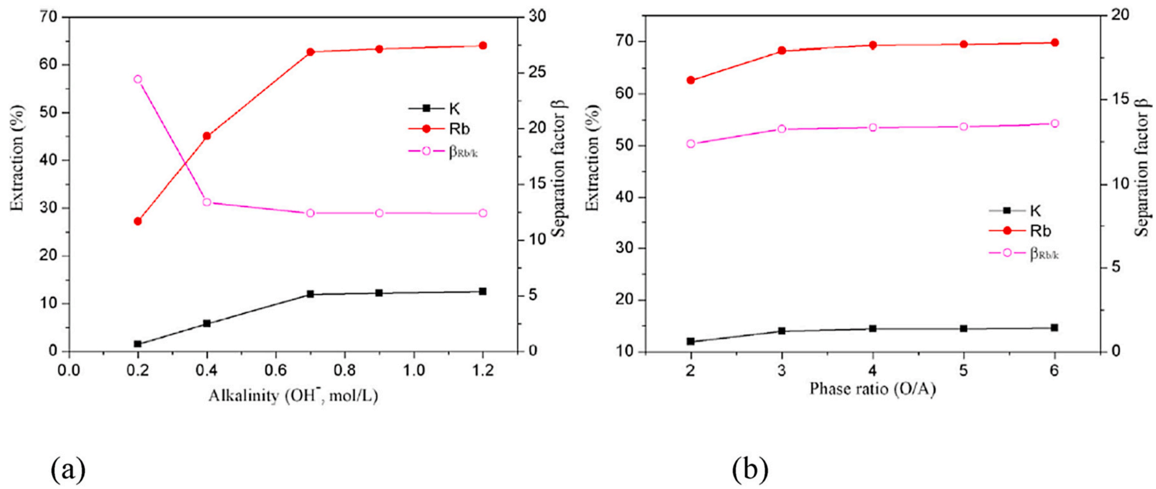


Fig. 13. Extraction-efficiency-based experimental evaluation of K(I) and Rb(I) by solvent extraction with t-BAMBP: (a) rubidium and potassium and (b) for (with an organic/aqueous phase ratio of 2, [t-BAMBP] of 1 mol/L, contact time of 5 min) (Zhang et al., 2020b).

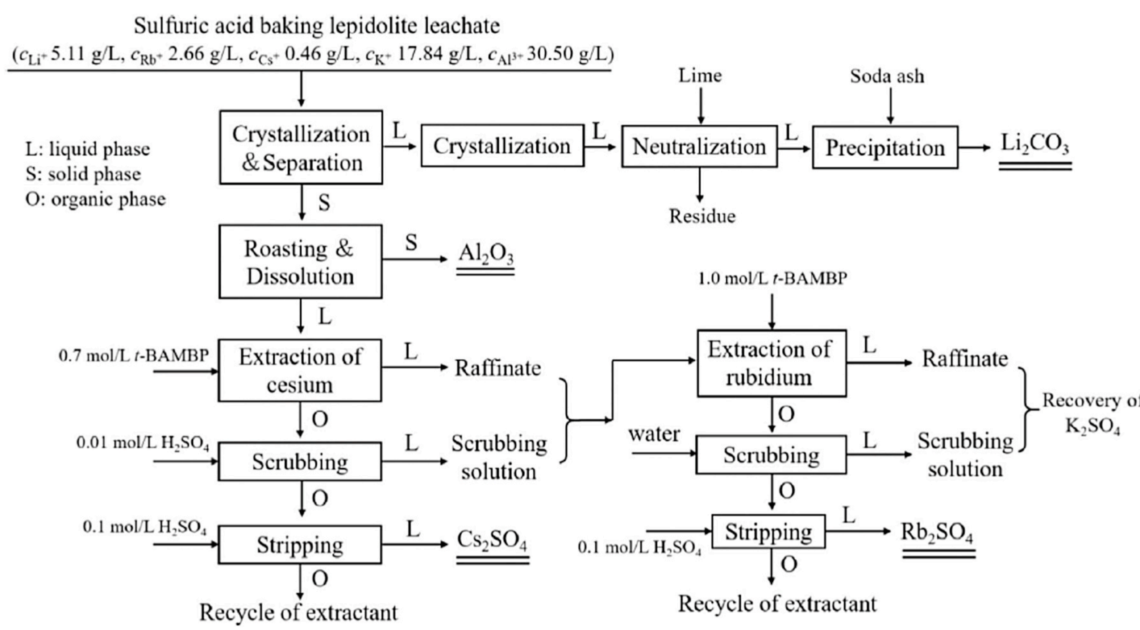


Fig. 14. Flowsheet for the recovery of lithium, caesium and rubidium from lepidolite via leaching with sulphuric acid, recovery of lithium as Li_2CO_3(s), and recovery of Cs(I) as Cs_2SO_4(s) and Rb(I) as Rb_2SO_4(s) by selective separation based on t- BAMBP (Zhang et al., 2020b).

roasting stage of the blend alum containing Rb(I) and Cs(I). Following the removal of the insoluble mineral (Al_2O_3(s)) and water leaching of the high-solubility minerals (Cs_2SO_4(s), Rb_2SO_4(s), and K_2SO_4(s)), a mixed sulphate solution of Rb(I), Cs(I), and K(I) is produced. The Cs(I) is separated from the Rb(I) and K(I) in the SX extraction stage by treating the aqueous solution containing sulphates with t-BAMBP, scrubbing, and stripping. The efficiency of Cs(I) recovery was found to reach 99 %, with a separation factor (Cs/Rb)) of 960 after the four-stage counter-current extraction, and a raffinate with 11 g/L Rb^+ and 51 g/L K^+ was obtained. Since Rb(I) is co-extracted during the extraction of Cs(I), the organic phase containing Cs(I) may contain up to 3.1 g/L Rb(I), despite the high SF (Cs(I)/Rb(I)). After the scrubbing stage with a diluted H_2SO_4 solution (e.g. 0.1 M), this Rb(I) content reached 6.9 g/L. When the raffinate and the washed solution were blended, a sulphate solution containing 10.3 g/L Rb^+, 0.24 g/L Cs^+, and 41 g/L K^+ was produced. Separation of Rb^+ and K^+ in this sulphate solution becomes more challenging due to these Rb(I) and K(I) concentrations. The richest streams obtained after the crystallisation of K_2SO_4(s) should be recirculated to the recovery stage of Rb(I).

2.3.2. Recovery and concentration of rubidium using sorption processes'

A limited number of studies have evaluated inorganic sorbents for Rb (I) recovery, and in most cases these have focused on the technological characterisation of materials in laboratory studies. The use of $(HK_3)O_2{\bullet}3SiO_2{\bullet}4TiO_2{\bullet}4H_2O$(s) as a cation exchanger (e.g. via exchange of K^+ by Rb^+ and Cs^+ ions) was described for the extraction of Rb and Cs from a solution after lepidolite processing (Yan et al., 2012a) **[32]**, in which two mesoporous biomass carbon sorbents were impregnated to immobilise particles of potassium titanosilicate in the surface by hydrothermal deposition. The sorption capacity of these two adsorbents reached values of 2.6 mmol Rb/g and 2.1 mmol Cs/g.

Other studies have considered the use of adsorbents that were initially developed for the extraction of Rb(I), Cs(I) and Sr(II) from nuclear waste streams. Most nuclear waste contains Rb(I), Cs(I) and Sr(II), and is characterised by a high salinity content, as it is often stored in concentrated solutions of caustic soda. Thus, the achieved selectivity factors in front of Na(I) are used as a reference for the mineral processing of streams with high salinity contents. **Taj et al.** (Taj et al., 2011) evaluated the recovery of Li^+, Rb^+ and Cs^+ ions using poly (methyl methacrylate) (PMMA) supported on $K_3Fe(Fe(CN)_6)$ (s), and found that the distribution coefficient (Kd) values followed the order $Cs^+ > Rb^+ > Li^+$. High removal ratios of Rb^+ and Cs^+ could also be achieved using 2.5 and 4.5 M HNO_3 solutions under optimised operating conditions. However, sorption studies were not complemented with re-extraction studies, as the system was developed as a stabilisation stage for Li^+, Rb^+ and Cs^+ in radioactive wastes. Lu et al. (2019) developed an interconnected superporous adsorbent (P-Yeast-PAA, porous polymer adsorbent) using a surfactant-free in water Pickering-High internal phase emulsions (HIPE) as a template. When applied to radioactive Rb^+, Cs^+, and Sr^{2+} ions, this porous adsorbent with a numerous carboxylic ($R\text{-}COO^-$) functional groups exhibited a high sorption capacity and fast kinetics. Under optimised conditions (a pH of 4–10 and an initial concentration of each metal ion of 0.3 g/L), the adsorption capacities for Rb^+, Cs^+ and Sr^{2+} ions were 178, 230 and 167 mg/g, respectively. Desorption was completed using a 0.5 mol/L HCl solution as an eluent, and the sorption/desorption reusability was maintained after five adsorption-desorption cycles.

Using a different approach to radioactive waste, a MOF-based adsorbent called UiO-66@iPCC5 was synthesised by hybridisation of 25,27-bis(iso-propoxyl)-calix [4]arene-26,28-crown-5 (iPCC5) into the pores of UiO-66 Zr-MOFs (Wang et al., 2021a). In this process, Rb was removed with an efficiency of 98 % from nuclear wastes by adsorption onto UiO-66@iPCC5 from 0.4 to 6.0 M HNO_3 containing solutions. Ammonium molybdophosphate (AMP) and zirconium phosphate (ZP) ion exchangers have also been efficiently embedded in alginate capsules as Rb extractants (Krys et al., 2013). Rb^+ sorption took place at pH 3, but this behaviour was found to be different for an AMP-based sorbent, as it was less sensitive to pH than its ZP-based counterpart. The sorption capacity ranged between 56 and 63 mg Rb/g, and a combination of ammonium chloride with nitric acid (ranging from 0.4 to 6 M) enabled desorption of Rb(I) from loaded sorbents. **Ertan et al.** (Ertan, 2023) evaluated the sorption of Li, Rb, and Cs from clay-containing boron waste using the same family of AMP-based sorbents, and reported sorption capacities of 299, 149 and 390 mg/kg, respectively. Using γ-ray irradiation, Khalil et al. (2023) synthesised a polyacrylonitrile-titanium oxide nanocomposite adsorbent and assessed how well it extracted Ga (III), Sr(II), and Rb(I) from groundwater and simulated radioactive wastewater. The Rb^+ sorption capacity reached 3.8 mg Rb/g, a moderate value, since the levels of interferent ions (e.g. Mg^{2+}, Na^+, K^+) were relatively low compared to mineral leachates or brines.

3. Recovery of rubidium from brines

Seawater has been identified in many national geological surveys as a potential source of elements. Due to the large amounts of water in oceans and seas ($1.3 \cdot 10^{18}$ tonnes), the total amount of dissolved salts (c. a. $5 \cdot 10^{16}$ tonnes) is much larger than is available from land reserves, which makes the option of recovering minerals from various brines an attractive one (Jones et al., 2019). One of the most important research and development efforts involves uranium as a resource for the nuclear industry and the uranium mining projects, from sea water, developed by Japan in the (Egawa et al., 1980) and more recently by USA in the last decade. In these two research programs, an on-site process was used in which U(VI) ions were extracted from seawater bodies using fibrous ion exchangers supported on floating structures or shipped by large boats to the areas with the highest ocean streams, as shown in Fig. 15.

Although these programs attracted very large investment efforts, involving billions of yen and dollars, both programs were stopped and were never implemented on an industrial scale. However, in the context of seawater desalination by RO as a solution to produce fresh water, the possibility of recovering minerals from the generated brines opens up new avenues. The metals will not be mined in the seawater bodies and the recovery plants will be allocated close to the fresh water producing plants. In practice, desalination plants represent one of the main sources of highly concentrated brines. According to published reports, information is available on nearly 20,000 seawater desalination plants across the world (Jones et al., 2019; DesalData, 2018). About 16,000 of these are currently operating in 177 countries and territories, with a combined desalination capacity of about 95.4 Mm^3/d (35 billion m^3/y). The main uses of desalinated seawater are in the industrial and municipal sectors (62.3 % and 30.2 %, respectively), although smaller amounts are also used in the irrigation and power sectors (4.8 % and 1.8 %, respectively).

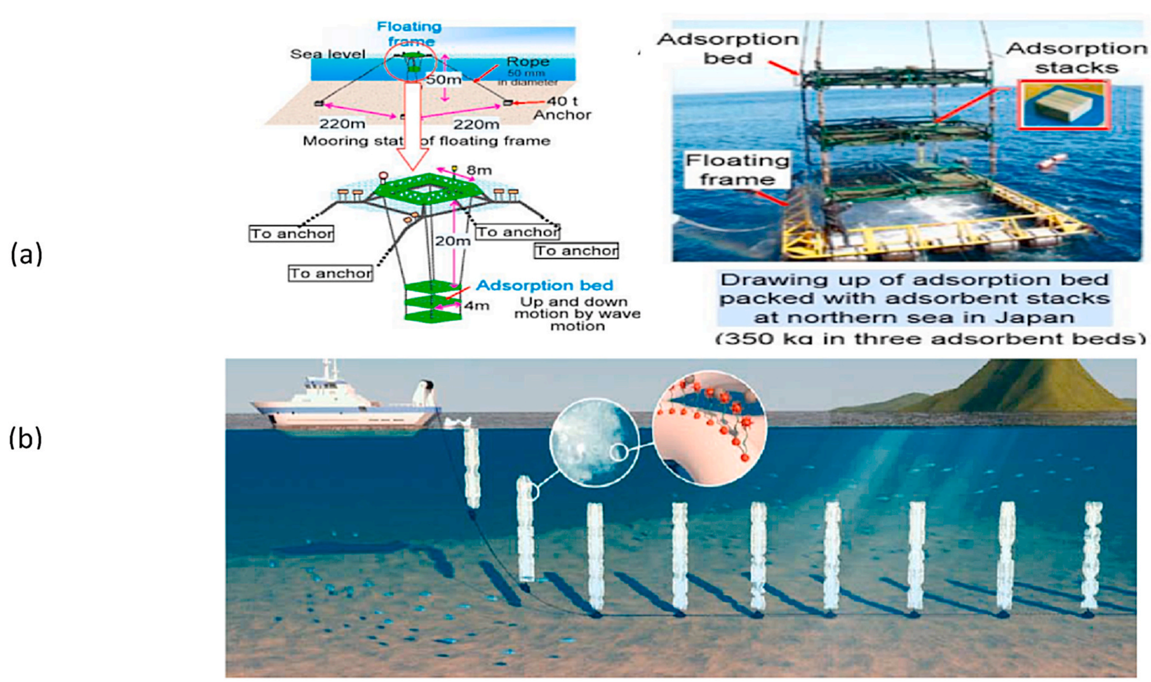


Fig. 15. On-site systems for uranium extraction from seawater: (a) U(VI) recovery project in Japan **[116]**; (b) U(VI) recovery project in the USA (Alexandratos,-dratos, 2016).

Municipal water desalination plants produce an average of 12,126 m^3/d, while the average production capacity of industrial desalination processes is 3712 m^3/d. Table 3 provides the numbers of desalination plants, desalination capacities and the brine production capacities for various geographic regions (Jones et al., 2019).

More recently, there has been growing interest in the residual brines generated from solar evaporation saltworks after the production of table salt (known as bitterns). Randazzo et al. (2024) evaluated the potential of these bitterns as a source of CRMs after characterising these brines in the Mediterranean region. These authors reported that several ions, such as Na, K, Mg, Cl^-, SO_4^{-2}, and Br^-, were present in large quantities (g/kg). B, Ca, Li, Rb and Sr were found in minor quantities (mg/kg), whereas cobalt, caesium, gallium, and germanium were found at trace levels (μg/kg). Fig. 16 shows the concentrations of the bitterns according to the nature of the source, i.e. sea, salt lake, ocean, river and subsoil. Of the minor elements, Rb^+ is present below 100 mg/kg. Compared to seawater, where Rb concentrations range from 0.1 to 0.2 mg/kg, the reported values for Rb are significantly greater (by a factor of 50–100).

Finally, the substantial increase in the worldwide use of geothermal power should also be considered, from 12.7 GWe in 2010 to 21 GWe in 2021 (Gao et al., 2017). This is expected to offer good prospects for the recovery of minerals from geothermal brines in the near future. There are several countries that operate geothermal power/energy plants, including North America, Asia, New Zealand, Mexico, Russia, and Iceland. A detailed discussion of geothermal brines has been presented by Sharma et al. (2023). Table 4 summarises the concentrations of the components of the bitterns, including rubidium, from the above-mentioned sources.

3.1. Salt lake brines

In recent years, a large number of natural salt lakes worldwide (e.g. in China, USA, and Peru) containing abundant contents of potassium, magnesium and lithium have attracted attention as resources for rubidium and caesium (Shvartsev et al., 2014; Shi et al., 2017). However, no trials have been carried out to recover rubidium and caesium at an industrial scale. It should be mentioned that the recovery of both these elements is relatively simple and cost-effective, since high concentrations (on the order of mg/L) can be found in these brines. This indicates that salt lake brines represent an important secondary source for rubidium and caesium (Shi et al., 2017). However, the recovery of rubidium involves a preliminary stage of extraction and concentration before any crystallisation process, using key technologies such as SX, adsorption and Ion Exchange IX), and employing both inorganic and polymeric sorbents (Arnold et al., 1965) **[139] .**

Table 3
Number/capacity of operational desalination plants and brine production by region (Jones et al., 2019).

	Number of desalination plants	Desalination capacity		Brine production	
		(Mm^3/d)	(%)	(M m^3/d)	(%)
Global	15,906	95.37	100	141.5	100
Geographic region					
Middle East and North Africa	4826	45.32	47.5	99.4	70.3
East Asia and Pacific	3505	17.52	18.4	14.9	10.5
North America	2341	11.34	11.9	5.6	3.9
Europe	2337	8.75	9.2	8.4	5.9
Latin America and Caribbean	1373	5.46	5.7	5.6	3.9
Southern Asia	655	2.94	3.1	3.7	2.6
Eastern Europe and Central Asia	566	2.26	2.4	2.5	1.8
Sub-Saharan Africa	303	1.78	1.9	1.5	1.0

3.1.1. Extraction via inorganic sorbents

As described in the previous section, inorganic sorbents have shown the best performance in the extraction of Rb^+ from aqueous solutions. Although KFeHCF has the highest selectivity factors, its low desorption efficiency has given rise to a need for the synthesis and evaluation of new cyano-based sorbents in which the Fe ion is replaced by a transition metal cation. Copper ferricyanide (CuHCF) and its analogues (PBAs) were used by Zhang et al. (2023) to selectively extract rubidium from salt lake brines via ion exchange. These authors employed an electrochemical technique to intercalate K^+ ions into copper hexacyanoferrate (Cu_3 $[Fe(CN)_6]_2$), and as a result, a K_2Cu_3 $[Fe(CN)_6]_2$(s) adsorbent was developed. The measured SF were $\beta_{Rb/K}$ (63), $\beta_{Rb/Na}$ (1635),$\beta_{Rb/Ca}$ (585) and $\beta_{Rb/Mg}$ (2543). Stripping of Rb was enhanced by electrochemical oxidisation of Rb_2Cu_3 $[Fe(CN)_6]_2$, with a reported efficiency of around 98 %, and a high rate of rubidium enrichment could be accomplished by repeating the above process over different cycles. This study was limited from the perspective of industrial applications, as the work was done using synthetic brines.

A similar study was conducted by Li et al. (2023b) with a new kind of hexacyanoferrate sorbent containing Mg(II) (RbCs–MgHCF)) and with a chemical composition of $Rb_{0.61}$ $Cs_{1.20}K_{0.17}Mg_{1.01}Fe(CN)_6$. In comparison with adsorbents containing conventional divalent transition metals (e.g. CuHCFs, ZnHCFs, NiHCFs, CoHCFs or FeHCFs), their sorbent had better extraction properties due to its large particle size, giving yields for Cs^+ and Rb^+ of 99 % and 95 %, respectively. For this newly developed sorbent, a cost analysis and an assessment of the long-term duration of the material are needed to understand its potential for use in full-scale applications. Using a different approach, Dagan-Jaldety et al. (2023) recovered pure RbCl(s) using polyethersulphone (PES)-coated Zn-HCF from solutions with large concentrations of Na^+ and K^+. The brine containing Rb^+ was treated in a two-column adsorption setup, and regeneration was done using 1 M NH_4Cl. Chromatography-based separation between Rb^+ and Na^+/K^+ was accomplished using the second column with the adsorbent in the ammonium form, with Rb^+ adsorption reaching values greater than 8 %. A small amount of Rb^+ was partially readsorbed in the second column after Na^+ and K^+ had been extracted in the first column using 0.05 M $NH4^+$ solution. The stripped solution generated from the second column was first concentrated by evaporation coupled with NH_3/HCl sublimation in order to obtain pure RbCl(s). The economic potential of this strategy was highlighted by an extensive cost study, which showed that the cost of producing RbCl(s) did not exceed about 25 % of the current salt price.

The following studies deal with the use of materials that are not related to cyanide groups to recover rubidium. Fang et al. (2023) evaluated the recovery of Rb^+/Cs^+ from brines using an integrated process of adsorption using AMP and flotation. The extraction yield of Rb^+ and Cs^+ reached 80 % after five cycles. Since the selectivity of AMP for Rb^+ and Cs^+ differs, these elements could be removed via a two-step adsorption process. The extraction efficiencies of Rb^+ and Cs^+ in the first step were 18 % and 91 %, respectively, while the values in the second step were 8 % and 80 %, respectively. The primary mechanism of Rb^+ extraction at the flotation stage was the interaction between the cetyltrimethylammonium bromide (CTAB) cations and the rubidium molybdophosphate (RMP)/caesium molybdophosphate (CMP) particles. At the flotation stage, the interaction between CTAB cations and RMP/CMP particles was the main Rb $^+$ extraction process. Fang et al. (2021) also reported an extraction efficiency of >99.9 % with the use of an integrated process of flotation and adsorption, using ammonium phosphowolframate (AWP) and CTAB as the collector and foaming agent for the extraction of Rb^+ and Cs^+ from aqueous solutions.

Some studies have also been conducted with a selective absorbent produced by grafting AMP onto a metal organic framework. Wang et al. (2022) prepared an adsorbent (AMP@MOF-76(Sm)) by grafting AMP onto a framework of MOF-76(Sm) via a solvothermal synthesis method.

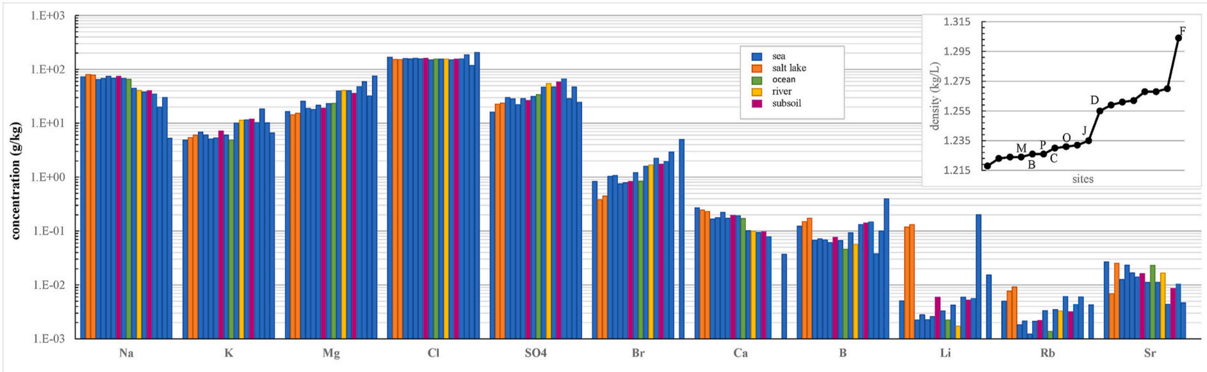


Fig. 16. Variation in the concentrations of elements present in saltwork bitterns, sorted according to density values from sample L (1.22 kg/L) to sample F (1.30 kg/L). Composition values are drawn from bitterns from 18 various solar sea saltworks located in the Mediterranean Sea (Randazzo et al., 2024).

Table 4
Chemical composition of brines.

Resources	Source	Concentration (mg/L)							Ref.
		Rb (mg/L)	K (mg/L)	Li (mg/L)	Cs (mg/L)	Mg (mg/L)	Na (mg/L)	Cl mg/L	
Seawater brine	El Prat de Llobregat, Spain	0.19	554	0.27	<0.001	2450	27,521	35,245	Peterskova et al., 2012
Salt lake brine	Qinhai, China	0.02	2.77	0.68				NA	Liu et al., 2015
Salt lake brine	Tibet, China	0.02–0.05			0.01–0.02				Li et al., 2014
Salt lake brine	Dead Sea, Israel	60	6670			39,600	2396		Ding et al. (2022); Reiss et al., 2021
Seawater brine	Perth, Australia	0.2	0.78	0.4		2.57	22,000	41,400	Naidu et al. (2017)
Seawater brine	La Skhira, Tunisia	0.12	380			1350	10,500	NA	Dirach et al. (2005)
Geothermal brine	Salton Sea, USA	170	16,700	194	20	33	53,000	NA	Warren, 2021
Geothermal brine	Cerro Prieto, Mexico	11	2210	27	39	0.50	8300	NA	Mercado and Hurtado, 1992

Over a range of pH from 4 to 10, the adsorbent extracted Rb^+ and Cs^+ ions from aqueous solution, whereas a coexisting ion such as Na^+ had little effect on the Rb^+ and Cs^+ capacity. The maximum measured capacities of Rb^+ and Cs^+ were 0.5 mmol/g and 0.44 mmol/g, and ions could be desorbed with 3.0; solution.

Although the studies described above (Fang et al., 2021, 2023; Wang et al., 2022) were carried out at a laboratory scale, there is not clear the potential to be applied to multicomponent mixtures for Rb recovery. The authors did not perform comparisons with other extraction techniques in terms of technoeconomic evaluations of the mineral processing cost under optimal conditions.

3.1.2. Extraction by organic sorbents

The main factors limiting the use of inorganic sorbents are related to the stripping stage and the difficulties in preparing adsorbents with the required particle size for column operation, and this has driven researchers to develop sorbents with a polymeric structure. On the basis of the polymeric structure, selective extractants, such as organic molecules, inorganic materials or nanomaterials, have been developed and evaluated, as described in the following sections. In most cases, the use of composite materials has been stressed.

Magnetic ion-imprinted polymers (MIIPs) for Rb^+ were developed by Jiang et al. (2022a). As the basic skeleton material, polydopamine-coated magnetic graphite oxide was used, modified with a silane coupling agent. 18-crown-6 (18C6) and methacrylic acid were used as ligands with cation recognition ability and functional monomers polymerised at the surface of the magnetic carrier. When tested on a solution containing 400 mg/L Rb^+ at pH = 10, the highest adsorption capacity of the sorbent was 186 mg Rb^+/g. The MIIP displayed selectivity for Rb against K^+, Li^+, Na^+, Mg^{2+}, and Ca^{2+}, with selectivity coefficients of 2.3, 2.5, 2.2, 1.9, and 1.8, respectively. This selectivity was found to be maintained after seven adsorption-desorption cycles, although the experiments were conducted using synthetic solutions.

Similarly, Zhou et al. (2021) used MAA as the functional monomer, 18-crown-6 (18C6) ether as the selective ligand, ethylene glycol dimethacrylate (EGDMA) as the cross-linking agent, and 2,20-azobisisobutyronitrile (AIBN) as the initiator to synthesise a Rb(I) ion-imprinted polymer (Rb^{In} -IIP). In this process, a promising Rb^+ adsorption capability was demonstrated by the Rb^+-IIPs (213 mg Rb^+/g), with high adsorption selectivity for Rb^+ ions against Li^+, Na^+ and K^+ ions. It should be mentioned that in neither of these studies was an economic evaluation of the cost of the prepared sorbents carried out.

Dai et al. (2018) developed a series of composite adsorbents that incorporated phosphomolybdic acid (PMA) and MOFs that contained Cu (II)-paddle wheel-type nodes and 1,3,5-benzenetricarboxylate struts (HKUST). Batch adsorption tests showed that these composites exhibited high Rb^+ capacities (ca. 99 mg Rb^+/g) through a sorption mechanism between the PMA group and Rb^+ ions. The co-ions (K^+, Na^+, and Cs^+) in the brines had a moderate impact on the Rb^+ sorption selectivity.

Using a PDMS-coating treatment on a bimetallic HKUST-1 surface, Tian et al. (2019a) developed a series of water-resistant adsorbents termed PDMS@HKUST-1 (PDMS, polydimethylsiloxane). These composites displayed good wettability, and had a high adsorption selectivity for Rb, with the initial Rb uptake capacity being maintained after five cycles. The coated adsorbents preserved the initial crystalline structure of HKUST-1 and the associated pore properties. A sulfhydryl group and Fe_3O_4(s) doped in the porous cavity of the adsorbent were used in a different study (HS-Fe_3O_4@MIL-53(Al)) (Wang et al., 2022); in this case, the composite material achieved efficient sorption for rubidium through the sulfhydryl functional group. The Rb^+ sorption selectivity was found to be high against K^+, Na^+ and Cs^+ due to doping with HS-Fe_3O_4. The Rb^+ uptake capacity of HS(50)-Fe_3O_4@MIL-53(Al) decreased from 85 to 57–64 mg/g in coexistence with Na^+, K^+ and Cs^+, but the large-scale production of this material would be challenging and expensive. A general synthetic strategy for sulfhydryl-Fe_3O_4@MIL-53(Al) adsorbent was developed by Tian et al. (2019b). This composite material exhibited a good adsorption performance for Rb^+. The interaction between the Rb^+ ions and the H^+ ions in the sulfhydryl group was the reason for the high Rb^+ uptake capacity over the HS-Fe_3O_4@MIL-53(Al). Due to the HS-Fe_3O_4 doped material, with high sorption capacity over these ions, the inclusion of other co-ions (K^+, Na^+ and Cs^+) had a moderate impact on the Rb^+ sorption selectivity.

With the aim of selectively adsorbing and enriching Rb^+ and Cs^+, Zhu et al. (2017) developed a recyclable magnetic porous adsorbent by grafting acrylic acid (AA) onto hydroxypropyl cellulose (HPC) and employing a Pickering emulsion that integrated precipitation and polymerisation. The grafting of modified Fe_3O_4(s) nanoparticles onto an absorbent surface stabilised the Pickering emulsion, thus generating a porous structure, and the separation of the adsorbent from the solution was simplified using a magnetic field. In this process, Rb^+ and Cs^+ had adsorption capacities of 310 and 448 mg/g, respectively, and regeneration was readily accomplished using acidic solutions.

A novel Rb^+ complexing structure based on potassium titanium silicate-calcium alginate was used by Li et al. (2014a) to extract Rb^+. Even for single electrolyte solutions, the maximum Rb^+ adsorption capacity was 1.6 mmol Rb/g, which was greatly reduced when the NaCl concentration was increased to 1 mol/L. Fang et al. (2017) prepared a phenol@MIL-101(Cr) composite material through impregnation to improve the uptake capacity and selectivity for Rb^+. The Rb^+ capacity of PH@MIL-101(Cr) composite sorbents was 93 mg/g, a value comparable or superior to those reported previously using potassium titanium silicate-calcium alginate composites. In addition to a high rubidium capacity, PH@MIL-101(Cr) sorbents could be easily regenerated by 5 M NH_4NO_3 and more than 90 % of the Rb^+ uptake capacity could be recovered after regeneration. In a similar study, an AMP, sodium alginate and calcium chloride adsorbent was prepared by Ye et al. (2009). Maximum adsorption capacities of 0.7 mmol Cs^+/g and 0.6 mmol Rb^+/g were achieved at pH values of 3.5–4.5.

In summary, although the results have been promising for the cases described above involving salt lake brines, further development will be necessary, including optimisation of strategies for sorbent regeneration, before they can be used as adsorbents for Rb^+ recovery. Most of the results reported in the literature are found to have a very limited technical feasibility when the sorption capacities are analysed. In addition, a detailed cost analysis of the synthesis of these materials will be required before moving to pilot-scale evaluations. In most cases, the production capacity of the selective ligands, ionic liquids, and nanoparticles is limited to a small scale, and there is a total lack of information on the cost of production for these types of sorbents.

3.1.3. Extraction via solvent extraction

As described in Section 2.1, the use of t-BAMBP as selective extractant for Rb^+ from lake brines has also been evaluated. In this scenario, the concentrations of Na^+ and K^+ are much higher than for the streams coming from mineral processing routes. For this reason, several studies have addressed the effects of both Na^+ and K^+ on the extraction of Cs^+ and Rb^+.

Li et al. (2017) reported a $\beta_{Cs/K}$ of 139 and $\beta_{Rb/K}$ of 11 with t-BAMBP from a synthetic brine solution containing 0.020 g/L Cs^+, 0.2 g/L Rb^+ and 5 g/L K^+. Using a single contact procedure, the extracted Cs^+, Rb^+, and K^+ organic complexes with t-BAMBP were stripped using 0.1 M HCl at an A/O ratio of 1:1. Under these conditions, a continuous batch test consisting of five extraction stages was carried out. Cs^+ and Rb^+ were extracted at a rate of more than 99 %, and K^+ was also co-extracted at a rate of 20 %. The concentration ratios achieved of K^+/Cs^+ and K^+/Rb^+ were decreased five times, although no discussion of the reduction mechanism was provided. Similarly, the influence of the nature of the organic solvent on the separation kinetics and efficiency of Cs^+ and Rb^+ extraction from a synthetic brine solution containing K^+ has also been evaluated (Zhang et al., 2018). t-BAMBP in dodecane was partially saponified by NaOH to promote the formation of ionised species of t-BAMBP (RO-Na + $_{org}$), and was directly used for extraction without adjusting the pH of the feed solution. Compared to cases involving protonated t-BAMBP (ROH_o) (Huang et al., 2018c), much higher separation factors (β) for Cs^+ and Rb^+ over K^+ ($\beta_{Cs/K}$ (1551), $\beta_{Rb/K}$ (45)) were achieved due to the enhanced difference in the kinetics during ion exchange and the ionised species of t-BAMBP (RO-Na + $_{org}$). In addition, since the NaOH saponifying agent was recyclable, significant amounts of strong alkaline wastewater were avoided and the usage of NaOH was decreased by 70–97 %. Using a three-stage extraction process, 99 % Cs^+ and 87 % Rb^+ were recovered from the synthetic brine solution along with high K^+ concentrations.

Other studies have more comprehensively covered the effects of both K^+ and Na^+ on the extraction of Cs^+ and Rb^+. For example, Li and Liu (2014b) reported that K^+ had a negative influence on the extraction of Rb^+ with t-BAMBP (in sulphonated kerosene (SK) as solvent), whereas Na^+ had no effect on extraction. With an increase in the K^+ concentration of the solution, the extraction rate of Rb^+ decreased significantly. In a second study, Liu et al. (2015) evaluated the sequential extraction of K^+, Rb^+ and Cs^+ using 1.0 M of t-BAMBP in SK, and identified the need to remove Mg(II) from the brine using a precipitation stage to achieve high concentration factors for Rb^+ and Cs^+. The co-extracted amounts of Na^+ and K^+ could be efficiently scrubbed with 10^{-4} mol/L NaOH solution, with losses for Rb^+ of about 11 % after three scrubbing stages for an O/A = 1:0.5. After a five-stage counter-current extraction process, the final extraction yields of Rb^+ and Cs^+ were 9 % and >99.8 %, respectively.

In similar work with real feed solutions (80 g/L K^+ and 200 g/L Na^+), the separation of Cs^+ and Rb^+ using t-BAMBP as an extractant was evaluated by Xie et al. (2023). The Cs^+ and Rb^+ extraction rates were 99.8 % and 98 %, respectively. Potassium was co-extracted with Cs^+ and Rb^+ in the organic phase after a five-stage counter-current extraction process with extraction rates of 19 %. After a five-stage counter-current scrubbing process, around 99 % of the K^+ in the organic phase was removed with deionised water at an O/A ratio of 2:1. Finally, stripping of caesium and rubidium was accomplished (>99 %) using 0.5 mol/L HCl via two-stage counter-current stages at an O/A ratio of 3:1.

The extraction of Rb^+ and Li^+ from a lithium-rich brine using t-BAMBP (in cyclohexane) was investigated by Wang et al. (2015). At a pH of 13 and with a 1 M t-BAMBP solution, t-BAMBP/cyclohexane was found to be effective and selective for Rb^+ extraction. Fig. 17 shows that with increasing t-BAMBP concentrations, the Rb^+ distribution coefficient and the Rb^+/Li^+ separation coefficient became larger. With $\beta_{Rb/Li}$ ratios of near 100, Rb recovery could exceed 90 % in this experimental

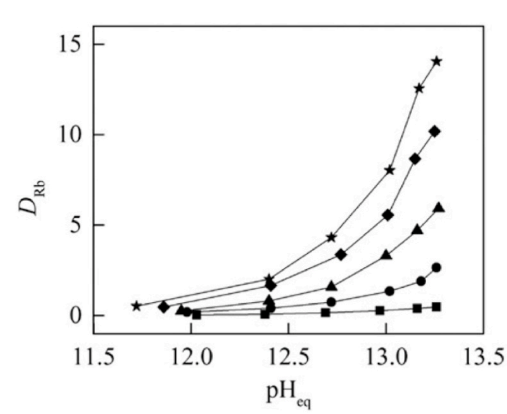


Fig. 17. Experimental data on the distribution coefficients (D_{Rb}) versus equilibrium pH for five different concentrations of t-BAMBP (■ 0.2 mol L^{-1}; ● 0.4 mol L^{-1}; ▲0.6 mol L^{-1}; ◆ 0.8 mol L^{-1}; ★ 1.0 mol L^{-1}), as reported by Wang et al.(Wang et al., 2015).

range.

Finally, there were preliminary attempts by Luo et al. (2019) to evaluate the use of ionic liquids (ILs), with a focus on C2mim$^+$ NTf2$^-$ as a solvent and bis(2-propyloxy)calix [4]crown-6 (BPC6) as an extractant for Rb$^+$ at a neutral pH. Characterisation of the extraction system indicated that BPC6 formed a 1:1 complex with Rb$^+$. Although the addition of C2mim$^+$ limited the extraction, the addition of NTf2$^-$ to the aqueous phase increased the extraction efficiency of Rb$^+$. In contrast, the extraction efficiency of Rb$^+$ by BPC6/C2mimNTf2 increased with an increase in the concentration of BPC6. When the concentration of BPC6 exceeded 12 mmol/L, the extraction efficiency was >99 % and the distribution coefficient of Rb (D_{Rb}) was 103. The extracted Rb$^+$ could be stripped (>84 %) with a 2 M NH_4Cl solution. Huang et al. (2018c) reported similar attempts to use an imidazolium-based IL, 1-ethyl-3-metyhlimidazaoliumbis (tri-fluoro-methylsulfonyl) imide ([C2mim$^+$][NTf2$^-$]), as part of a solvent used for extraction and DCH18C6 as a selective extractant to separate Rb from brines. The extraction mechanism was based on a cation exchange reaction related to the formation of [Rb·DCH18C6]$^+$ complexes. For Rb$^+$ and Cs$^+$ ions, the single extraction efficiencies were 84 % and 95 %, respectively, while the stripping efficiency was 82 % using 2.5 M NH_4NO_3. The selectivity factors for Rb against K$^+$ and Cs$^+$ were 93 and 104, respectively. The main limitation of this system was related to the potential for recycling and reusability of the solvent and extractant. In any case, the limited success of IL in industrial applications is reducing the potential interest of this option when thinking about industrial recovery schemes of Rb from salt brines.

Table 5 presents the results of an assessment of different technologies and methodologies for extracting Rb from brine, with SX being the most commonly used approach.

3.2. Seawater brine

Although the total Rb$^+$ content in seawater (156,000 × 10^6 tonnes) is several orders of magnitude higher compared to land mineral reserves (0.08× 10^6 tonnes), Rb$^+$ is present at lower concentrations (0.19–0.24 mg/L) compared to the other prevailing elements (Na, Ca, Mg and K) (Jones et al., 2019). Due to its lower concentration, Rb$^+$ is generally more difficult to extract from seawater brine than from salt lake brine. Nevertheless, the increase in desalination capacity worldwide is continuously increasing in thelast decade. In view of this, the recovery of large amounts of minerals such as rubidium from brines is under evaluation. In most seawater reverse osmosis desalination brines (Seawater reverse osmosis desalination brines) (SWROs) (www.sea4value.ue) and the seawater brines generated during the production of table salt in saltworks (www.searcualrmine.eu) (Production of table salt), the concentration of Rb$^+$ is in the lower range (mg/L scale) (2–20 mg/L), whereas that of Cs$^+$ is on the μg/L scale. Hence, the challenges associated with extracting rubidium from these types of brines are similar to those described in Section 3.1. A few studies have been conducted on the application of SX, but the concentration levels of Rb$^+$ are far from feasible for the practical use of SX in terms of the large-scale recovery of Rb. Following the main efforts developed are presented.

3.2.1. Extraction of rubidium via solvent extraction

Chen et al. (2020) examined the recovery of Rb$^+$ and Cs$^+$ from brine. In order to overcome the main limitations of the process, the brine was concentrated via evaporation prior to SX, and the main competing ions (Li$^+$, Na$^+$, K$^+$, Ca^{+2} and Mg^{+2}) were removed under different conditionsn. More precisely, K$^+$ was precipitated as $KClO_4$ by adding perchloric acid. In the next step, t-BAMBP was used as an extractant (Rb extraction >50 % at 1 M t-BAMBP) and ammonia was used as a stripping agent to obtain RbOH and CsOH solutions (with a stripping efficiency for Rb$^+$ of >90 %). Furthermore, Rb_2CO_3(s) and Cs_2CO_3(s) were obtained by adding ammonium carbonate. Using a similar approach, a solution of t-BAMBP in xylene was used to recover Rb$^+$ and Cs$^+$ from a raffinate with a high concentration of K$^+$, to simulate an application involving seawater (Lv et al., 2023). Following a four-stage counter-current extraction process, yields of Rb$^+$ of above 95 % were obtained, and K$^+$

Table 5
Studies reporting the use of solvent extraction of Rb from brine.

Experimental conditions	Metal ions	Initial/raffinate concentration (g/L)	Extraction (%)/metal product	Ref.
Extractant: 1 mol/L t-BAMBP (in sulphonated kerosene) Conditions: 0.5 mol/L NaOH, O/A ratio of 3:1, and 1 min contact time	K$^+$ concentration (exceeding 80 g/L), 200 g/L alkali metal ions	Raffinate K 77.1 Rb 5.5 Cs 2.3	> 99 % of Cs and 98 % of Rb via a five-stage counter-current extraction	Xie et al. (2023)
Extractant: DCH18C6 in imidazolium-based IL, 1-ethyl-3-metyhlimidazaoliumbis (trifl uoromethylsulfonyl)imide ([C2mim+][NTf2–]) Conditions: 2.5 mol/L NH_4NO_3 as a stripping agent	Rb, Mg, K K/Rb ratio: 1.58	In mg/L range	94 % of Rb in a single extraction	Huang et al. (2018b)
Extractant: 1 mol/L t-BAMBP, 1 mol/L alkalinity OH^-, phase ratio (O/A) 1:1; scrubbing: NaOH 10^{-4}mol/L, phase ratio 1:0.5	Rb Cs Na Mg K Li	Rb 0.020 Cs 0.002 Na 19.39 Mg 28.38 K 2.77 Li 0.68	>95 % Rb and Cs	Liu et al. (2015)
Extractant: t-BAMBP in dodecane solution partially saponified by NaOH, feed pH > 13	Rb Cs K	Rb 20 mg/L Cs 2 mg/L Na 19.4 g/L Mg 28.4 g/L K 2.8 g/L, Li 0.7 g/L	SF(Cs/K) = 1551, SF(Rb/K) = 45, in presence of high K, extraction 99 % Cs and 87 % Rb	Zhang et al. (2018)
Extractant: 30 % t-BAMBP	Rb Cs K	0.2 0.02 5.0	>99 % Rb and Cs, 19 % K	Li et al. (2017)
Extractant: t-BAMBP in sulphonated kerosene 0.1 mol/L, NaOH, at 28 °C.	Rb, Cs, Na, Mg K, Li	Rb 1.1 × 10^{-3} M	Not available	Huang et al. (2018c)
$HClO_4$ added to brine at pH 2 at −5 °C to precipitate $KClO_4$ in order to reduce the potassium, 0.1 M t-BAMBP, pH 8, O/A ratio 0.1, 3 min reaction time, stripping agent: 1 M NH_3 under O/A ratio 2	Li, Na, K, Ca, Mg and Rb	–	Metal extraction: >90 %	Chen et al. (2020)
Extractant: t-BAMBP in xylene Conditions: deionised water as a scrubbing agent at phase ratio of 5:1, and 1 M HCl as a stripping agent	Li, Na, K, Ca, Mg and Rb	Seawater composition	Rb > 80 % (K 19 %, high-purity products -RbCl(s) and Rb_2CO_3(s)/ 99.5 %)	Lv et al. (2023).

was co-extracted at a rate of 19 % in the organic phase. The co-extracted potassium was removed (>99 %) after five scrubbing stages with water for an O/A (5/1). An RbCl solution was obtained by using HCl as a stripping solution, which was then used to prepare high-purity Rb products such as RbCl(s) and Rb_2CO_3(s), with a purity of 99.5 %. Although the authors claimed that this processing route opened up a new pathway for Rb^+ recovery from seawater, a cost analysis would be needed in order to compare this method with current industrial processes for rubidium production.

3.2.2. Extraction of rubidium using inorganic sorbents

Potassium cobalt hexacyanoferrate (KCoFC) has also been evaluated for the recovery of rubidium from seawater desalination brines (Naidu et al., 2016). Ion exchange between Rb^+ and K^+ in the crystal lattice and the adsorption at the surface are responsible for the observed adsorption capacity of KCoFC for Rb^+ (96 mg Rb^+/g), which was significantly higher than for Li^+, Na^+, and Ca^{+2} (<2 mg/g). The Rb^+ desorption rate was fairly high (74 %) with 0.1 mol/L KCl. To reduce the competence of K^+, a modified structure of KCoFC was created by grafting zeolitic imidazole frameworks (ZIF) with KCoFC (KCoFC@ZIF) (Dai et al., 2023). A concentration of 2-methylimidazole (HMIM) on the new composite material resulted in a decreased pore diameter but a larger surface area. A reasonable balance was achieved with the synthesised sorbent, which increased the material surface area by 63 % while reducing the pore diameter by only 30 % when the HMIM content was reduced to a ratio of KCoFC:HMIM:Zn 1:12:5. By increasing the surface area and ion dehydration of KCoFC@ZIF, the grafted ZIF served as a catalytic layer that improved the diffusion of Rb^+ into the KCoFC core structure. For the composite, the Rb^+ sorption capacity increased to 143/1279 mg Rb^+/g). The sorption kinetics was also improved compared to KCoFC in single Rb^+ solutions. The Rb uptake capacity of KCoFC@ZIF(d) (1279 mg/g) was eight times higher, with accelerated kinetics compared to KCoFC (143 mg/g). The capacity of both sorbents, KCoFC and KCoFC@ZIF(d) (where d represents KCoFC:HMIM:Zn with a molar ratio of 1:20:5), was reduced by about 45 % in seawater due to the presence of K^+. Nevertheless, KCoFC@ZIF(d) maintained a relatively high uptake of 236 mg/g Rb^+ in seawater, which was maintained after five cycles of sorption/desorption. The authors proposed the following step-by-step mechanism for adsorption. First, the fully hydrated Rb ions are partially dehydrated at the water–ZIF interface. According to measurements, ZIF crystals have cavities with a pore diameter of ~5.4 Å, a value smaller than the hydrated ionic diameter of Rb (6.58 Å) but larger than the dehydrated ionic diameter of Rb (2.96 Å). These partially dehydrated cations lead to enhanced ion transport in the confined channel, thereby improving the adsorption kinetics. Subsequently, the Rb ion reaches the KCoFC@ZIF interface, where ion exchange occurs between Rb and K. Studies that have dealt with KCoFC (Naidu et al., 2016a) and KCoFC@ZIF(Dai et al., 2023) have demonstrated its potential for Rb recovery, although the synthesis cost of large-scale production of KCoFC@ZIF and the comparative perfmance of KCoFC@ZIF and KCoFC was not discussed by these authors.

Naidu et al. (2016b) considered a KCuFC sorbent, which exhibited higher selectivity for the extraction of Rb^+ than for KniFC, KcoFC, KfeFC and other metal combinations (1.7 mmol/g for KcuFC, with values for other metal combinations of KNiFC (1.4 mmol/g); KCoFC (1.2 mmol/g); and KFeFC, (0.61 mmol/g)). An organic polymer (PAN)-encapsulated KCuFC (KCuFC-PAN) material achieved a maximum Rb sorption capacity of 1.2 mmol/g at pH 7.0, which was maintained over a wide concentration range of co-existing ions and salinity conditions when treating SWRO brines. When used in a fixed-bed KCuFC-PAN column treating 0.06 mM Rb^+ (similar to seawater), the sorption capacity was maintained at around 1 mmol Rb^+/g after two complete sorption/desorption cycles. Using 0.2 M KCl, around 95 % of the Rb^+ was desorbed from the column.

Jiang et al. (2022b) developed a magnetic potassium cobalt hexacyanoferrate composite (Mag-KCoFC) that provided a desorption capacity of 95 % for Rb^+ with 0.2 M KCl solution. This new sorbent, which yielded good performance of the KCoFC thanks to the grafting technique, gave an Rb^+ sorption rate of 209 mg/g and allowed for rapid separation from the liquid phase via magnetic fields. The Mag-KCoFC sorbent displayed promising selectivity values, with a high adsorption yield for Rb^+ and limited adsorption values for Na^+ (0.8 mg/g), Mg^{2+} (1 mg/g), Li^+ (2 mg/g) and Ca^{2+} (3 mg/g). This magnetic sorbent could be regenerated with 1 mol/L KCl with a desorption efficiency of 98 %, while achieving a 92 % adsorption efficiency.

One of the most sustained studies is the work done by Gibert et al. (Gibert et al., 2010) in evaluating three sorbents/exchangers (CsTreat, ZrP) for the extraction of Cs^+, Rb^+, and Li^+ via column operation from a seawater reverse osmosis brine. CsTreat (KCoFeCN) yielded a high sorption capacity for Rb^+ (238 mg/g) and Cs^+(44 mg/g), whereas ZrP had a poor extraction capacity for Li^+ (2 mg/g). In a similar study, Peterskov́á et al. (Peterskov́á et al., 2012) evaluated and compared the efficiency of the same sorbents for the extraction of Cs^+, $Rb^+$and Li^+ from brine via batch processing. The hexacyanoferrate-based adsorbent (CsTreat) worked better for both Cs^+ (32 mg/g) and Rb^+ (47 mg/g) in comparison with the zirconium phosphate-based material (ZrP). Conversely, salinity did not affect the sorption of Cs^+ onto CsTreat. Remarkable efforts in regard to Rb^+ extraction from seawater brine were also made by Dirach et al. (2005), using titanium silicates ($M_2Ti_2O_3$.SiO_4·nH_2O(s)) as a cation exchanger. The extraction efficiency of alkali metals with this cation exchanger followed the order: $Rb^+ > Cs+ > K^+ > > Na^+ > Li^+$, while sorbent regeneration could be achieved with HCl and NaCl mixtures.

The potential value of brines generated in seawater saltworks in terms of Rb^+ recovery is under evaluation. For example, Vallès et al. (2023) evaluated the extraction of trace elements (Li, B, Co, Ga, Ge, Rb, Sr, and Cs) from synthetic saltwork brines using both polymeric and inorganic sorbents, in order to determine the selectivity pattern and distribution coefficients. No polymeric sorbents were suitable for Rb^+ extraction, although the inorganic sorbent, CsTreat, achieved the most promising performance, with high values of Cs^+ and Rb^+ adsorption, despite its poor desorption efficiency.

3.2.3. Extraction of rubidium using organic sorbents

More limited work has been carried out on the application of polymeric adsorbents, as there is a lack of commercially available sorbents with selective functional groups for Rb^+. In one interesting study, Xu et al. (2018) developed a Li^+/Rb^+ imprinted hierarchical mesoporous silica (Li/Rb-IHPS) for simultaneous selective extraction of these ions. Using 12-C-4 and N-[(3-trimethoxysilyl) propyl] ethylendiaminetriacetic acid trisodium salt (TMS-EDTA) as functional monomers, this material was developed via ion imprinting technology. The adsorption capacities of Li/Rb-IHPS were 166 mg/$L_{sorbent}$ for Li^+ and 141 mg/$L_{sorbent}$ for Rb^+, and it showed good selectivity against other common ions in sea water such as Na^+, K^+, Mg^{2+} and Cs^+. This was reported as having industrial applications to the extraction of lithium and rubidium, especially from seawater, but in fact, there was no preliminary techno-economic evaluation of the possibility of large-scale production of this material or a comparison of its cost with other conventional absorbents/extractants, apart from long-term reusability assays to evaluate their lifespan.

4. Integration of membrane-based techniques for rubidium recovery

Membrane separation processes are regarded as novel and powerful strategies for the recovery of rubidium from secondary sources and its selective separation from other impurities present in the solution. Moreover, combinations of membrane-based techniques with other separation processes (e.g. adsorption or ion exchange) in hybrid separation systems have attracted significant attention in the last few years (Bello et al., 2021; Stankiewicz and Moulijn, 2000; Gao et al., 2017;

Ahmed et al., 2021). This is because integrating these techniques with other approaches qualifies the criteria to follow process intensification, which can deliver several advantages such as higher efficiency, a small footprint, environmental friendliness, ease of scaling, and cost economy. Table 6 summarises some important publications on the use of different membrane processes for Rb^+ extraction/recovery.

The management of SWRO brine is essential for desalination. In order to improve both the brine recovery and resource recovery rates, the overall desalination process needs to be optimised to give its maximum output. The integration of membranes for Rb^+ recovery in the case scenarios described in Sections 3 and 4 can be related to the integration of pressure-driven technologies such as RO and nanofiltration (NF), as is the case in many projects involving lithium recovery from salt-lake brines (Shi et al., 2023; Figueira et al., 2023). The RO process is used in many projects in South America (e.g. Peru, Chile, Argentina) as a pre-concentration technology before the evaporation stages. However, since these projects are mainly devoted to lithium recovery, little information has been reported in relation to rubidium. In contrast, NF can be used for the selective removal of Ca(II) and Mg(II), giving a permeate that is rich in single-charged ions (Rb^+, Li^+), which can simplify the purification and concentration stages.

It is also worth mentioning the use of electro-membrane processes to concentrate Li-containing brines, as is the case in electrodialysis, including the production of LiOH solutions using electrodialysis with bipolar membranes. In this case, since the main focus is on lithium, limited amounts of information have been reported on the fate of Rb^+ ions in these processing schemes. The possibility of promoting the selective separation of rubidium from brines has been reported in the literature as part of efforts to synthesise polymeric membranes with selective functional groups to promote the selective transport of highly added value ions such as Li, Rb, or Cs. Efforts have also been made to integrate membrane processes into sorption processes where adsorbents (typically inorganic) are used in powder form. The integration of ultrafiltration (UF) and microfiltration (MF) in order to develop hybrid configurations is one of the more active research and development fields.

4.1. Membrane filtration and membrane sorption

One of the first studies to consider a membrane adsorption hybrid system (MAHS) for Rb^+ extraction from seawater using KCoFC and AMP adsorbents was carried out by Nur et al. (Nur et al., 2018). At a pH of 6.5–7.5, the reported sorption capacities were 145 and 77 mg/g Rb, respectively, with Rb^+ removal ratios ranging from 90 % to 96 %. With an increase of 25 % in the initial dose of 0.2 g/L, the highest adsorption capacity for KCoFC in the MAHS was found to be 69 mg/g, five times more than the value obtained with AMP. The cumulative rate of Rb^+ recovery by KCoFC in the MAHS was reduced by 33 % when high concentrations of competing ions were present in seawater. Rb desorption was not very efficient, with a 30 % yield using a 1 M KCl solution. The authors did not discuss or evaluate any other options for the effective desorption of Rb from sea water using the MAHS. In order to overcome some of the disadvantages faced by previous reseachers, further studies were done by Naidu et al. (2018) with a more advanced membrane process, namely a submerged membrane sorption reactor (SMSR). This reactor coupled sorption with an MF membrane process, giving an integrated low-energy system that delivered an unique solution to overcome the limitations of fixed bed column sorption. The authors proposed an alternative option for Rb recovery via seawater RO, in which the K^+ content hinders the extraction of Rb^+. Powdered clinoptilolite, a natural zeolite, was used to selectively remove K^+ from the brine with a maximum sorption capacity of 58 mg Rb^+/g. An integrated SMSR, consisting of a reactor containing SWRO brine maintained at a constant volume of 4 L with submerged hollow-fiber membrane distillation (MD),

Table 6
Summary of studies in which membrane processes have been used to achieve Rb^+ separation, concentration extraction and recovery.

Name of membrane process	Metal ions	Experimental conditions	Removal efficiency (%)/extraction/recovery/flux	Ref.
Membrane adsorption hybrid system (MAHS)	Rb in synthetic seawater conditions (Na, Mg, Ca, K)	KCoFC and AMP adsorbents in MAHS, pH 6.5–7.5 for extraction and desorption by 1 M KCl and 1 M HCl and purification using RF	93–96 % Rb	Nur et al. (2018)
Polysulphone (PSf)-based rubidium ion-imprinted membrane	Rb, Na, Ca, and Mg, K	pH 9.0, eluent 0.1 M HCl, PSf/P(S-co-VPh)-6 % (IIP), surface area = 25.6 m^2/gm, temp. 15 °C	>90 % Rb	Shi et al. (2023)
Integrated submerged membrane sorption reactor (SMSR)	Ca, Mg, Na, K, Sr, Li, Rb	Natural clinoptilolite in powder form (dose 300 g/L)	65 % K removal, 83 % Rb sorption efficiency by KCuFC(PAN) (K removed by zeolite)	Zhong et al. (2007)
Integrated submerged MD-adsorption system	Seawater brine conditions (Na, Mg, Ca, K, Rb)	Granular KCuFC, pH_{eq} 7.0, MD process at 55 °C, desorption solution 0.2 M NH_4Cl	69 % at 0.05 g/L Rb and 98 % at 0.24 g/L Rb KCuFC, adsorption capacity 216 mg/g Rb	Choi et al. (2019)
Mixed matrix membranes using Rb ion separation membranes (Rb-ISMs)	Mg, Ca, Cs and Rb	Immersion of a PVDF/GO casting solution into coagulating bath containing 18-crown-6 (18C6) grafted mesoporous silica, ethanol and water	Rb-ISMs rebinding capacities (49 mg/g)	Yu et al. (2021)
Membrane crystallisation (polypropylene (PP) and poly-(vinylidene fluoride) (PVDF) membranes)	Pure RbCl	Concentrated solution of RbCl (900 g/L)	PP membranes enabled crystallisation of Rb flux 0.3–1.2 LMH, and rejections >99.97 % crystal size 149 μm RbCl production	Macedonio et al. (2023)
AMP/PSf mixed matrix membranes	Na, K, Ca, and Mg	Na, K, Ca, and Mg ions set at 9.6, 0.5, 0.4, and 1.28 g/L, respectively	Flux from 210 to 370 L m^2 h^{-1}, Rb (98 %) Rb capacity (99 mg/g)	Sun et al. (2022)
Polymer inclusion membranes (PDA-PIM and PDT-PIM): PVC/DCH18C6 (carrier), and Aliquat 336/TOPO (plasticisers)	Rb, Na and K	Rb(I) concentration: 0.03 g/L	At 180 V, the P_{Rb} values for PDA-PIM and PDT-PIM were 12.1 and 2.4 μm/s, and the $S_{Rb/Na}$ values were 11 and 15 for both membranes	Meng et al. (2021)
NF membranes for the fractionation of monovalent/divalent ions	Seawater brine conditions (Na, Mg, Ca, K and Rb)	Na(I) 21,690; Cl (-I): 39,635; B(III) 8.45; S(VI) 1889; Li(I) 0.5; K(I) 802; In(III) 0.5; Rb(I) 0.5; V (V) 0.5; Mg(II) 2,8; Ga(III) 0.5; Ca(II) 86; Sc(III) 0.5; Mo(VI) 0.5 (mg/L)	NF270, Fortilife XC-N and PRO-XS2	Sun et al. (2025)
NF membranes for the fractionation of monovalent/divalent ions	Seawater brine conditions (Na, Mg, Ca, K and Rb)	Na(I) 21,690; Cl (-I): 39,635; B(III) 8.45; S(VI) 1889; Li(I) 0.5; K(I) 802; In(III) 0.5; Rb(I) 0.5; V (V) 0.5; Mg(II) 2.8; Ga(III) 0.5; Ca(II) 86; Sc(III) 0.5; Mo(VI) 0.5 (mg/L)	NF90, NFS, NFX, VNF1, and DK	Figueira et al. (2023)

was then used to remove K from the brine. The SMSR containing the selected zeolite powder achieved a removal rate of 65 % for K. Fig. 18(a) depicts the performance of zeolite when treating the brine, where the influence on the sorbent capacity of competing ions, such as K^+, can easily be seen. The efficiency of the recovery of K^+, Na^+ and Rb^+ as a function of the clinoptilolite dose is shown in Fig. 18(b). Periodic replacement of clinoptilolite in the SMSR coupled with membrane backwashing cleaning stages was found to be effective in maintaining a high efficiency for K recovery and a stable water flux. Rb losses of less than 5 % occurred during K sorption, demonstrating the high selectivity of the clinoptilolite for K in the brine. The authors reported that the application of K-loaded clinoptilolite as a slow-release fertiliser would be beneficial for agriculture. In a SWRO brine with lower K concentrations, the efficiency of Rb^+ sorption of the encapsulated form of KCuFC increased significantly from 18 % to 83 %. Although there was no discernible trace of membrane abrasion, we note that the short testing period may not have been sufficient to prove this possibility. To assess the relationship between zeolite and membrane abrasion, a longer-term investigation would be required. Membrane fouling should also be examined through long-term operation under the same experimental conditions.

4.2. Membrane distillation and membrane crystallisation

As a potential implementation of rubidium sorbents, various efforts have been made towards integration with membrane filtration processes. An integrated submerged MD (S-MD) process using KCuFC as adsorbent was evaluated by Choi et al. (2019) for improving water recovery from SWRO brine and promoting the recovery of Rb, as shown in Fig. 19. In comparison to a traditional fixed bed column, the MD-adsorption integration was shown to be advantageous, since it gave the encapsulated KCuFC a longer contact time to extract Rb. Meanwhile, the adsorption capacity of KCuFC was improved by the MD heat source, which was added to the feed solution for vapor transfer (without the requirement for an extra heat source). The membrane-based setup was operated at 55 °C, with continuous delivery of an Rb-rich SWRO brine to the sorption reactor containing the granular KCuFC (with 99 % Rb recovery efficiency) while producing water.

The reactor was operated with a concentration of 0.24 g KCuFC/L, and achieved water recovery ratios of 85 %. The highest water recovery ratios promoted scaling events, with the deposition of gypsum on both the membrane surfaces and the KCuFC particles, thereby reducing both the Rb recovery and the water recovery rates. The global efficiency of the S-MD system improved after the removal of Ca^{2+}, with a total of sorption capacity of 6.7 mg of Rb/g KCuFC. Rubidium recovery reached values of 81 %, while the water recovery rate was 81 %, and the sorbent yielded an adsorption capacity of 87 mg Rb^+/g. Once the KCuFC sorbent was saturated, Rb recovery was carried out using 0.2 M NH_4Cl solutions. Although complete regeneration was not achieved, the sorbent remained suitable for reuse in the sorption reactor. When SWRO brine was used as the feed solution, $CaSO_4$ crystallisation and deposition were observed on the membrane surface and within the reactor. As a result, the adsorbents and S-MD performance deteriorated. It was therefore recommended that Ca be removed from SWRO brine using an integrated approach. In another study, a more effective strategy was adopted by Naidu et al. (2017), in which they evaluated the same configuration for a similar application using an encapsulated form of KCuFC with PAN. The encapsulated sorbent (KcuFC-PAN) gave an Rb^+ sorption capacity of 109 mg Rb^+/g at 25 °C and 125 mg Rb^+/g at 55 °C. The integrated system with periodic membrane cleaning achieved a 65 % water recovery rate when treating SWRO brine with a permeate quality of 15–20 mS/cm. Continuous cycles of sorption and filtration with SWRO brine gave rise to the possibility of producing 1.9 kg of Rb per day for a plant producing 10,000 m^3 per day of SWRO brine. Similarly to non-encapsulated sorbents, the loaded Rb could be eluted using a 0.2 M NH_4Cl solution.

Using a different approach, the use of vacuum membrane crystallisation (V-MCr) to produce RbCl(s) was described by Macedonio et al. (Macedonio et al., 2023). Two membranes made of polypropylene (PP) and polyvinylidene fluoride (PVDF) were used, with equal pore size and porosity (0.2 μm and 70 %, respectively). In the experiments with the PVDF membrane, RbCl(s) crystals were not obtained due to wetting phenomena, whereas those performed with the PP membrane resulted in efficient crystal formation. The authors claimed that the membrane material was an important factor determining the success or failure of the membrane crystallisation operation. Experiments were carried out with synthetic rubidium containing brines, and then the results were extrapolated on performance with real samples. As a preliminary proof of concept, the proposed scheme showed potential as a viable method for recovering high-purity RbCl crystals.

4.3. Development of rubidium-selective membranes

The development of selective ion exchange membranes through the use of imprinting technologies is an area attracting growing interest fron researchers looking for a different approach for developing membrane-based systems (Shi et al., 2023). An Rb $^+$ -imprinted membrane was synthesised using a non-solvent inversion phase separation method, and a membrane casting solution was prepared by blending the RbCl template with poly (styrene-co-4-hydroxylstyrene) (PSf/P(S-co-VPh) and polysulphone. The presence of the phenolic hydroxyl groups of P (S-co-VPh) promoted coordination with Rb cations in the casting solution to form an imprinted Rb site (Shi et al., 2023). The maximum Rb sorption capacity (70 mgRb/g) was measured at basic pH values (e.g. 9).

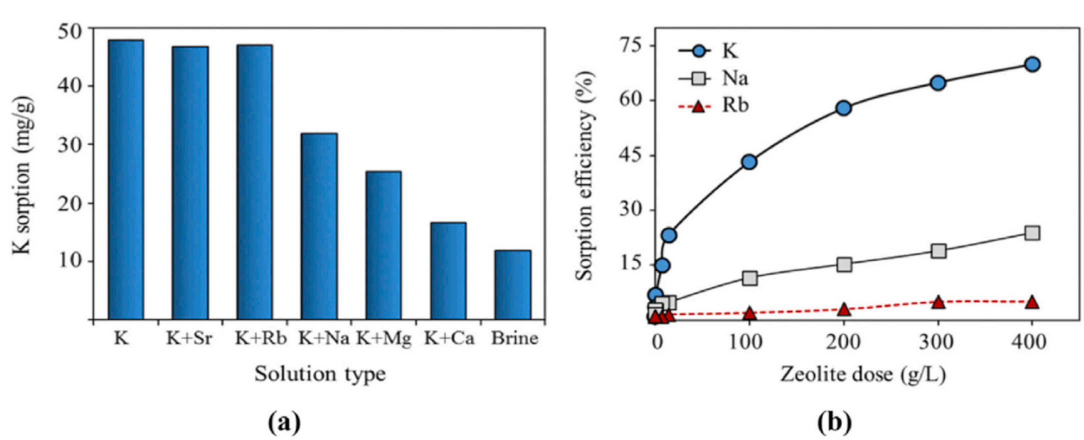


Fig. 18. Performance of zeolite in seawater reverse osmosis brine: (a) variation in K sorption as a function of the ions present in SWRO, for a clinoptilolite dose of 1.2 g/L; (b) variation in the sorption efficiency of the K, Na and Rb as a function of clinoptilolite dose (Naidu et al. 2018) .

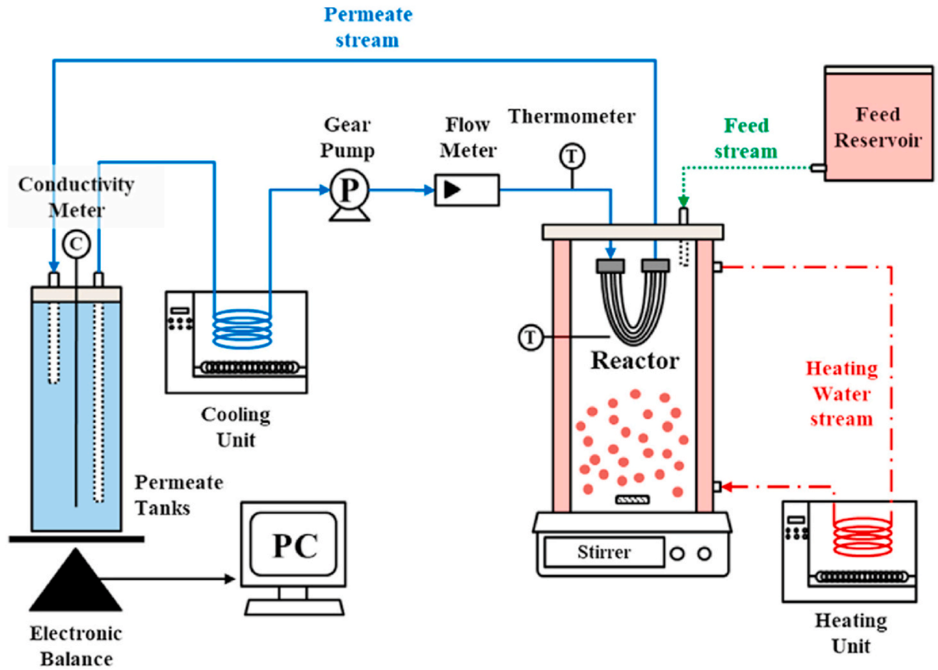


Fig. 19. Schematic diagram of the integrated S-MD-adsorption process including the main components of the membrane module inside the sorption reactor, the auxiliary components for addition of the brine, and the heating and cooling units used to operate the membrane distillation unit (Choi et al., 2019).

The results for the interference in the Rb sorption capacity and yield (Fig. 20) indicated that K^+ causes higher interference than Na^+, Ca^2+, and Mg^{2+}, with a reduction of up to 60 % in the Rb sorption capacity for non-imprinted membranes and only 10 % for imprinted membranes. The elution of Rb from the membranes was shown to be efficient with an 0.1 M HCl solution, as the recyclability of the membrane was maintained after five sorption/desorption cycles (Shi et al., 2023). Before moving to large-scale production, the issues of membrane fouling and lifetime would need to be evaluated.

Using a different approach, Yu et al. (2021) evaluated the recovery of Rb with synthesised Rb ion separation membranes (Rb-ISMs) via a delayed phase inversion method. Selected sorbents with a high specific surface area and high Rb selectivity were prepared by impregnating mesoporous silica with Rb-selective ligands (e.g. 18C6). An analysis of the cross-section and top surface morphology of the Rb-ISMs showed the presence of uniform and dense nanoparticles on the membrane surface. The authors observed an asymmetric structure containing a uniform particle layer and a porous sublayer with finger-like macrovoids. Due to the high affinity of the selective ligand and the good dispersity of the recognition sites, the obtained Rb-ISMs exhibited enhanced extraction efficiencies for Rb (49 mg Rb^+/g). In addition, high selectivity factors were measured for Rb^+/Ca^{2+} (128), Rb^+/Cs^+ (7), and Rb^+/Mg^{2+} (9) as well as high perm-selectivity factors for Ca^{2+}/Rb^+ (2), Cs^+/Rb^+ (3), and Mg^{2+}/Rb^+ (2), demonstrating the good selectivity of the Rb-ISMs for Rb recovery. Most importantly, the membranes showed good recyclability after five sorption and desorption cycles. Several other new membranes have been considered by researchers aiming to achieve more promising performance. For instance, AMP/PSf mixed matrix membranes were prepared using a nonsolvent-induced phase separation (NIPs) method from a N,N-dimethylacetamide (DMAc) solution (Sun et al., 2022). With an increase in the AMP content, the active surface area of the membrance and its hydrophilicity increased, providing higher water fluxes (up to 370 L/(m^2·h)) and Rb sorption capacities (e.g. 98 mg Rb/g). Finally, reusability tests of AMP/PSf porous membranes showed that they maintained up to 95 % of their sorption capacity after five adsorption–desorption recycles. Since this is a newly developed membrane there is a need for membrane fouling and membrane abrasion dat. In addition, a cost analysis of AMP/PSf porous membranes compared to

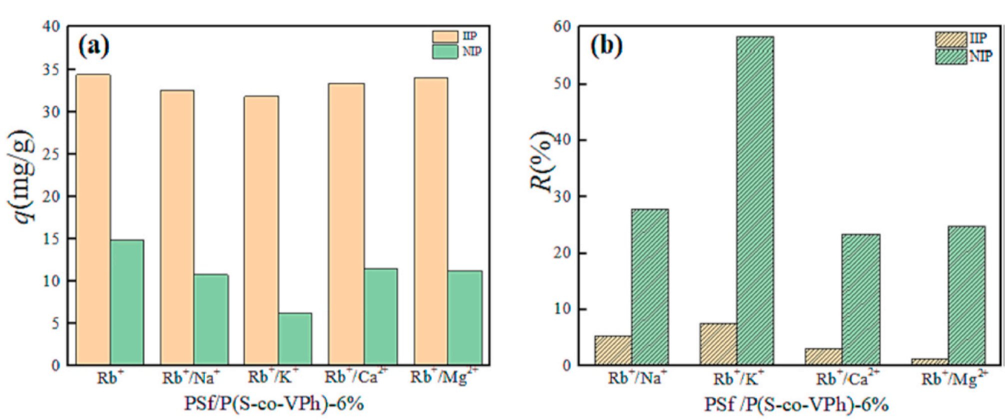


Fig. 20. Performance of non-imprinted (NIP) and imprinted (IIP) membranes poly (styrene-co-4-hydroxylstyrene) (PSf/P(S-co-VPh) used for recovery of Rb from brines: (a) adsorption capacity (q (mg/g) as a function of the interferent cations in the brines; (b) metal extraction efficiencies R (%) for the cations present in the brines. The experimental conditions were C_0 = 0.05 g/L, V = 20 mL, T = 25 °C, t = 24 h, pH = 9.0 (Shi et al., 2023).

other conventional membranesis also lacking as preliminary information in order to scale up this scheme. In order to selectively extract Rb, more specific carriers such as crown ethers and some anionic extractant have been used with a polymer inclusion membrane. Two types of polymer inclusion membranes (PDA-PIM and PDT-PIM) were developed by Meng et al. (2021). PDA-PIM and PDT-PIM used polyvinyl chloride (PVC) and dicyclohexano 18 crown 6 (DCH18C6) as carriers, and tricaprylmethylammonium chloride (Aliquat 336) and trioctylphosphine oxide (TOPO) were used as plasticisers, respectively. For PDA-PIM and PDT-PIM, the ideal mass ratios (measured in weight/weight) were P (25) D (3.5)A (25) and P (25)D (3.5)T (1) [(P: PVC, D: DCH18C6, and T: TOPO)], respectively. The transport of Rb was electrically driven, and the permeability coefficient of Rb^+ reached values of 12 and 2.4 μm/s for PDA-PIM and PDT-PIM, whereas the values of the selectivity factor against Na^+ were 11 and 14. The removal performance of PDT-PIM for Rb^+ is shown in Fig. 21, where the concentration in the feed phase was lowered from 30 to 4 mg Rb/L and the concentration of Na^+ was reduced from 30 to 23 mg Na/L. After 82 h, the enrichment ratio of Rb^+ was three.

In a similar scheme, a phenolic hydroxyl-rich poly (styrene-co-4-hydroxylstyrene) copolymer (P(S-co-VPh)) was used to synthesise an Rb^+-imprinting nanofiber membrane (P(S-co-VPh)-(IIP)) by Sun et al. using electrostatic spinning technology. The membrane prepared in this way had an Rb sorption capacity of 140 mgRb/Lwhen used in a packing bed mode (Sun et al.,2023).

Although membrane processes have shown promise in terms of promoting process intensification for Rb recovery from secondary sources, further studies are still needed to evaluate their long-term performance at a demonstration scale under real conditions. Most of the studies reported here describe proof-of-concept schemes for the synthesis of new membranes at a small scale, using complex synthesis processes and expensive reagents. It should also be mentioned that key performance indicators such as the kinetic parameters, previously described are not promising. As discussed previously, the efforts that have been reported are far from robust technological solutions when both the technical aspects and economic viability are considered.

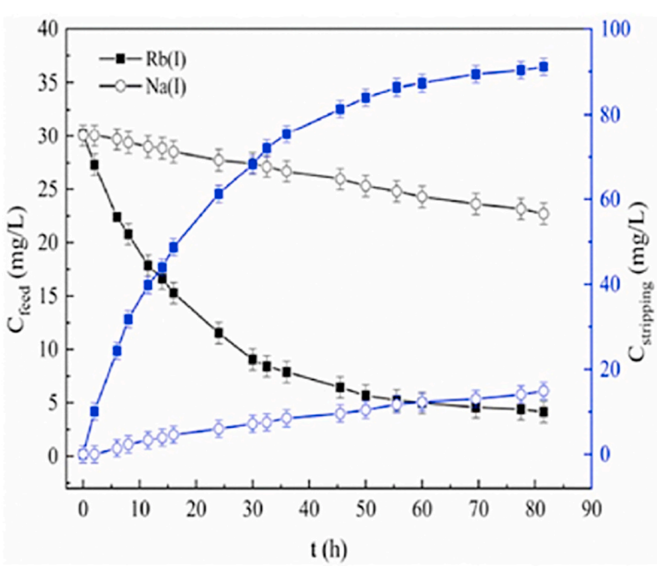


Fig. 21. Variation in Rb and Na in filtration experiments with feed and stripping streams. Enrichment of Rb(I) was achieved using PDT-PIM P25D3.5T10 (i.e., Aliquat 336/DCH18C6/PVC-PIM (PDA-PIM) and TOPO/DCH18C6/PVC-PIM (PDT-PIM)). Feed phase: 350 mL, 30 mg/L Rb(I), Na(I), pH 7. Stripping phase: 100 mL, deionised water, voltage: 180 V(Meng et al., 2021).

5. Economic factors affecting rubidium recovery

The steadily increasing price of rubidium, coupled with its critical role in a range of applications, including biomedical technologies, speciality glass, quantum computing, electronics, solar cells, atomic clocks, and pyrotechnics, underscores the urgent need for effective recovery strategies. Although rubidium is not as widely used as other alkali metals, it is gaining in strategic importance for use in emerging advanced technologies, as discussed in Section 1. However, the recovery of rubidium is often constrained by both technical and economic challenges. The economic viability of rubidium production is closely tied to the nature of the source material, the concentration of Rb(I), and the associated extraction and purification costs. In this section, we examine the key economic considerations involved in recovering rubidium from three main sources: (i) primary lithium-bearing minerals, such as lepidolite and zinnwaldite, from which Rb can be co-extracted during lithium processing; (ii) brines from seawater desalination, which are emerging as unconventional but potentially valuable sources of dissolved rubidium; and (iii) saltwork bitterns, residual brine streams rich in alkali metals that may serve as a low-cost and underutilised reservoir for Rb(I) recovery.

Evaluation of the economic feasibility of these sources requires consideration of factors such as resource availability, rubidium concentration, recovery efficiency, infrastructure demands, and the potential for integration with existing industrial processes.

5.1. Primary lithium-bearing minerals from which Rb can be co-extracted during lithium processing

Rubidium is typically found in lithium-bearing minerals such as lepidolite and zinnwaldite. However, current industrial efforts primarily focus on the production of lithium carbonate and lithium hydroxide, and rubidium recovery is only considered in a marginal number of cases. Despite the relatively high rubidium content of these minerals, its recovery is often overlooked, resulting in low overall rates of Rb recovery across the value chain.

Minerals used as primary sources for lithium production should also be recognised as significant potential sources for rubidium extraction. Greater emphasis on the co-recovery of Rb could improve the resource efficiency and economic value of these operations. As discussed in Section 2, roasting and acid digestion methods have been reported as efficient techniques for leaching lithium and rubidium from minerals such as lepidolite and pollucite, and are already being applied at an industrial scale. Of these, H_2SO_4 digestion combined with roasting has proven particularly advantageous in terms of energy consumption and metal extraction efficiency. However, one major drawback is the co-leaching of aluminium, which leads to complications such as lithium being entrained in the subsequent precipitation of aluminium. Furthermore, to obtain pure rubidium alum, multiple fractional crystallisation steps are required, increasing the complexity and cost. These processes also face challenges related to: (i) high energy demands; (ii) the generation of solid and liquid waste; and (iii) the complexity of rubidium separation, due to its chemical similarity to potassium and caesium.

Despite these limitations, recovering rubidium from lithium-rich minerals remains an area of increasing interest due to its strategic importance in electronics, speciality glass, atomic clocks, and biomedical applications. Although rubidium is relatively rare, and is not mined as a primary element, its co-recovery (especially when coupled with lithium and caesium extraction) offers a potentially viable pathway for industrial valorisation. Rubidium typically occurs in low concentrations (0.5–2 %), making standalone recovery economically unfeasible. Its separation and purification require highly selective reagents and multi-step processing, which further increases the operational costs. The limited global demand for rubidium also reduces the incentive for dedicated recovery processes, making integrated valorisation strategies more appealing.

Most existing economic assessments have focused on lithium recovery, and data on the economics of rubidium processing are scant, particularly at the industrial scale. Nevertheless, leaching techniques involving acid digestion and roasting have been confirmed as effective for decomposing Rb-bearing minerals, and have been used in industrial settings (Sharma et al., 2023; Xing et al., 2021; Macedonio et al., 2023).

5.2. Seawater desalination brines as potential source of Rb(I)

Seawater represents a significant potential source of minerals, and various types have already been produced at an industrial scale. One of the most prominent techniques is the use of solar sea saltworks, particularly in the Mediterranean Basin and, more recently, in northwestern Australia. In the latter case, the deployment of solar evaporation for NaCl production followed the discontinuation of table salt production in Japan, which had relied for decades on electrodialysis and mechanical vapor compression. Another notable example is the Dead Sea, located between Israel and Jordan, where advanced solar evaporation technologies are employed. In addition to NaCl, other commercially valuable minerals such as $MgCl_2(s)$, K_2SO_4, and Br_2 are also produced on a large scale.

The growing interest in the use of oceans as an alternative source for mineral production is driving major global research initiatives aimed at evaluating the techno-economic feasibility of this approach. These efforts are especially focused on elements that are present in low concentrations, such as lithium, gallium, germanium, cobalt, and rare earth elements (REEs), which are critical for supporting the ongoing ecological transition [29]. As a result, current efforts are primarily directed toward developing processes and recovery routes for extracting metals and minerals from SWRO brines, which are being assessed in terms of their cost competitiveness compared to conventional mining operations (Naidu et al., 2017). In this context, rubidium has attracted considerable attention in terms of extraction from salt lake brines and seawater brines. Its industrial relevance, as briefly outlined in Table 1, gives rise to a need to ensure a reliable supply for various applications.

Furthermore, the development and implementation of established or emerging technologies for mineral and metal recovery from salt lake brines and SWRO concentrates will be crucial in determining the scalability of Rb extraction for industrial use. In particular, the availability of cost-effective membrane systems for recovering Rb from these sources will be vital for enhancing future commercialisation. This could also contribute to reducing the price of Rb, thereby promoting its broader use in industrial applications (Sharkh et al., 2022).

In the future, depletion of these mineral resources for lithium production may take place, which highlights the importance of looking for alternative rubidium sources, such as salt lakes and seawater brines. Nevertheless, the extremely low concentration of rubidium remains the main challenge for its extraction/recovery from these alternate sources. Although some preliminary proofs of concept have been presented, such as the integration of MD with adsorption, the maturity of this approach remains at the laboratory scale. Efforts should be made towards: (i) the preparation of sorbents in non-encapsulated and encapsulated forms to extract Rb rather than a conventional fixed-bed column operation; and (ii) improvements to the regeneration and reusability of sorbents (KCuFC). Trials of the simultaneous water recovery and rubidium extraction from SWRO brine indicate that this is a promising approach but still far from deployment at an industrial scale.

The obstacles to making the process of extracting rubidium from brine feasible for full-scale industrial application are the low concentration of rubidium and the problem of interfering elements such as potassium and sodium. As an established technique, SX is known for its high capacity and good rubidium recovery rates. In a later step of the SX process, sodium hydroxide is added to adjust the alkalinity of the brine, to ensure optimal rubidium recovery. Adsorption of rubidium with various adsorbents (KMFC, and AMP) has been performed in laboratory-scale experiments, which exhibited high rubidium adsorption capacities. However, these studies were mainly conducted using synthetic solutions of rubidium, and in many cases without assessing the potential for interference from potassium and sodium. Hence, further studies are needed to evaluate the adsorption efficiency of rubidium to develop real solutions. Another important aspect of studies of rubidium adsorption is the low desorption efficiencies achieved during regeneration of the loaded adsorbent. Overall, the studies reviewed here highlight promising avenues for rubidium recovery from salt lakes, seawater brines and other important secondary resources, which are expected to play an important role in achieving sustainable scale-up and full-scale implementation of rubidium recovery projects.

As discussed previously, the demand for rare metals that are essential for developing the technologies needed for the green transition has, in recent decades, directed attention toward recovering these critical metals from marine sources. Many publications and reports have declared this option as being more cost-competitive than traditional mining processing, but any sustained techno-economic assessment has been provided by experts (Gibert et al., 2010; Morgante et al., 2024; Bardi, 2010; Diallo et al, 2015). Typical economic analyses have been based on the study of the actual concentrations in seawater and without a clear definition of the “element or compound” to be recovered and/or produced. As an example, we consider the scheme proposed by Sharkh et al. (2022) in Fig. 22, although similar overviews of brine mining possibilities have presented similar graphical analyses (Loganathan et al., 2017; Shahmansouri et al., 2015; Kumar et al., 2021). The economic discussion is based on the inclusion of a line separating ‘economically feasible’ from ‘economically challenging’ target elements, with the different proposals postulated over the last decade shown in the graph. In these published studies, the element prices used are typically estimated based on elements in the metallic form, whereas in most cases these are not significant commercial products. An effort to correct this limitation was made by Sharkh et al. (2022), where prices of the most common industrial used chemicals were selected for analysis. Price estimates were generally requested for chemical compounds which are not typically traded, and if the information was difficult to obtain, prices were estimated based on data from major traders in China or India.

In this scenario, and assuming access to seawater brine from desalination plants, Sharkh et al. (2022) estimated the approximate gross value in USD per 1000 m^3 treated. For Rb(I), its recovery as RbCl(s) was integrated with the recovery of K in the form of K_2SO_4, and the authors estimated a potential economic value per 500–1000 USD/1000 m^3 of brine. A summary of the most relevant minerals that can be recovered and the associated economic value is given in Table 7.

The recovery of mineral byproducts from SWRO desalination plant concentrates has been widely discussed as a potential strategy for resource recovery. These facilities are considered to be attractive from the point of view of sustainability, due to: (i) their lower impact compared to terrestrial mining of the same compounds; (ii) the essentially inexhaustible scale of the ocean; (iii) the constant composition of seawater; (iv) the vast capacity of the ocean to dilute treated waste streams; and (v) the stable and fixed footprint of the mining operation. However, although new projects involving SWRO brine valorisation at close to full scale are planned in countries such as Saudi Arabia, Indonesia or Qatar, the recovery process streams are centred solely on the recovery of NaCl(s), $Br_2(g)$ and $Mg(OH)_2(s)$. There are no defined proposals for the recovery of elements at low concentrations, such as Rb (I) or even Li(I). Studies are centred around laboratory work, as described in this review, and there are serious doubts about the technoeconomic feasibility of seawater brines as a secondary resource for the industrial production of rubidium-based compounds.

5.3. Saltwork bitterns as potential sources of Rb(I)

As part of research undertaken through the European Union and Development Programs for Critical and Strategic Raw Materials, in the

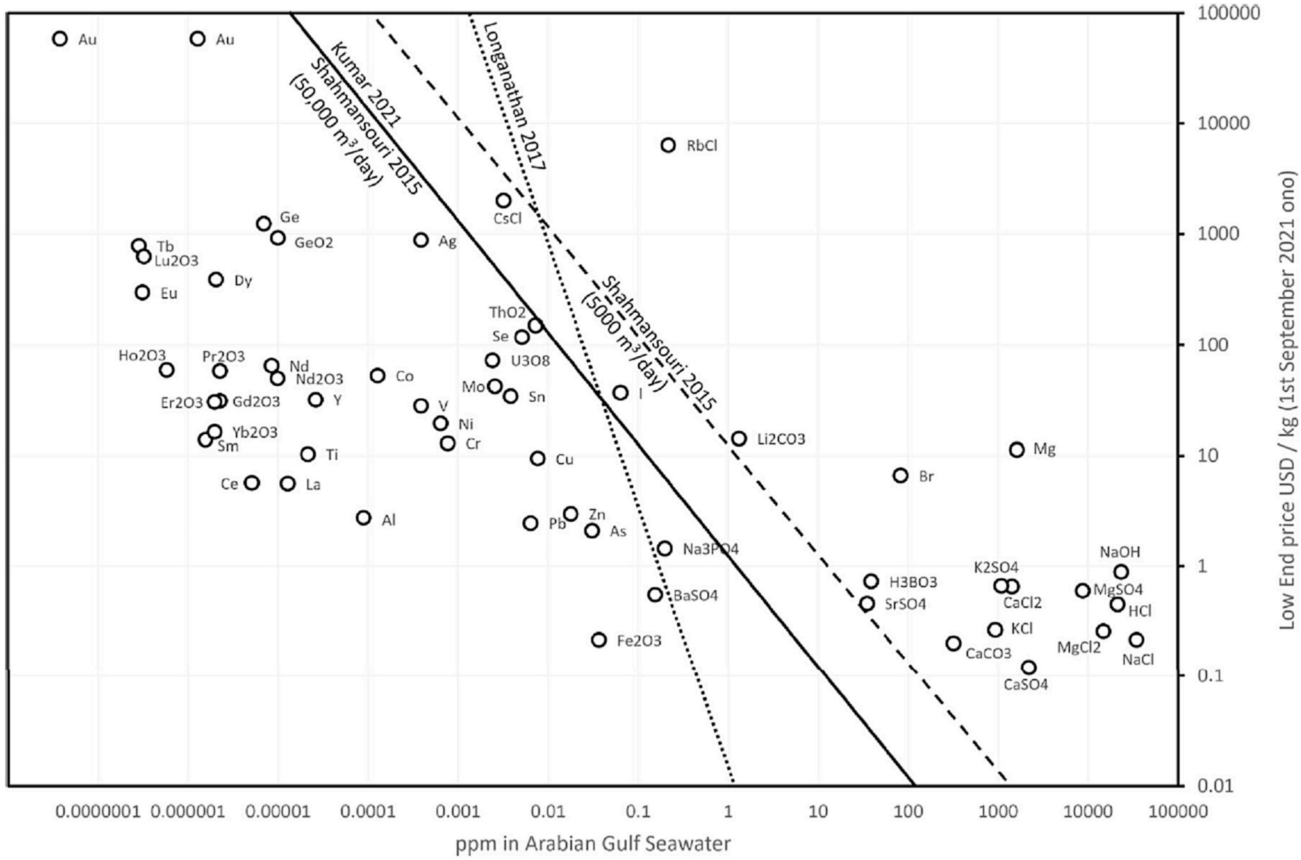


Fig. 22. Market values for elements present in seawater as a function of seawater concentration and price (as estimated in September 2021). Compositions and concentrations are referred to (https://web.stanford.edu/group/Urchin/mineral.html) for Standard Sea Water (SSW) composition (Sharkh et al., 2022; Market value estimations, 2024).

Table 7
Potential gross value USD per 1000 m^3 of seawater components (Sharkh et al., 2022).

Final product(s)	Approximate gross value in USD per 1000 m^3 Arabian Gulf seawater
NaOH, HCl	8000–12000
NaCl, $MgCl_2.2H_20$, Mg, $MgSO_4.7H_2O$	2000–4000
K_2SO_4, RbCl	500–1000
Br_2, $CaCl_2$	300–500
KCl, $CaSO_4.2H_2O$	200–300
H_3BO_3, $CaCO_3$	20–40
Li_2CO_3, $SrSO_4$	10–20
CsCl, I_2	1–2

SEArcularMINE project (SEArcularMINE value estimations, 2024) (SEArcularMINE, 2021, www.searcularmine.eu), a set of relevant minerals and compounds that could potentially be produced from the treatment of bitterns were selected (see Table 8). Rb(I) was selected, although this is not considered by the EU to be a critical or strategic mineral, since its limited availability and the elevated cost of Rb compounds made it a target element to be analysed. In the economic methodology implemented in this project, the expected economic benefit generated from the recovery of a given mineral was considered to be proportional to both its market price and the amount recovered (Sharkh et al., 2022). Table 8 presents a summary of the main minerals and compounds that have the potential to be recovered from saltwork bitterns, along with their economic value, based on prices reported in the 2022 Commodity Summary by the USGS (2022) (Randazzo et al., 2024).

Table 8
Minerals and compounds with potential for recovery from seawater saltwork bitterns and their specific value calculated according to the Commodity Summary of USGS (Randazzo et al., 2024).

Mineral	Specific value [€/kg]
NaCl(s)	0.23
Na_2SO_4(s)	0.17[a]
KCl(s)	0.36
K_2SO_4(s)	0.55
$Mg(OH)_2$	1.01[b]
$MgSO_4$(s)	0.34[b]
Br_2(g)	2.77
B_2O_3(s)	0.41
Li_2CO_3(s)	17.68
RbCl(s)	6520.80
$SrSO_4$(s)	93.60
Co(s)	50.44
GaAs/GaN (Ga pricing)	592.80
CsCl(s)	2329.60
Ge(s)	1248.00
HCl	0.07[b]

[a] Price obtained from https://www.chemanalyst.com/Pricing-data/sodium-sulfate-1480 (Quotations, 2023).
[b] Price obtained from the Eurostat customs database (Intra European trade value, 2021).

We consider the bittern concentrations shown in Fig. 16 (Section 3), and take as a reference for the economic evaluation a volume of one cubic metre of bittern. Initially, we consider: (i) quantitative (100 %) recovery efficiencies for the elements analysed, and (ii) the possible competition and mutual exclusion between the different elements. The latter issue was resolved by diallocating the target elements in the different mineral forms to be potentially recovered according to their relative content; for example, Rb(I) was assumed to be recovered in the form of RbCl(s) in proportion to the amount of Rb(I) present in the saltwork bittern. Taking into account these working hypotheses, the values of the minerals in the Mediterranean saltwork bitterns are summarised in Fig. 23, based on the prices in Table 8 and the average compositions of the bitterns. For the selected group of elements and minerals, the values of the potentially recovered elements range from 35 €/ton of bittern for Mg(II) in the form of $Mg(OH)_2$(s) to less than 1€ per ton of bittern for the case of Li(I) as Li_2CO_3(s).

For Rb(I), even though this is present in much lower concentrations in comparison with Na, Cl, Mg and SO_4, it is associated with a potential recovery value of 25 €/m^3 of bittern due to the extremely high value of RbCl(s) (6.5 k€/kg RbCl(s) in Table 8). It should also be remembered that its cost has increased over the last few years due to its application in quantum computing (Wang et al., 2021b). Other elements such as gallium, germanium and caesium without primary resources shown also high market prices due to their applications in semiconductors and photovoltaic cells (see Table 8). However, although the content of these minerals in bittern is one order of magnitude lower than for Rb(I), their market value for one volume unit of treated bittern (1 m^3) becomes negligible compared to the others (data not shown). In an analysis of the potential economic impact of seawater saltwork bitterns, Randazzo et al. (2024) took into account the following factors: (a) the combined production of table salt in Europe and North Africa is estimated as approximately 53 Mt/y (Idoine et al., 2022); (b) 10 % of the salt produced in Europe comes from saltworks (e.g. solar salt) (Eusalt, 2023); and (c) 1.7 m^3 of bittern can be extracted for every ton of salt produced by evaporation. According to the European Solar Salt Producers Association (EUSALT), the Mediterranean region is estimated to have a total bittern production capacity of approximately 9 Mm^3/y. Based on the composition of elements provided in Table 8, the Euro-Mediterranean production potential is shown in Table 9.

The corresponding values associated with elements present in concentrations of μg/Kg, such as Co, Ga, Cs and Ge, indicated that although postulated by many organisations worldwide it could not be considered as economically reliable secondary sources. This conclusion could was already achieved by Japan (Kim et al., 2013) and USA for the uranium recovery projects to produce uranium fuels for nuclear power stations as U(VI) it is also present in seawater in the order of μg/Kg. When analysing the case of Li the potential recovery could be considered as negligible if it is analysed from the global market needs. For the case of Li, the production reported is 1.12 kton/y of lithium carbonate (Table 9) that represents 0.3 % of the total world production capacity. A similar situation holds in the case of rubidium, as the current global rubidium market is estimated at 2.6 kt (Ogunbiyi et al., 2021; Mordor intelligence, 2022) while the potential production capacity of the Euro-Mediterranean saltworks bittern is 0.06 kt/y, representing 2.3 % of global production. Although this 2.3 % could be considered as a potential secondary resource, the techno-economic feasability of developing a large number of installation plants across the Euro-Mediterranean basin is questionable. Accordingly, even though this has been suggested in many publications and public reports, it is probably a not reliable option. A clearer analysis needs to be performed to guide new initiatives, such as options for the production of rubidium-based compounds. Despite the current lack of information, the recovery of Rb from streams generated in the processing of Li minerals should be evaluated as an option.

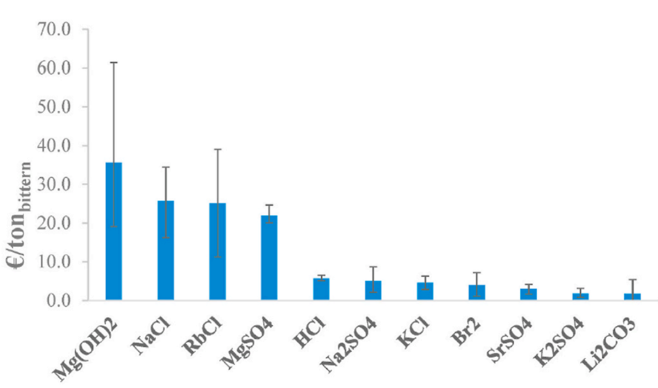


Fig. 23. Potential value of minerals recovered from seawater saltwork bitterns (€/ton of bittern) using the market prices summarised in Table 8 (Randazzo et al., 2024).

Table 9
Estimated production capacities of minerals and elements (t/y) from seawater saltwork bitterns in the Mediterranean basin (Randazzo et al., 2024).

Mineral	Production (tons/year)
B_2O_3	4441
Li_2CO_3	1116
$SrSO_4$	354
RbCl	61
Co	0.25
Ga	0.28
Cs	0.96
Ge	0.70

6. Conclusions and recommendations

There is a growing need for practical applications supporting the use of cleaner technologies for metal recovery from secondary and alternative resources. Several approaches have been suggested, with the aim of minimising environmental impact, reducing energy consumption, and promoting the sustainable use of resources. The concept of cleaner technologies for metal recovery focuses on reducing the environmental footprint of metal extraction processes by the integration of innovative methods. This concept has been developed for the recovery of valuable metals from various waste streams, although in most cases, projects are still at the stage of development and implementation at pre-industrial scale (e.g. new leaching processes with greener solvents, new flotation stages with bio-based agents, new materials, new separation and concentration technologies incorporating membranes).

A critical assessment of the state of the art in the potential for cleaner production of Rb enables the following conclusions to be drawn:

1. An analysis of the relevance of sources for Rb production compounds indicates that the processing of Li-containing minerals may be one of the most reliable solutions.
2. An initial techno-economic analysis of the use of SWRO brines and solar seawater saltwork bitterns as secondary resources for Rb production is not very positive, due to factors such as (i) the typically low levels of such elements in these type of brines, and (ii) the lack of selective materials for concentration and separation (e.g. inorganic and organic adsorbents).
3. Inorganic formulations of adsorbents provide the highest sorption capacities, but these are characterised by very limited desorption ratios. When evaluating the possibility of desorption by thermal oxidation of the sorbents, is associated with a very high operational cost, and could be criticised from the point of view of the circularity of materials.
4. When processing Rb-containing minerals through solvent extraction, the excellent performance of t-BAMBP was noted in regard to Rb recovery and concentration, and future work should be initiated to reduce its cost. More research is needed to study its compatibility with other diluents to make the process environmentally friendly. Researchers have also utilised batch extraction techniques for Rb, although further work is required on integrating various technologies, including membrane adsorption.

5. As reported in by several publications, more work is required on the desorption process. More attention should be given on developing effective desorption agents, or to designing the sorbent in such a way as to facilitate an easy desorption process. This will open a new path towards developing effective schemes for the recovery of rubidium by using adsorption/desorption methodology.
6. Considering the promising performance of MOFs, the grafting of MOFs with KCoFC should be an option to be evaluated to assess the potential performance enhancement with respect to metal uptake. Therefore, future research on MOFs for this application could be an area to be addressed to explore the most optimum materials to avoid pore blocking.
7. When evaluating the work and proposals done in the integration of sorption and membrane process as a process intensification approach the maturity of the work done is very limited as it is related to the lack of robust inorganic adsorbents and the lack of reliable sorption-desorption cycles.
8. More attention should be paid to the development of a suitable precipitant to selectively recover Rb, with the main objective to of improving the purity of the final solid. The availability of information on the most widely used Rb(I) compounds and RbCl(s) and Rb_2CO_3(s) is very limited.
9. Due to the economic importance of Rb, a techno-economic analysis will be required to assess the feasibility of recovering this mineral. Future investigations should focus on feasibility analyses to evaluate the potential for implementation of recovery schemes on an industrial scale.

In summary, a careful analysis of the state of the art indicates that the proposed cleaner routes, based on the valorisation of waste streams (e.g. mineral processing stages of lithium production or lakes and seawater brines), are faced with large thermodynamic barriers. In general, such sources are characterised by the following features: (i) Rb is typically in the range of mg/L; and (ii) there are large concentrations of other monovalent and divalect cations in the same solution. This requires the development of processes in which the desired materials should have a higher selectivity than the other cations, and higher capacity. This second property will be always limited in a thermodynamic sense by the levels of Rb in the solutions to be treated. Thermodynamic barriers such as these are currently limiting both the technical and economical feasibility of new, cleaner alternatives for Rb recovery. It should also be pointed out that despite the launch of national programmes in the area of critical and strategic raw materials to promote such initiatives, the efforts made towards the recovery of uranium from sea water in Japan and the USA, or towards the recovery of lithium in Japan over recent decades, have demonstrated that there is no techno-economic feasibility for such inititatives. The case studies reviewed here indicate that recovery options such as the waste processing streams generated by industries producing Li_2CO_3(s) and LiOH(s) and the brines generated by saltworks (e.g. exploiting the fact that these brines have been concentrated naturally by up to 30 times) could be cleaner alternatives for Rb production.

CRediT authorship contribution statement

Anil K. Pabby: Writing – original draft, Visualization, Software, Data curation, Conceptualization. **Ana María Sastre:** Writing – review & editing, Visualization, Data curation, Conceptualization. **José Luis Cortina:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Development of upcycling approaches in the agri-food and process industries to promote on-site and sustainable chemicals production (Upcycling) project (PID 2023-147160OB-C21), Integration of waste processing technologies to ensure a safe, circular and sustainable battery value chain by promoting urban and industrial mining (MET4LIBS), (TED 2021-131583 B-I00) and SUstainable Pathways for the energy transition through the efficient REcovery and upclycing of strategic Metals (SUPREME) (PID 2024-158241OB-I00) financed by the Spanish Research Agency (AEI). Additionally, the authors acknowledge the support of the Catalan Government through the R2EM group (2021-SGR-GRC-00596). Moreover, this work was carried out within the framework of the Multiscale Center of Excellence, funded by the María de Maeztu Program for Units of Excellence (CEX 2023-001300-M), supported by MCIN/AEI/Ministerio de Universidades, Spain. Finally, J.L. Cortina was also received through the "ICREA Academia" recognition for excellence in research funded by the Generalitat de Catalunya.

Data availability

Data will be made available on request.

References

Ahmed, F.E., Khalil, A., Hilal, N., 2021. Emerging desalination technologies: current status, challenges and future trends. Desalination 517. https://doi.org/10.1016/j.desal.2021.115183, 11518.3.
Alexandratos, S.D., 2016. Preface to the special issue: uranium in seawater. Ind. Eng. Chem. Res. 55, 4101–4102. https://doi.org/10.1021/acs.iecr.6b01293.
American Mineralogist. Crystal structure database. https://rruff.geo.arizona.edu/AMS/amcsd.php (date accessed; 03-12–2023).
Arnold, W.D., Crouse, D.J., Brown, K.B., 1965. Solvent extraction of cesium (and rubidium) from ore liquors with substituted phenols. Ind. Eng. Chem. Process Des. Dev. 4, 249–254.
Awwad, N.S., Ahmed, M.K., Afifi, M., Ibrahim, H.A., 2021. Rb(I)/Se(IV) co-dopant into hydroxyapatite; their structural, morphological, and antibacterial effectiveness for biomedical applications. Appl. Phys. A Mater. Sci. Process. 127, 1–10. https://doi.org/10.1007/s00339-021-04337-8.
Bardi, U., 2010. Extracting minerals from seawater: an energy analysis. Sustainability 2, 980–992. https://doi.org/10.3390/su2040980.
Bauch, A., 2003. Caesium atomic clocks: function, performance and applications. Meas. Sci. Technol. 14 (8), 1159–1173. https://doi.org/10.1088/0957-0233/14/8/301.
Bello, A.S., Zouari, N., Daana, D.A., Hahladakis, J.N., Al-Ghouti, M.A., 2021. An overview of brine management: emerging desalination technologies, life cycle assessment, and metal recovery methodologies. J. Environ. Manag. 288, 112358. https://doi.org/10.1016/j.jenvman.2021.112358.
Bian, Z., Zhang, H., Ye, J., Ning, Z., 2023. Flotation behavior of oleate and dodecylamine as mixed collector for recovery of lithium and rubidium from low-grade spodumene tailings: experiment, characterization and DFT calculation. Appl. Surf. Sci. 638, 158117. https://doi.org/10.1016/j.apsusc.2023.158117.
Burt, R.O., 1993. Cesium and cesium compounds. In: Kirk-Othmer Encyclopedia of Chemical Technology, fourth fifth ed. John Wiley & Sons, Inc, New York, pp. 749–764.
Butterman, W.C., Reese, R.G., 2003. Rubidium. Mineral commodity profiles. U. S. Geolog. Surv. 03–045.
Cernigliaro, A., Valloreia, S., Galleani, L.P., 2013. GNSS Space Clocks: Performance Analysis IEEE, pp. 1–5. https://doi.org/10.1109/ICL-GNSS.2013.6577251.
Chen, W.-S., Lee, C.-H., Chung, Y.-F., Tien, K.-W., Chen, Y.-J., Chen, Y.-A., 2020. Recovery of rubidium and cesium resources from brine of desalination through t-BAMBP extraction. Metals 10 (607), 1–15. https://doi.org/10.3390/met10050607.
Cheng, X., Shi, D., Zhang, Y., Peng, X., Xie, S., Li, L., Niu, Y., Wang, Y., Song, F., 2024. Cesium extraction from an alkali-free solution using a 4-tert-butyl-2-(α-methylbenzyl) phenol–di-(2-ethylhexyl) phosphoric acid synergistic system: a conceptual process flowsheet for separating cesium salt from salt-lake brine. Hydrometallurgy 228, 106353. https://doi.org/10.1016/j.hydromet.2024.106353.
Choi, Y., Ryu, S., Naidu, G., Lee, S., Vigneswaran, S., 2019. Integrated submerged membrane distillation-adsorption system for rubidium recovery. Sep. Purif. Technol. 218, 146–155. https://doi.org/10.1016/j.seppur.2019.02.050.
Dagan-Jaldety, C., Nativ, P., Cristal, Y.S., Lahav, O., 2023. A prussian-blue analogue (PBA) ion-chromatography-based technique for selective separation of Rb^+ (as RbCl) from brines. Water Res. 247, 120757. https://doi.org/10.1016/j.watres.2023.120757.
Dai, W., Fang, Y., Yu, L., Zhao, G., Yan, X., 2018. Rubidium ion capture with composite adsorbent PMA@HKUST-1. J. Taiwan Inst. Chem. Eng. 84, 222–228. https://doi.org/10.1016/j.jtice.2018.01.023.

Dai, Q.T., Choo, Y., Akther, N., Roobavannan, S., Norouzi, A., Gupta, V., Blumenstein, M., Tien, V., Nguyen, T.V., Naidu, G., 2023. Selective rubidium recovery from seawater with metal-organic framework incorporated potassium cobalt hexacyanoferrate nanomaterial. Chem. Eng. J. 454, 140107. https://doi.org/10.2139/ssrn.4203035.

Davoodbeygi, Y., Askari, M., Salehi, E., Kheirieh, S., 2023. A review on hybrid membrane-adsorption systems for intensified water and wastewater treatment: process configurations, separation targets, and materials applied. J. Environ. Manag. 335, 117577. https://doi.org/10.1016/j.jenvman.2023.117577.

De, A.K., 2003. A Textbook of Inorganic Chemistry, ninth ed. New Age International (P) Limited, Publishers, pp. 251–254.

DesalData, 2018. GWI/DESALDATA. https://www.desaldata.com.

Diallo, M.S., Kotte, M.R., Cho, M., 2015. Mining critical metals and elements from seawater: opportunities and challenges. Environ. Sci. Technol. 49, 9390–9399. https://doi.org/10.1021/acs.est.5b00463.

Ding, T., Zheng, M., Nie, Z., Ma, L., Ye, C., Wu, Q., Zhao, Y., Yang, D., Wang, K., 2022. Impact of regional climate change on the development of lithium resources in zabuye salt Lake, Tibet. Front. Earth Sci. 10, 1–10. https://doi.org/10.3389/feart.2022.865158.

Dirach, J.L., Nisan, S., Poletiko, C., 2005. Extraction of strategic materials from the concentrated brine rejected by integrated nuclear desalination systems. Desalination 182, 449–460. https://doi.org/10.1016/j.desal.2005.02.037.

Egawa, H., Harada, H., Nonaka, T., 1980. Adsorption resins for uranium in sea water. Nippon Kagaku Kaishi 11, 1767–1772.

Ertan, B., 2023. Extraction of high-value metals from boron waste by bioleaching using Aspergillus niger. Trans. Indian Inst. Met. 76, 3137–3745. https://doi.org/10.1007/s12666-023-03013-0.

Eusalt, 2023. https://eusalt.com. (Accessed 26 December 2024).

Fang, Y., Zhao, G., Dai, W., Ma, L., Ma, N., 2017. Enhanced adsorption of rubidium ion by a phenol@MIL-101(Cr) composite material. Microporous Mesoporous Mater. 251, 51–57. https://doi.org/10.1016/j.micromeso.2017.05.048.

Fang, D., Wang, Y., Liu, H., Zhang, H., Yea, X., Li, Q., Li, J., Wu, Z., 2021. Efficient extraction of Rb+ and Cs^+ by a precipitation flotation process with ammonium phosphowolframate as precipitant. Colloids Surf., A 608, 125581. https://doi.org/10.1016/j.colsurfa.2020.125581.

Fang, D., Lu, M., Wang, Y., Ma, L., Li, K., Liu, H., Zhang, H., Shi, G., Wu, Z., Ye, X., 2023. Extraction of rubidium and cesium from oilfield brine by the two-step adsorption–flotation method. Miner. Eng. 201, 108161. https://doi.org/10.1016/j.mineng.2023.108161.

Figueira, M., Rodríguez-Jimenez, D., Lopez, J., Reig, M., Cortina, J.L., Valderrama, C., 2023. Evaluation of the nanofiltration of brines from seawater desalination plants as pre-treatment in a multimineral brine extraction process. Sep. Purif. Technol. 322, 124232. https://doi.org/10.1016/j.seppur.2023.124232.

Fortier, S.M., Nassar, N.T., Graham, G.E., Hammarstrom, J.M., Day, W.C., Mauk, J.L., Seal, R.R., 2021. Annual Review: USGS Critical Minerals Review. U.S. Geological Survey.

Gao, L., Yoshikawa, S., Iseri, Y., Fujimori, S., Kanae, S., 2017. An economic assessment of the global potential for seawater desalination to 2050. Water 9 (Same as 5), 763. https://doi.org/10.3390/w9100763.

Gibert, O., Valderrama, C., Peterková, M., Cortina, J.L., 2010. Evaluation of selective sorbents for the extraction of valuable metal ions (Cs, Rb, Li, U) from reverse osmosis rejected brine. Solvent Extr. Ion Exch. 28, 543–562. https://doi.org/10.1080/07366299.2010.480931.

Grohol, M., Veeh, C., 2023. Study on the Critical Raw Materials for the EU 2023 – Final Report. Publications Office of the European Union. European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. https://data.europa.eu/doi/10.2873/7255.

Guggenheim, S., 1981. American Mineralogist 66 (1981). Cation Ordering in Lepidolite Type 2M2 from Radkovice Database_Code_Amcsd 0000850, pp. 1221–1232.

Harikesh, P.C., Mulmudi, H.K., Ghosh, B., Goh, T.W., Teng, Y.T., Thirumal, K., Mathews, N., 2016. Rb as an alternative cation for templating inorganic lead-free perovskites for solution-processed photovoltaics. Chem. Mater. 28 (20), 7496–7504. https://doi.org/10.1021/acs.chemmater.6b03310.

Hosoe, M., Oi, T., Kawada, K., Kakihana, H., 1988. Separation of rubidium isotopes by ion exchange chromatography. J. Chromatogr. A 435, 253–255. https://doi.org/10.1016/S0021-9673(01)82184-1.

Hosseini, M., Sparkes, B.M., Campbell, G., Lam, P.K., Buchler, B.C., 2011. High efficiency coherent optical memory with warm rubidium vapour. nat Commun. 2, 174. https://doi.org/10.1038/ncomms1175.

HSC Chemistry Software, Version 7.1, Outotec, 2011.

Huang, M., Stapleton, A., Camparo, J., 2018a. Lamplight stabilization for GPS Rb atomic clocks via RF-Power control. IEEE Trans. Ultrason. Ferroelectrics Freq. Control 65, 1804–1809. https://doi.org/10.1109/TUFFC.2018.2856246.

Huang, D., Zheng, H., Liu, Z., Bao, A., Li, B., 2018b. Extraction of rubidium and cesium from brine solutions using a room temperature ionic liquid system containing 18-crown-6. Pol. J. Chem. Technol. 20 (2), 40–46. https://doi.org/10.2478/pjct-2018-0021.

Huang, D., Zheng, H., Li, B., Ma, G., 2018c. Extraction kinetics of rubidium in brine solutions by 4-tert-butyl-2-(α-methylbenzyl)phenol/sulfonated kerosene using a single drop method. Chem. Phys. Lett. 713, 105–110. https://doi.org/10.1016/j.cplett.2018.10.021.

Idoine, N.E., Raycraft, E.R., Shaw, R.A., Hobbs, S.F., Deady, E.A., Everett, P., Evans, E.J., Mills, A.J., 2022. World mineral production 2016–2020. https://nora.nerc.ac.uk/id/eprint/534464.

Intra European trade value, 2021. From Eurostat Customs Database Using Harmonized System Code, 28.16.10, 28.33.21 and 28.06.10.

Jandova, J., Vu, H.N., 2008. Processing of zinnwaldite wastes to obtain lithium and rubidium compounds. Minerals. Metals & Materials Society, pp. 923–929.

Jandova, J., Dvorak, P., Vu, H.N., 2010. Processing of zinnwaldite waste to obtain Li_2CO_3. Hydrometallurgy 103, 12–18. https://doi.org/10.1016/j.hydromet.2010.02.010.

Jandova, J., Dvorák, P., Formánek, J., Vu, H.N., 2012. Recovery of rubidium and potassium alums from lithium-bearing minerals. Hydrometallurgy 119, 73–76. https://doi.org/10.1016/j.hydromet.2012.02.010.

Jiang, C., Fang, M., Huang, A., Han, S., Jin, G.P., 2022a. Fabrication of a novel magnetic rubidium ion-imprinted polymer for selective separation. New J. Chem. 46, 6343. https://doi.org/10.1039/D1NJ06207G.

Jiang, C., Ni, J., Jin, G.-P., 2022b. Magnetic potassium cobalt hexacyanoferrate nanocomposites for efficient adsorption of rubidium in solution. Sep. Purif. Technol. 296, 121383. https://doi.org/10.1016/j.seppur.2022.121383.

Jiao, F., Zhang, Z., Wei, Q., Qin, W., 2024. Key technologies and development trends for efficient flotation recovery of lepidolite. Green Smart Min. Eng. 1, 273–288. https://doi.org/10.1016/j.gsme.2024.08.002.

Jones, E., Qadir, M., van Vliet, T.H.M., Smakhtin, V., Kang, S., 2019. The state of desalination and brine production: a global outlook. Sci. Total Environ. 657, 1343–1356,. https://doi.org/10.1016/j.scitotenv.2018.12.076.

Jordens, A., Marion, C., Grammatikopoulos, T., Waters, K.E., 2016. Understanding the effect of mineralogy on muscovite flotation using QEMSCAN. Int. J. Miner. Process. 155, 6–12. https://doi.org/10.1016/j.minpro.2016.08.003.

Khalil, M., Sayed, M.A., Kotp, Y.H., 2023. Gamma radiation-induced synthesis of TiO2 immobilized on polyacrylonitrile nanocomposite for gallium, strontium and rubidium ions separation from aqueous solutions. Radiat. Phys. Chem. 212, 111085. https://doi.org/10.1016/j.radphyschem.2023.111085.

Kim, J., Tsouris, C., Mayes, R.T., Oyola, Y., Saito, T., Janke, C.J., Dai, S., Schneider, E., Sachde, D., 2013. Recovery of uranium from seawater: a review of current status and future research needs. Separ. Sci. Technol. 48, 367–387. https://doi.org/10.1080/01496395.2012.712599.

Kot, F.S., 2018. On the rubidium and lithium content and availability in the sub-arid south-eastern mediterranean: potential health implications. Environ. Geochem. Health 40, 1841–1851. https://doi.org/10.1007/s10653-018-0134-8.

Kristianova, E., Vu, N.H., Dvorak, P., Tomasko, T., Vaculíkov, L., 2023. Extraction of Li, K, and Rb from zinnwaldite by chlorination roasting with CaCl2. Miner. Eng. 204, 108420. https://doi.org/10.1016/j.mineng.2023.108420.

Krys, P., Testa, F., Trochimczuk, A., Pin, C., Taulemesse, J.-M., Vincent, T., Guibal, E., 2013. Encapsulation of ammonium molybdophosphate and zirconium phosphate in alginate matrix for the sorption of rubidium(I). J. Colloid Interface Sci. 409, 141–150. https://doi.org/10.1016/j.jcis.2013.07.046.

Kumar, A., Naidu, G., Fukuda, H., Du, F., Vigneswaran, S., Drioli, E., Lienhard, J.H., 2021. Metals recovery from seawater desalination brines: technologies, opportunities, and challenges. ACS Sustain. Chem. Eng. 9, 7704–7712. https://doi.org/10.1021/acssuschemeng.1c00785.

Lenher, V., Kemmerer, G., Whitford, E., 1924. Extraction of cesium from pollucite. Ind. Eng. Chem. 16 (12), 1280.

Lepidolite. https://crystallography365.wordpress.com/2014/09/12/where-does-all-the-lithium-come-from-lepidolite/ date accessed; 03-12–2023.

Li, B., Liu, H., Ye, X., Li, S., Wu, Z., 2014. Rubidium and cesium ion adsorption by a potassium titanium silicate-calcium alginate composite adsorbent. Sep. Sci. Technol. 49, 1076–1085.

Li, R., Liu, C., 2014. Research of extraction and influencing factor of Rb and Cs with t-BAMBP. J. Salt Chem. Ind. 43 (1), 17–19.

Li, K., Yin, Z., Dong, J., Xu, L., Liu, Y., Ma, R., Li, S., 2023a. Separation of Cs^+ from Rb+ and Cs^+ co-existing solution by Cs3Bi2I9 precipitation to produce CsI. Chem. Eng. Res. Des. 197, 372–379. https://doi.org/10.1016/j.cherd.2023.07.023.

Li, K., Xie, J., Shi, Z., Min, C., Yin, Z., Ruixin, Ma, Shina, Li, 2023b. Cheap and easily separate Cs(I) and Rb(I) from high salty solution by forming RbCs–MgHCF. J. Clean. Prod. 425, 139007. https://doi.org/10.1016/j.jclepro.2023.139007.

Li, Z., Pranolo, Y., Zhu, Z., Chen, C.Y., 2017. Solvent extraction of cesium and rubidium from brine solutions using 4-tert-butyl-2-(α-methylbenzyl)-phenol. Hydrometallurgy 171, 1–7. https://doi.org/10.1016/j.hydromet.2017.03.007.

Liu, J.-L., Yin, Z.-L., Li, X.-H., Hu, Q.-Y., Liu, W., 2019. Recovery of valuable metals from lepidolite by atmosphere leaching and kinetics on dissolution of lithium. Trans. Nonferrous Met. Soc. China 29, 641–649. https://doi.org/10.1016/S1003-6326(19)64974-5.

Liu, J.-L., Yin, Z., Liu, W., Li, X., Hu, Q., 2020. Treatment of aluminum and fluoride during hydrochloric acid leaching of lepidolite. Hydrometallurgy 191, 105222. https://doi.org/10.1016/j.hydromet.2019.105222.

Liu, S., Liu, H., Huang, Y., Yang, W., 2015. Solvent extraction of rubidium and cesium from salt lake brine with t-BAMBP kerosene solution. Trans. Nonferrous Metals Soc. China 25, 329–334. https://doi.org/10.1016/S1003-6326(15)63608-1.

Liu, Y., Ma, B., Lv, Y., Wang, C., Chen, Y., 2022a. Thorough extraction of lithium and rubidium from lepidolite via thermal activation and acid leaching. Miner. Eng. 178, 107407. https://doi.org/10.1016/j.mineng.2022.107407.

Liu, Y., Ma, B., Lv, Y., Wang, C., Chen, Y., 2022b. Selective recovery and efficient separation of lithium, rubidium, and cesium from lepidolite ores. Sep. Purif. Technol. 288, 120667. https://doi.org/10.1016/j.seppur.2022.120667.

Loganathan, P., Naidu, G., Vigneswaran, S., 2017. Mining valuable minerals from seawater: a critical review. Environ. Sci. Water Res. Technol. 3, 37–53. https://doi.org/10.1039/c6ew00268d.

Lu, T., Zhu, Y., Wang, W., Qi, Y., Wang, A., 2019. Interconnected superporous adsorbent prepared via yeast-based pickering HIPEs for high-efficiency adsorption of Rb+, Cs+ and Sr2+. Chem. Eng. J. 361, 1411–1422. https://doi.org/10.1016/j.cej.2018.11.006.

Luo, Y., Chen, Q., Shen, X., 2019. Complexation and extraction investigation of rubidium ion by calixcrown-C2mimNTf2 system. Sep. Purif. Technol. 227, 115704. https://doi.org/10.1016/j.seppur.2019.115704.

Luong, V.T., Kang, D.J., An, J.W., Kim, M.J., Tran, T., 2013. Factors affecting the extraction of lithium from lepidolite. Hydrometallurgy 134–135, 54–61. https://doi.org/10.1016/j.hydromet.2013.01.015.

Luong, V.T., Kang, D.J., An, J.W., Dao, D.A., Kim, M.J., Tran, T., 2014. Iron sulphate roasting for extraction of lithium from lepidolite. Hydrometallurgy 141, 8–16. https://doi.org/10.1016/j.hydromet.2013.09.016.

Lv, Y., Xing, P., Ma, B., Liu, Y., Wang, C., Zhang, W., Chen, Y., 2020. Efficient extraction of lithium and rubidium from polylithionite via alkaline leaching combined with solvent extraction and precipitation. ACS Sustain. Chem. Eng. 8 (38), 14462–14470. https://doi.org/10.1021/acssuschemeng.0c04437.

Lv, Y., Ma, B., Liu, Y., Wang, C., Zhang, W., Chen, Y., 2022. Selective extraction of cesium from high-concentration rubidium chloride leach liquor of lepidolite. Desalination 530, 115673. https://doi.org/10.1016/j.desal.2022.115673.

Lv, Y., Liu, Y., Ma, B., Wang, B., Chen, Y., 2023. Preferential extraction of rubidium from high concentration impurity solution by solvent extraction and preparation of high-purity rubidium salts. Desalination 545, 116162. https://doi.org/10.1016/j.desal.2022.116162.

Macedonio, F., Frappa, M., Alessandro, F., Drioli, E., 2023. Formation of solid RbCl from aqueous solutions through membrane crystallization. Desalination 566, 116903. https://doi.org/10.1016/j.desal.2023.116903.

Market values estimations for elements present in seawater, Visit:https://web.stanford.edu/group/Urchin/mineral.html (accessed the November 19, 2024).

Meng, X., Tian, Y., Tang, M.Y., Qiao, R., 2021. Separation and enrichment of Rb(I) in dicyclohexano 18 crown 6 (DCH18C6)/PVC polymer inclusion membrane assisted by electric field. J. Environ. Chem. Eng. 9, 106712. https://doi.org/10.1016/j.jece.2021.106712.

Mercado, S., Hurtado, R., 1992. Potash extraction from cerro prieto geothermal brine. Geothermics 21, 759–764. https://doi.org/10.1016/0375-6505(92)90028-8.

Mohammadi, M.R.T., Koleini, S.M.J., Javanshir, S., Abolghasemi, H., Abdollahy, M., 2015. Extraction of rubidium from gold waste: process optimization. Hydrometallurgy 151, 25–32. https://doi.org/10.1016/j.hydromet.2014.10.016.

Mohite, B.S., Burungale, S.H., 1999. Separation of rubidium from associated elements by solvent extraction with dibenzo-24-crown-8. Anal. Lett. 32, 173–183.

Momiera, R., Sargsyan, A., Tonoyan, A., Leroy, C., Sarkisyan, D., 2023. Micrometric-thin cell filled with Rb vapor for high-resolution atomic spectroscopy. Opt. Mem. Neural Network. 32 (Suppl. 3), S349–S355.

Mordor intelligence, 2022. Rubidium market - growth, trends, covid-19 impact, and forecasts (2022 - 2027). https://www.ptonline.com/articles/how-to-get-better-mfi-results.

Morgante, C., Lopez, J., Cortina, J.L., Tamburini, A., 2024. New generation of commercial nanofiltration membranes for seawater/brine mining: experimental evaluation and modelling of membrane selectivity for major and trace elements. Sep. Purif. Technol. 340, 126758. https://doi.org/10.1016/j.seppur.2024.126758.

Mulwanda, J., Senanayake, G., Oskierski, H., Altarawneh, M., Dlugogorski, B.Z., 2021. Leaching of lepidolite and recovery of lithium hydroxide from purified alkaline pressure leach liquor by phosphate precipitation and lime addition. Hydrometallurgy 201, 105538. https://doi.org/10.1016/j.hydromet.2020.105538.

Naidu, G., Nur, T., Loganathan, P., Kandasamy, J., Vigneswaran, S., 2016a. Selective sorption of rubidium by potassium cobalt hexacyanoferrate. Sep. Purif. Technol. 163, 238–246. https://doi.org/10.1016/j.seppur.2016.03.001.

Naidu, G., Loganathan, P., Jeong, S., Johir, M.A.H., To, V.H.P., Kandasamy, J., Vigneswaran, S., 2016b. Rubidium extraction using an organic polymer encapsulated potassium copper hexacyanoferrate sorbent. Chem. Eng. J. 306, 31–42. https://doi.org/10.1016/j.cej.2016.07.038.

Naidu, G., Jeong, S., Johir, M.A.H., Fane, A.G., Kandasamy, J., Vigneswaran, S., 2017. Rubidium extraction from seawater brine by an integrated membrane distillation-selective sorption system. Water Res. 123, 321–331. https://doi.org/10.1016/j.watres.2017.06.078.

Naidu, G., Jeong, S., Choi, Y., Song, M.H., Oyunchuluun, U., Vigneswaran, S., 2018. Valuable rubidium extraction from potassium reduced seawater brine. J. Clean. Prod. 174, 1079–1088. https://doi.org/10.1016/j.jclepro.2017.11.042.

Nie, W., Wen, S., Xian, Y., Li, Y., Han, G., Jiang, Y., 2021. Leaching rubidium from a low-grade rubidiumbearing aluminosilicate ore. J. Mater. Res. Technol. 13, 1546–1554. https://doi.org/10.1016/j.jmrt.2021.05.052.

Nur, T., Loganathan, P., Johir, M.A.H., Kandasamy, J., Vigneswaran, S., 2018. Removing rubidium using potassium cobalt hexacyanoferrate in the membrane adsorption hybrid system. Sep. Purif. Technol. 191, 286–294. https://doi.org/10.1016/j.seppur.2017.09.048.

Ogunbiyi, O., Saththasivam, J., Dal-Masri, D., Manawi, Y., Lawler, J., Zhang, X., Liu, Z., 2021. Sustainable brine management from the perspectives of water, energy and mineral recovery: a comprehensive review. Desalination 513, 115055. https://doi.org/10.1016/j.desal.2021.115055.

Pabby, A.K., Wickramasinghe, S.R., Sirkar, K.K., Sastre, A.M., 2020. Hollow Fiber Membrane Contactors: Module Fabrication, Design and Operation, and Potential Applications. Boca Raton, FL,USA, ISBN 9780367025786.

Peterková, M., Valderrama, C., Gibert, O., Cortina, J.L., 2012. Extraction of valuable metal ions (Cs, Rb, Li, U) from reverse osmosis concentrate using selective sorbents. Desalination 286, 316–323.

Production of table salt in saltworks (www.searcualrmine.eu) [acceseed on 19th Dec. 2023].

Qian, Y., Ding, D., Li, K., Fang, D., Wang, Y., Hu, Y., Ye, X., Liu, H., Wu, Z., Li, J., 2022. Extraction of rubidium and cesium from a leach solution of lepidolite with biomass carbon adsorbents. Hydrometallurgy 208, 105802. https://doi.org/10.1016/j.hydromet.2021.105802.

Quotations from. https://www.chemanalyst.com/Pricing-data/sodium-sulfate-1480 (accessed the November 25, 2023).

Randazzo, S., Vicari, F., Lopez, J., Salem, M., Brutto, R.L., Azzouz, S., Chamam, S., Cataldo, S., Muratore, N., Fernandez de Labastida, M., Valles, V., Pettignano, A., Dalì Staiti, G., Pawlowski, S., Hannachi, A., Cortina, J.L., Cipollina, A., 2024. Unlocking hidden mineral resources: characterization and potential of bitterns as alternative sources of critical raw materials. J. Clean. Prod. 436, 140412. https://doi.org/10.1016/j.jclepro.2023.140412.

Reiss, A.G., Ganor, J., Hamawi, M., Gavrieli, I., 2021. Dynamics of turbidity in gypsumprecipitating brines: the case of the red sea-dead sea project. J. Environ. Manag. 288, 9. https://doi.org/10.1016/j.jenvman.2021.112369.

Saliba, M., Matsui, T., Domanski, K., Seo, J.Y., Ummadisingu, A., Zakeeruddin, S.M., Gratzel, M., 2016. Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance. Science 354 (6309), 206–209. https://doi.org/10.1126/science.aah5557.

SEArcularMINE, 2021. www.searcularmine.eu. (Accessed 19 December 2023).

Seawater reverse osmosis desalination brines (SWRO). www.sea4value.ue. (Accessed 19 December 2023).

Shahmansouri, A., Min, J., Jin, L., Bellona, C., 2015. Feasibility of extracting valuable minerals from desalination concentrate: a comprehensive literature review. J. Clean. Prod. 100, 4–16. https://doi.org/10.1016/j.jclepro.2015.03.031.

Shan, Z.-Q., Shu, X.-Q., Feng, J.-F., Zhou, W.-N., 2013. Modified calcination conditions of rare alkali metal Rb-containing muscovite ($KAl_2[AlSi_3O_{10}](OH)_2$). Rare Met. 32 (6), 632–635. https://doi.org/10.1007/s12598-013-0068-3.

Sharkh, B.A., Al-Amoudi, A.A., Farooque, M., Fellows, C.M., Ihm, S., Lee, S., Li, S., Voutchkov, N., 2022. Seawater desalination concentrate—a new frontier for sustainable mining of valuable minerals. npj Clean Water 5, 9. https://doi.org/10.1038/s41545-022-00153-6.

Sharma, S.K., Truong, D.Q., Guo, J., An, A.K., Naidu, G., Deka, B.J., 2023. Recovery of rubidium from brine sources utilizing diverse separation technologies. Desalination 556, 116578. https://doi.org/10.1016/j.desal.2023.116578.

Shi, Z., Du, X., Wang, S., Guo, Y., Deng, T., 2017. Application status of rubidium, cesium and research situation of its separation from brine with solvent extraction. IOP Conf. Ser. Mater. Sci. Eng. 274, 12081. https://doi.org/10.1088/1757-899X/274/1/012081.

Shi, Q., Wang, Y., Liu, W., Zhang, H., Li, Y., Jiao, J., Zhao, W., Shen, Y., 2023. Preparation of polysulfone-based rubidium ion imprinted membrane and its selective adsorption and separation performance. React. Funct. Polym. 191, 105696. https://doi.org/10.1016/j.reactfunctpolym.2023.105696.

Shvartsev, S.L., Kolpakov, M.N., Isupov, V.P., Vladimirov, A.G., Ariunbileg, S., 2014. Geochemistry and chemical evolution of saline Lakes of Western Mongolia. Geochem. Int. 52 (5), 388–403. https://doi.org/10.1134/S0016702914030070.

Siame, E., Pascoe, R.D., 2011. Extraction of lithium from micaceous waste from China clay production. Miner. Eng. 24, 1595–1602. https://doi.org/10.1016/j.mineng.2011.08.013.

Stankiewicz, A.I., Moulijn, J.A., 2000. Process intensification: transforming chemical engineering. Chem. Eng. Prog. 96, 22–34.

Sun, X., Wang, Y., Zhan, H., Shen, Y., Shi, Q., Chen, H., Sun, J., Zhang, Z., 2022. Fabrication and performance of the ammonium molybdophosphate/polysulfone mixed matrix membranes for rubidium adsorption in aqueous solution. J. Appl. Polym. Sci. 139, 51798. https://doi.org/10.1002/app.51798.

Sun, X., Shi, Q., Liu, W., Su, L., Shen, Y., Lei, W., Li, J., Meng, X., Zhang, Z., 2025. Efficiently selective adsorption Rb(I) based on ion-imprinted membrane chromatography: batch adsorption and dynamic filtration. Sep. Purif. Technol. 354 (1), 128707. https://doi.org/10.1016/j.seppur.2024.12870.

Taj, S., Muhammad, D., Chaudhry, M.A., Mazhar, M., 2011. Lithium, rubidium and cesium ion removal using potassium iron(III) hexacyanoferrate(II) supported on polymethylmethacrylate. J. Radioanal. Nucl. Chem. 288, 79–88. https://doi.org/10.1007/s10967-010-0873-1, 2011.

Tang, H., Zhao, L., Sun, W., Hu, Y., Ji, X., Han, H., Guan, Q., 2018. Extraction of rubidium from respirable sintering dust. Hydrometallurgy 175, 144–149. https://doi.org/10.1016/j.hydromet.2017.11.003.

Thompson, C.A., 2011. Mineral commodity summaries, 2011 Miner. Commod. Summ. 1–295.

Tian, L., Gong, A., Wu, X., Li, J., Liu, C., Yu, X., Xu, Z., 2020. Non-isothermal kinetic studies of rubidium extraction from muscovite using a chlorination roasting-water leaching process. Powder Technol. 373, 362–368. https://doi.org/10.1016/j.powtec.2020.06.015.

Tian, N., Gao, Y., Wu, J., Luo, S., Dai, W., 2019a. Water-resistant HKUST-1 functionalized with polydimethylsiloxane for efficient rubidium ion capture. New J. Chem. 43, 15539. https://doi.org/10.1039/c9nj03632f.

Tian, N., Dai, Y., Liu, Q., Dai, W., 2019b. Highly efficient capture of rubidium ion by a novel HS-Fe_3O_4@MIL-53(Al) composite material. Polyhedron 166, 109–114. https://doi.org/10.1016/j.poly.2019.03.052.

USGS, 2023. Mineral commodity summaries. https://pubs.usgs.gov/periodicals/mcs2023/mcs2023.pdf.

USGS, 2020. Mineral commodity summaries. https://pubs.usgs.gov/periodicals/mcs2020/mcs2020.pdf.

Vallès, V., López, J., Fernández de Labastida, M., Gibert, O., Leskinen, A., Koivula, R.T., Cortina, J.L., 2023. Polymeric and inorganic sorbents as a green option to recover critical raw materials at trace levels from sea saltwork bitterns. Green Chem. 25, 700–719. https://doi.org/10.1039/D2GC02338E.

Vieceli, N., Nogueira, C.A., Pereira, M.F.C., Dias, A.P.S., Durão, F.O., Guimarães, C., Margarido, F., 2017. Effects of mechanical activation on lithium extraction from a lepidolite ore concentrate. Miner. Eng. 102, 1–14. https://doi.org/10.1016/j.mineng.2016.12.001.

Vieceli, N., Nogueira, C.A., Pereira, M.F.C., Durao, F.O., Guimaraes, C., Margarido, F., 2018. Recovery of lithium carbonate by acid digestion and hydrometallurgical processing from mechanically activated lepidolite. Hydrometallurgy 175, 1–10. https://doi.org/10.1016/j.hydromet.2017.10.022.

Vicari, F., Randazzo, S., López, J., Fernández de Labastida, M., Vallès, V., Micale, G., Tamburini, A., D'Alì Staiti, G., Cortina, J.L., Cipollina, A., 2022. Mining minerals and critical raw materials from bittern: understanding metal ions fate in saltwork ponds. Sci. Total Environ. 847, 157544. https://doi.org/10.1016/j.scitotenv.2022.157544.

Vu, H., Bernardi, J., Jandova, J., Vaculíkova, L., Golias, V., 2013. Lithium and rubidium extraction from zinnwaldite by alkali digestion process: sintering mechanism and leaching kinetics. Int. J. Miner. Process. 123, 9–17. https://doi.org/10.1016/j.minpro.2013.04.014.

Wagner, F.S., 2011. Rubidium and Rubidium Compounds, Kirk-Othmer Encyclopedia of Chemical Technology, pp. 1–11, 2011 Available from: http://onlinelibrary.wiley.com/doi/10.1002/0471238961.1821020923010714.a01.pub3. (Accessed 18 October 2024).

Wang, J., Hu, H., Wu, K., 2020. Extraction of lithium, rubidium and cesium from lithium porcelain stone. Hydrometallurgy 191, 105233. https://doi.org/10.1016/j.hydromet.2019.105233.

Wang, Z., Zhang, A., Suen, J., 2021a. Selective adsorption of potassium and rubidium by metal–organic frame works supported supramolecular materials. J. Radioanal. Nucl. Chem. 329, 359–370. https://doi.org/10.1007/s10967-021-07779-9.

Wang, Y., Li, K., Fang, D., Ye, X., Liu, H., Tan, X., Li, Q., Li, J., Wu, Z., 2022. Ammonium molybdophosphate/metal-organic framework composite as an effective adsorbent for capture of Rb^+ and Cs^+ from aqueous solution. J. Solid State Chem. 306, 122767. https://doi.org/10.1016/j.jssc.2021.122767.

Wang, C., Yu, Y., Chen, Y., Cao, M., Wang, J., Yang, X., Qiu, S., Wei, D., Gao, H., Li, F., 2021b. Efficient quantum memory of orbital angular momentum qubits in cold atoms. Quantum Sci. Technol. 6, 45008. https://doi.org/10.1088/2058-9565/ac120a.

Wang, J., Che, D., Qin, W., 2015. Extraction of rubidium by t-BAMBP in cyclohexane. Chinese J. Cem. Eng. 23, 1110–1113. https://doi.org/10.1016/j.cjche.2015.04.005.

Warren, I., 2021. Techno-Economic Analysis of Lithium Extraction from Geothermal Brines. Golden, National Renewable Energy Laboratory. https://www.nrel.gov/docs/fy21osti/799178.pdf.

Wills, B.A., Finch, J.A., 2015. Wills' Mineral Processing Technology: an Introduction to the Practical Aspects of Ore Treatment and Mineral Recovery, eighth ed. Elsevier. Butterworth-Heinemann, ISBN 978-0-08-097053-0.

Wright, S.W., Hageman, D.L., McClure, L.D., 1994. Fluoride-mediated boronic acid coupling reactions. J. Org. Chem. 59, 6095–6097. https://doi.org/10.1021/jo0.

Xie, J., Li, K., Shi, Z., Min, C., Li, S., Yin, Z., Ma, R., 2023. Separation of ZCesium and rubidium from solution with high concentrations of potassium and sodium. Separations 10 (42), 1–12. https://doi.org/10.3390/separations10010042.

Xie, S., Shi, D., Zhang, Y., Peng, X., Lu, H., Li, L., 2024. Research status of separation and extraction of rubidium and cesium by inorganic adsorbents. Chin J. Rare Metals 47 (7), 1020–1042. https://doi.org/10.13373/j.cnki.cjrm.XY22120033.

Xing, P., Wang, C., Ma, B., Wang, L., Zhang, W., Chen, Y., 2018a. Rubidium and potassium extraction from granitic rubidium ore: process optimization and mechanism study. ACS Sustain. Chem. Eng. 6, 4922–4930. https://doi.org/10.1021/acssuschemeng.7b04445.

Xing, P., Wang, C., Wang, L., Ma, B., Chen, Y., Wang, G., 2018b. Clean and efficient process for the extraction of rubidium from granitic rubidium ore. J. Clean. Prod. 196, 64–73. https://doi.org/10.1016/j.jclepro.2018.06.041.

Xing, P., Wang, C., Chen, Y., Ma, B., 2021. Rubidium extraction from mineral and brine resources: a review. Hydrometallurgy 203, 105644. https://doi.org/10.1016/j.hydromet.2021.105644.

Xu, L., Wu, H., Dong, F., Wang, L., Wang, Z., Xiao, J., 2013. Flotation and adsorption of mixed cationic/anionic collectors on muscovite mica. Miner. Eng. 41, 41–45. https://doi.org/10.1016/j.mineng.2012.10.015.

Xu, X., Li, Y., Yang, D., Zheng, X., Wang, Y., Pan, J., Zhang, T., Xu, J., Qiu, F., Yan, Y., Li, C., 2018. A facile strategy toward ion-imprinted hierarchical mesoporous material via dual-template method for simultaneous selective extraction of lithium and rubidium. J. Clean. Prod. 171, 264–274. https://doi.org/10.1016/j.jclepro.2017.10.023.

Yan, Q., Li, X., Wang, Z., Wu, X., Guo, H., Hu, Q., Peng, W., Wang, J., 2012a. Extraction of valuable metals from lepidolite. Hydrometallurgy 117–118, 116–118. https://doi.org/10.1016/j.hydromet.2012.02.004.

Yan, Q., Li, X., Wang, Z., Wang, J., Guo, H., Peng, W., Wu, X., 2012b. Extraction of lithium from lepidolite using chlorination roasting–water leaching process. Trans. Nonferrous Metals Soc. China 22, 1753–1759. https://doi.org/10.1016/S1003-6326(11)61383-6.

Yan, Q., Li, X., Wang, Z., Wu, X., Wang, J., Guo, H., Hu, Q., Peng, W., 2012c. Extraction of lithium from lepidolite by sulfation roasting and water leaching. Int. J. Miner. Process. 110–111, 1–5. https://doi.org/10.1016/j.minpro.2012.03.005.

Yan, Q., Li, X., Yin, Z., Wang, Z., Guo, H., Peng, W., Hu, Q., 2012d. A novel process for extracting lithium from lepidolite. Hydrometallurgy 121–124, 54–59. https://doi.org/10.1016/j.hydromet.2012.04.006.

Yang, H., Ma, B., Lv, Y., Wang, C., Chen, Y., 2022. Novel technology for synergistic extraction of Li and Rb from a complex lithium concentrate. ACS Sustainable Chem. Eng. 10, 12030–12040. https://doi.org/10.2478/pjct-2023-0009.

Ye, X., Wu, Z., Li, W., Liu, H., Li, Q., Qing, B., Guo, M., Ge, F., 2009. Rubidium and cesium ion adsorption by an ammonium molybdophosphate–calcium alginate composite adsorbent. Colloids Surf. 342, 76–83. https://doi.org/10.1016/j.colsurfa.2009.04.011.

Yu, C., Lu, J., Hou, Z., Ma, Z., Lin, X., Dong, Z., Nie, W., Yan, Y., Wu, Y., 2021. Mixed matrix membranes for rubidium-dependent recognition and separation: a synergistic recombination design based on electrostatic interactions. Sep. Purif. Technol. 255, 117727. https://doi.org/10.1016/j.seppur.2020.117727.

Zeng, Q., Huang, L., Ouyang, D., Hu, Y., Zhong, H., He, Z., 2019. Process optimization on the extraction of rubidium from rubidium-bearing biotite. Miner. Eng. 137, 87–93. https://doi.org/10.1016/j.mineng.2019.03.020.

Zeng, Q., Li, S., Sun, W., Hu, L., Zhong, H., He, Z., 2021. Eco-friendly leaching of rubidium from biotite-containing minerals with oxalic acid and effective removal of Hg^{2+} from aqueous solution using the leaching residues. J. Clean. Prod. 306, 127167. https://doi.org/10.1016/j.jclepro.2021.127167.

Zhang, J., Yang, L., Dong, T., Pan, F., Xing, H., Liu, H., 2018. Kinetics-controlled separation intensification for cesium and rubidium isolation from salt Lake brine. Ind. Eng. Chem. Res. 57, 4399–4406. https://doi.org/10.9719/eeg.2020.53.5.529.

Zhang, X., Tan, X., Li, C., Yi, Y., Liu, W., Zhang, L., 2019. Energy-efficient and simultaneous extraction of lithium, rubidium and cesium from lepidolite concentrate via sulfuric acid baking and water leaching. Hydrometallurgy 185, 244–249. https://doi.org/10.1016/j.hydromet.2019.02.011.

Zhang, X., Aldahri, T., Tan, X., Liu, W., Zhang, L., Tang, S., 2020a. Efficient co-extraction of lithium, rubidium, cesium and potassium from lepidolite by process intensification of chlorination roasting. Chem. Eng. Process 147, 107777. https://doi.org/10.1016/j.cep.2019.107777.

Zhang, X., Qin, Z., Aldahrid, T., Ren, S., Liu, W., 2020b. Solvent extraction of rubidium from a sulfate solution with high concentrations of rubidium and potassium using 4-tert-butyl-2-,(a-methylbenzyl)-phen, ol. J. Taiwan Inst. Chem. Eng. 116, 43–50. https://doi.org/10.1016/j.jtice.2020.11.014.

Zhang, X.Y., Lin, H., Huang, H., Reckmeier, C., Zhang, Y., Choy, W.C.H., Rogach, A.L., 2016. Enhancing the brightness of cesium lead halide perovskite nanocrystal based greenlight-emitting devices through the interface engineering with perfluorinated ionomer. Nano Lett. 16, 1415–1420. https://doi.org/10.1021/acs.nanolett.5b04959.

Zhang, Z., Zhao, T., He, L., Zhao, Z., Sun, F., Xu, W., Liu, D., 2023. Electrochemical extraction of rubidium from salt lake by using cupric, ferrocyanide based on potassium shuttle. Desalination 549, 116331. https://doi.org/10.1016/j.desal.2022.116331.

Zheng, S., Li, P., Tian, L., Cao, Z., Zhang, T., Chen, Y., Zhang, Y., 2016. A chlorination roasting process to extract rubidium from distinctive kaolin ore with alternative chlorinating reagent. Int. J. Miner. Process. 157, 21–27. https://doi.org/10.1016/j.minpro.2016.09.006.

Zhong, J., Yao, W.G., Lee, W.H., 2007. Cesium chloride protects cerebellar granule neurons from apoptosis induced by low potassium. Int. J. Dev. Neurosci. 25, 359–365. https://doi.org/10.1016/j.ijdevneu.2007.07.003.0099a049.

Zhong, S., Liang, D., Weng, W., Chi, X., Zhang, W., Tan, W., 2024. Efficient extraction of Li and Rb from zinnwaldite via thermal activation and acid leaching. Miner. Eng. 216, 108898. https://doi.org/10.1016/j.mineng.2024.108898.

Zhou, L., Yuan, T., Li, R., Zhong, Y., Lei, X., 2015. Extraction of rubidium from kaolin clay waste: process study. Hydrometallurgy 158, 61–67. https://doi.org/10.1016/j.hydromet.2015.10.010.

Zhou, H., Liu, Y., Ma, B., Wang, B., Chen, Y., 2023. Strengthening extraction of lithium and rubidium from activated α-spodumene concentrate via sodium carbonate roasting. J. Ind. Eng. Chem. 123, 248–259. https://doi.org/10.1016/j.jiec.2023.03.040.

Zhou, Z., Hu, Y., Wang, Z., Zhang, H., Zhang, B., Ren, Z., 2021. Facile preparation of a rubidium ion-imprinted polymer by bulk polymerization for highly efficient separation of rubidium ions from aqueous solution. New J. Chem. 45, 9582. https://doi.org/10.1039/D1NJ00568E.

Zhu, Y., Wang, W., Zhang, H., Ye, X., Wu, Z., Wang, A., 2017. Fast and high-capacity adsorption of Rb+ and Cs+ onto recyclable, magnetic porous spheres. Chem. Eng. J. 327, 982–991. https://doi.org/10.1016/j.cej.2017.06.169.

Zinnwaldite. https://crystallography365.wordpress.com/2014/09/12/where-does-all-the-lithium-come-from-lepidolite/ date accessed; 03-12–2024.